



(12) **United States Patent**
Yanagida

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(45) **Date of Patent:** ***Aug. 12, 2025**

(54) **ELECTRONIC DEVICE WITH MULTIPLE
RESIN LAYERS REDUCING SUPPRESSION
IN RELIABILITY**

(71) Applicant: **ROHM CO., LTD.**, Kyoto (JP)

(72) Inventor: **Hideaki Yanagida**, Kyoto (JP)

(73) Assignee: **ROHM CO., LTD.**, Kyoto (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 332 days.

This patent is subject to a terminal disclaimer.

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(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation of application No. 16/704,961, filed on Dec. 5, 2019, now Pat. No. 11,417,624.

(30) **Foreign Application Priority Data**

Feb. 26, 2019 (JP) 2019-032444

(51) **Int. Cl.**

H01L 23/00 (2006.01)

H01L 21/56 (2006.01)

H01L 23/31 (2006.01)

(52) **U.S. Cl.**

CPC **H01L 24/45** (2013.01); **H01L 21/56** (2013.01); **H01L 23/3192** (2013.01); **H01L 24/43** (2013.01); **H01L 24/85** (2013.01)

(58) **Field of Classification Search**

CPC H01L 24/45; H01L 24/85; H01L 24/16; H01L 24/81; H01L 24/13;

(Continued)

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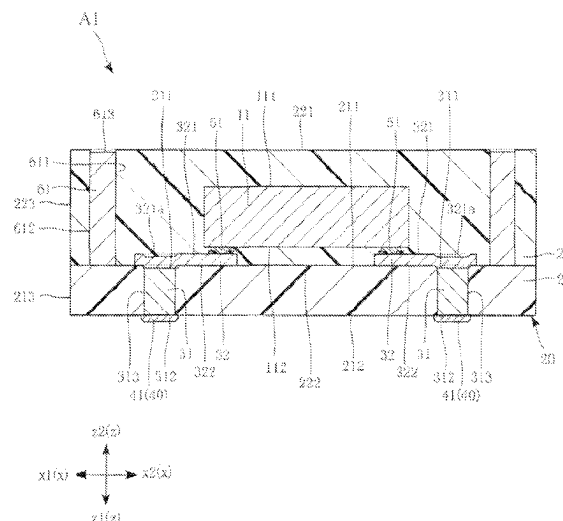
Primary Examiner — Didarul A Mazumder

(74) *Attorney, Agent, or Firm* — HSML P. C.

(57) **ABSTRACT**

The present disclosure provides an electronic device. The electronic device includes a first resin layer, having a first resin layer main surface and a first resin layer inner surface; a first conductor, having a first conductor main surface and a first conductor inner surface; a first wiring layer, formed adjacent to the first resin layer main surface and connected to the first conductor main surface; a first electronic component, electrically connected with the first wiring layer; a second resin layer, having a second resin layer main surface facing same direction as the first resin layer main surface and a second resin layer inner surface being in contact with the first resin layer main surface; an external electrode; and a second conductor, penetrating the second resin layer, wherein the second conductor is disposed on a periphery of the first electronic component.

14 Claims, 44 Drawing Sheets



(58) **Field of Classification Search**

CPC H01L 2224/16227; H01L 21/56; H01L
 23/3192; H01L 23/552; H01L
 2924/15311; H01L 23/3114; H01L
 2924/19102; H01L 25/0657; H01L 24/97;
 H01L 2224/97; H01L 2224/81; H01L
 23/3121

USPC 257/786; 438/612, 666
 See application file for complete search history.

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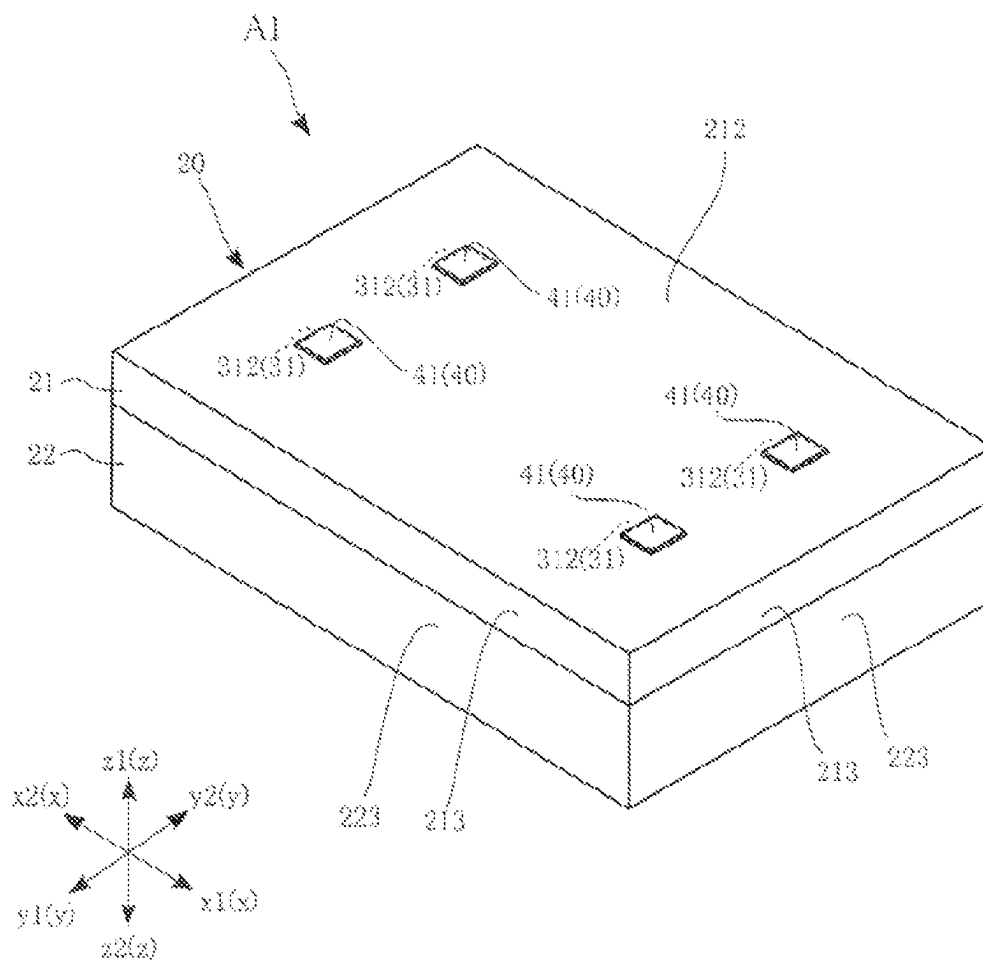


FIG. 1

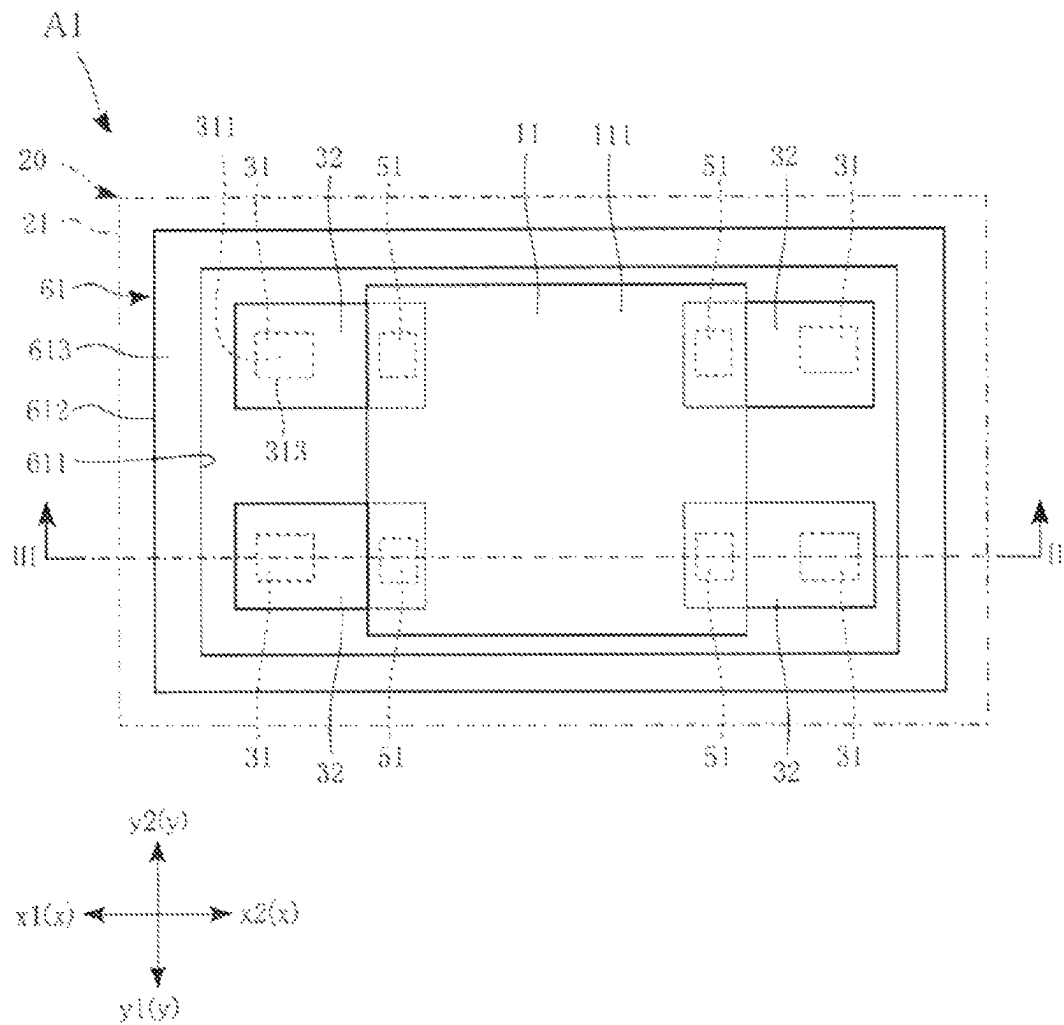


FIG.2

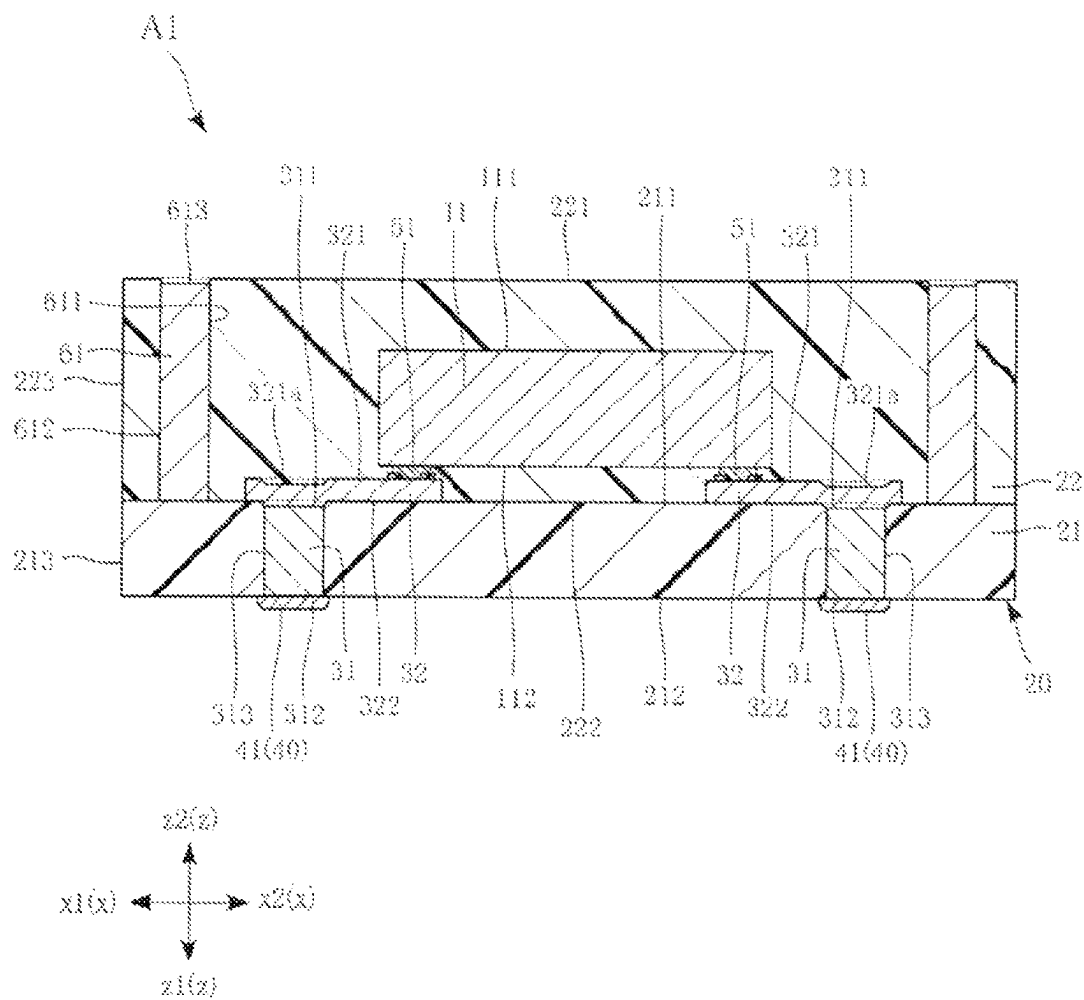


FIG. 3

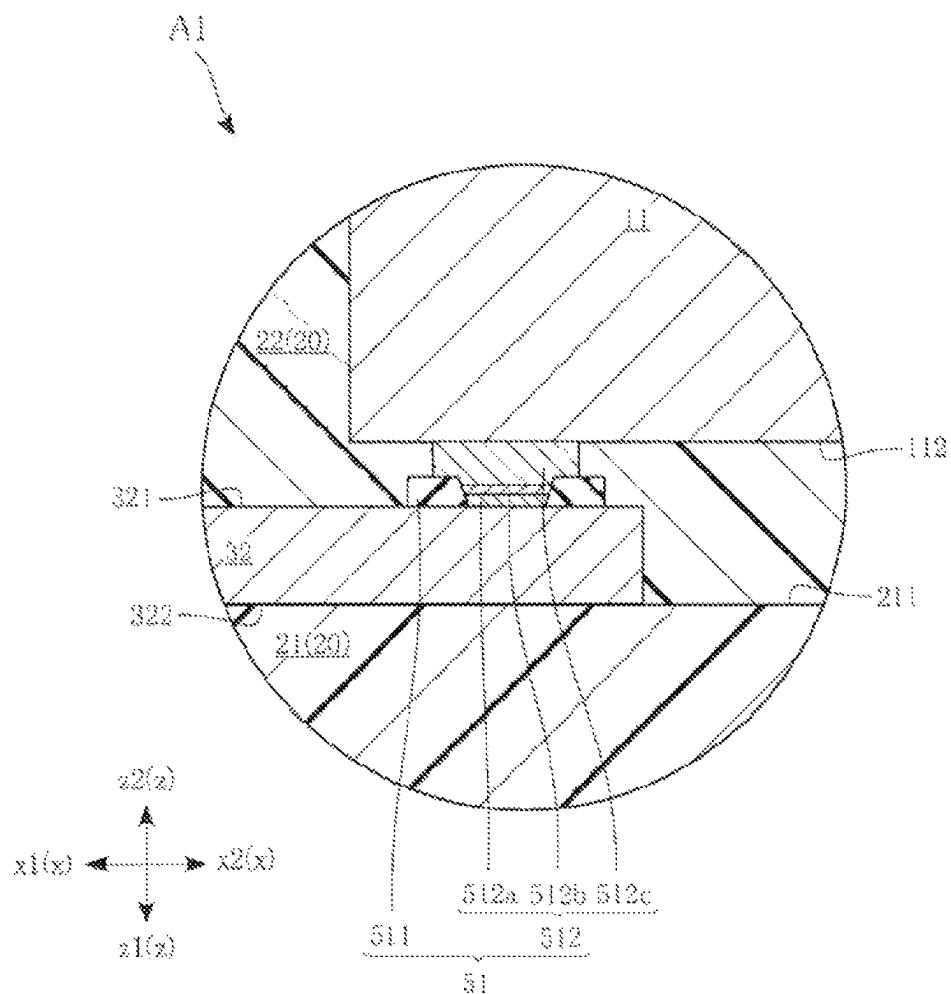


FIG.4

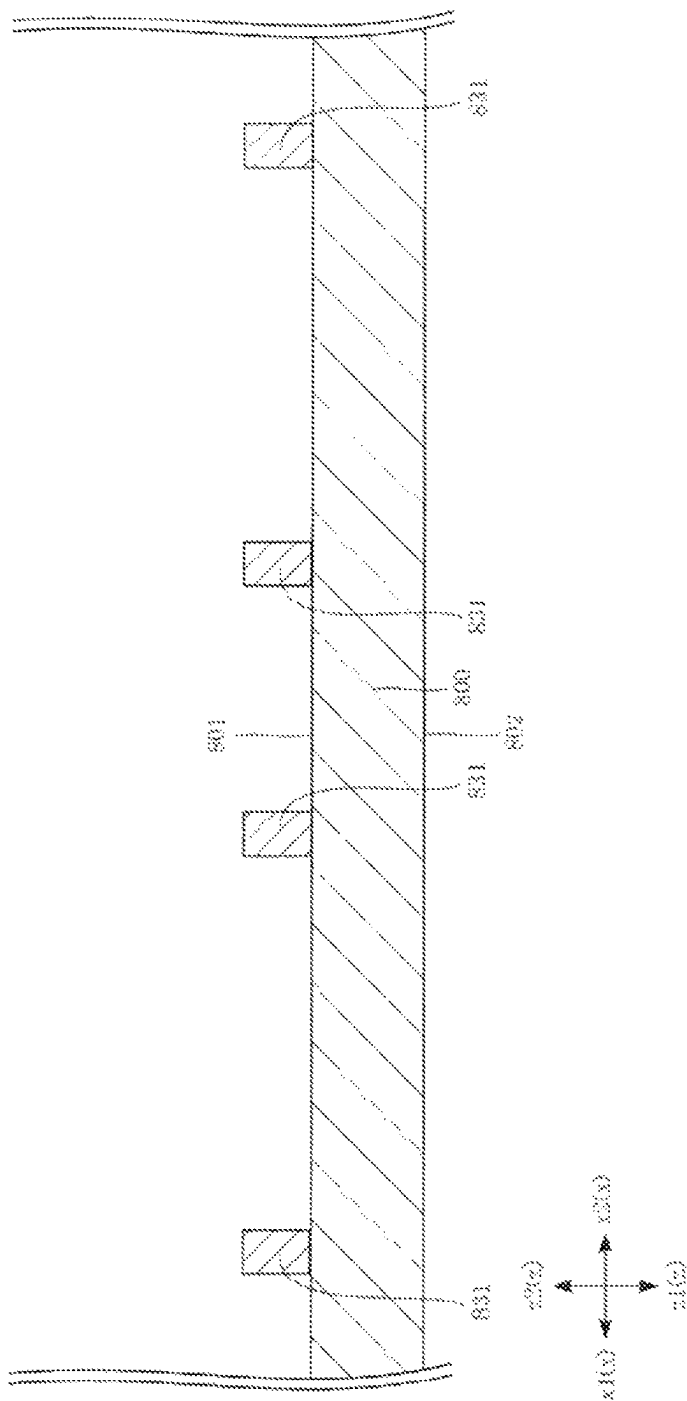


FIG.5

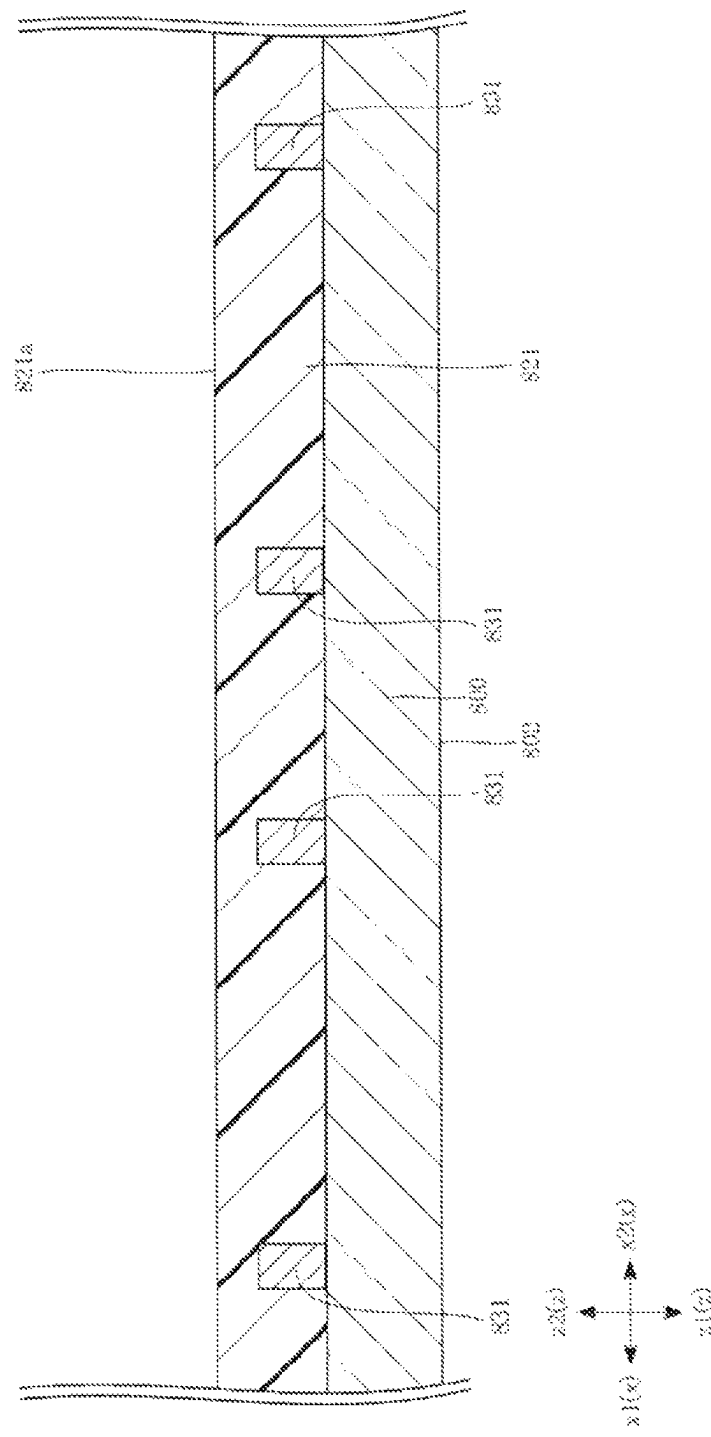


FIG.6

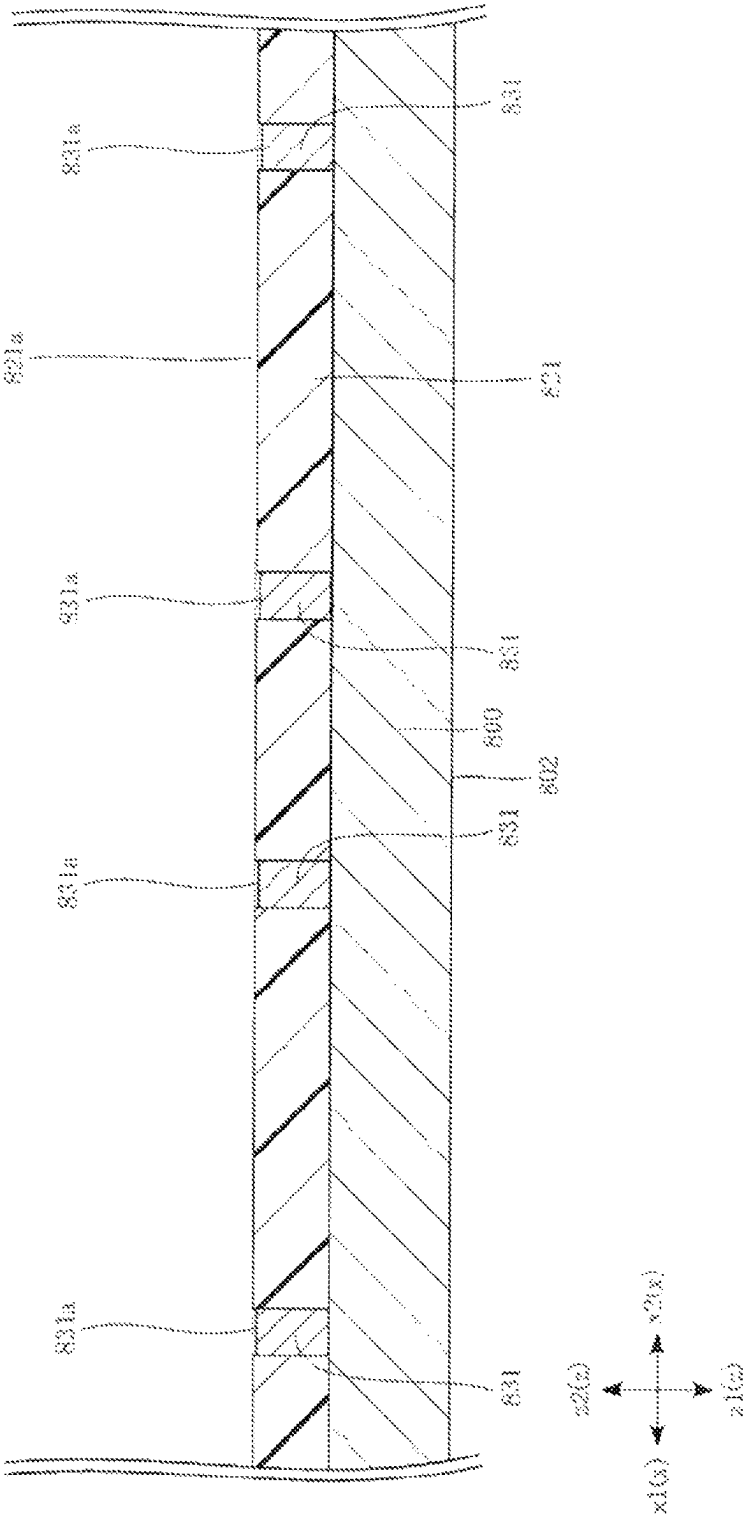


FIG. 7

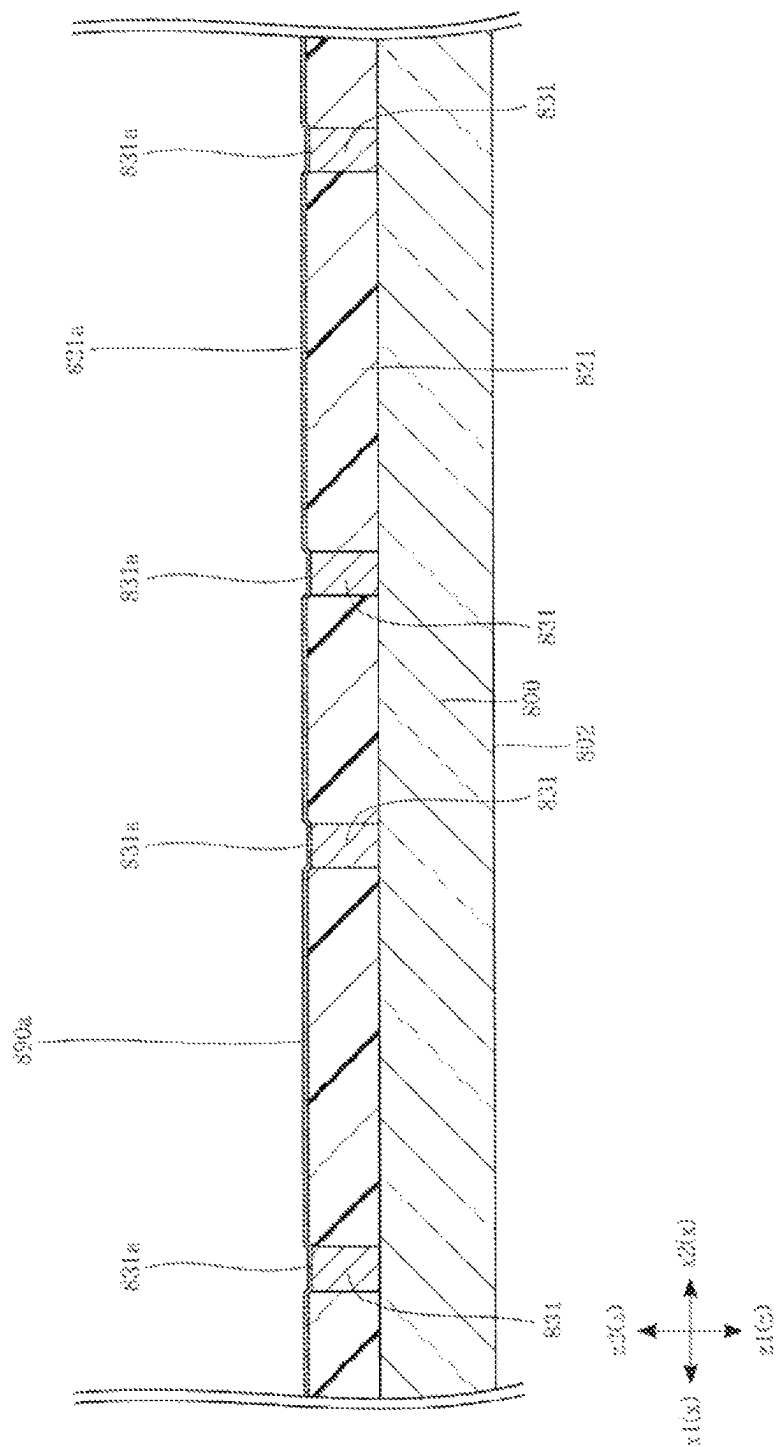


FIG. 8

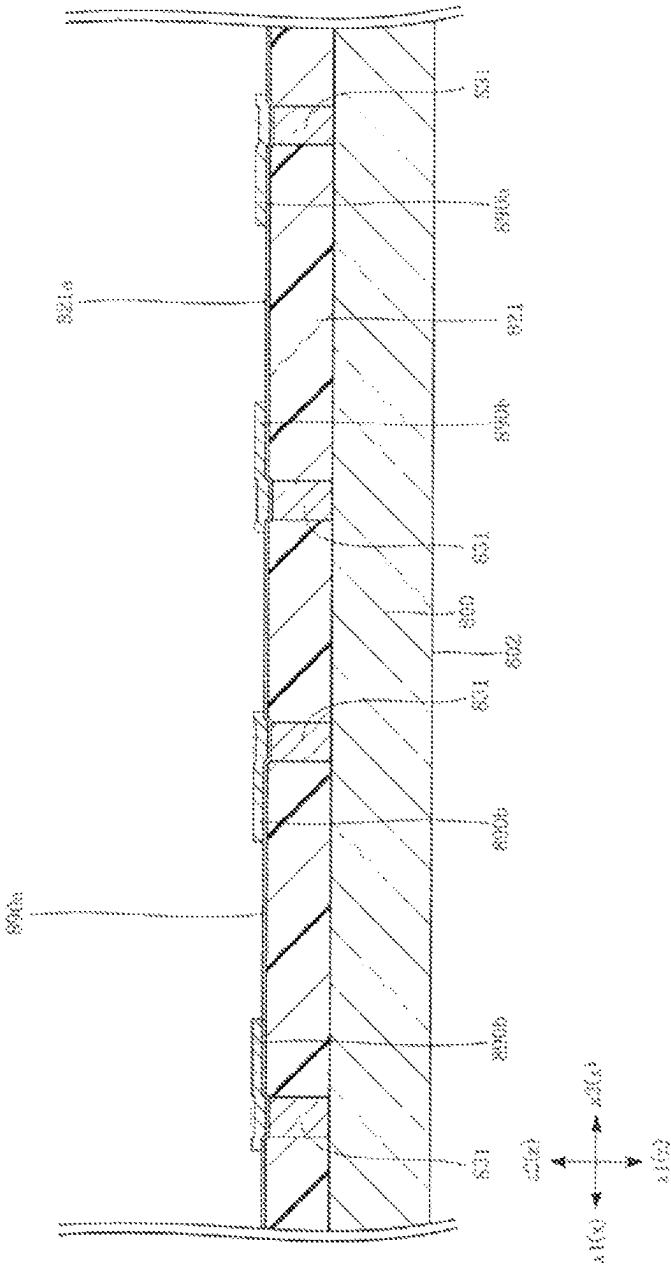


FIG.9

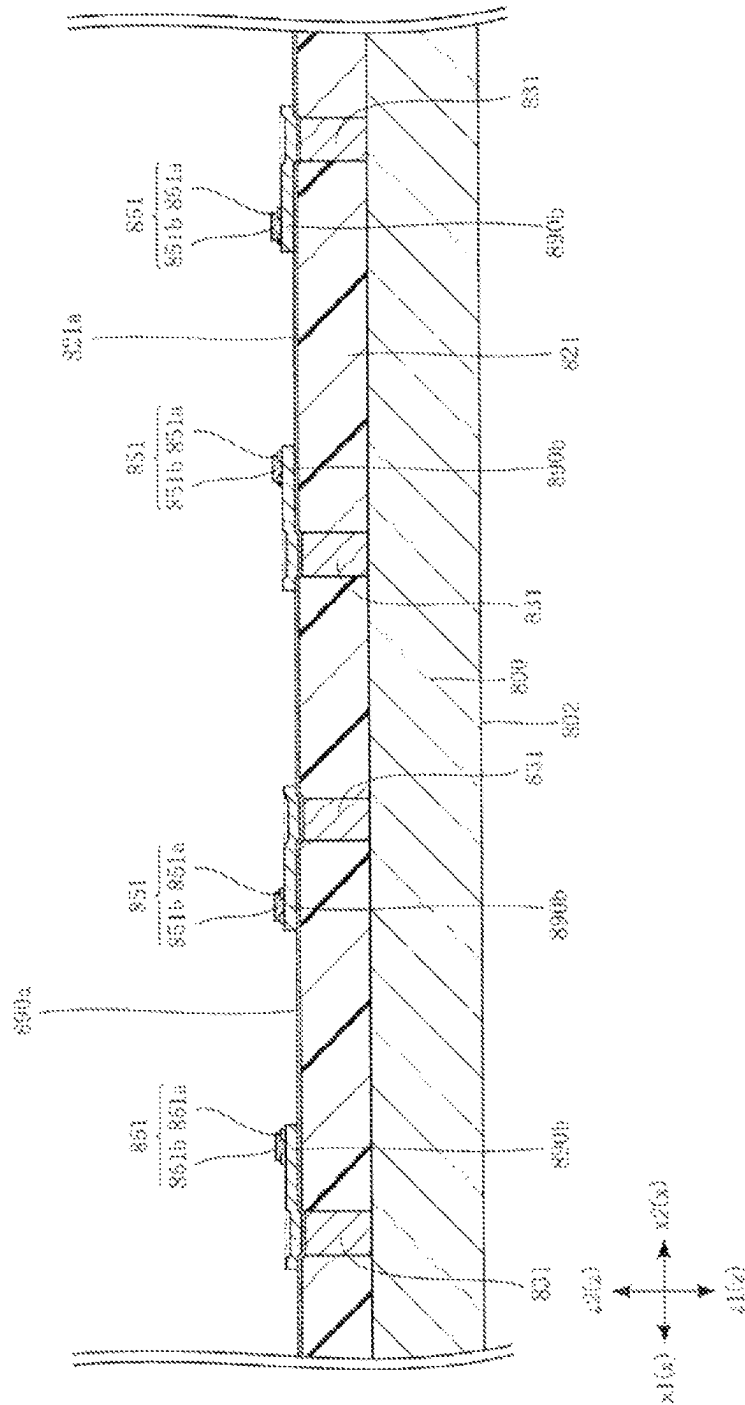


FIG. 10

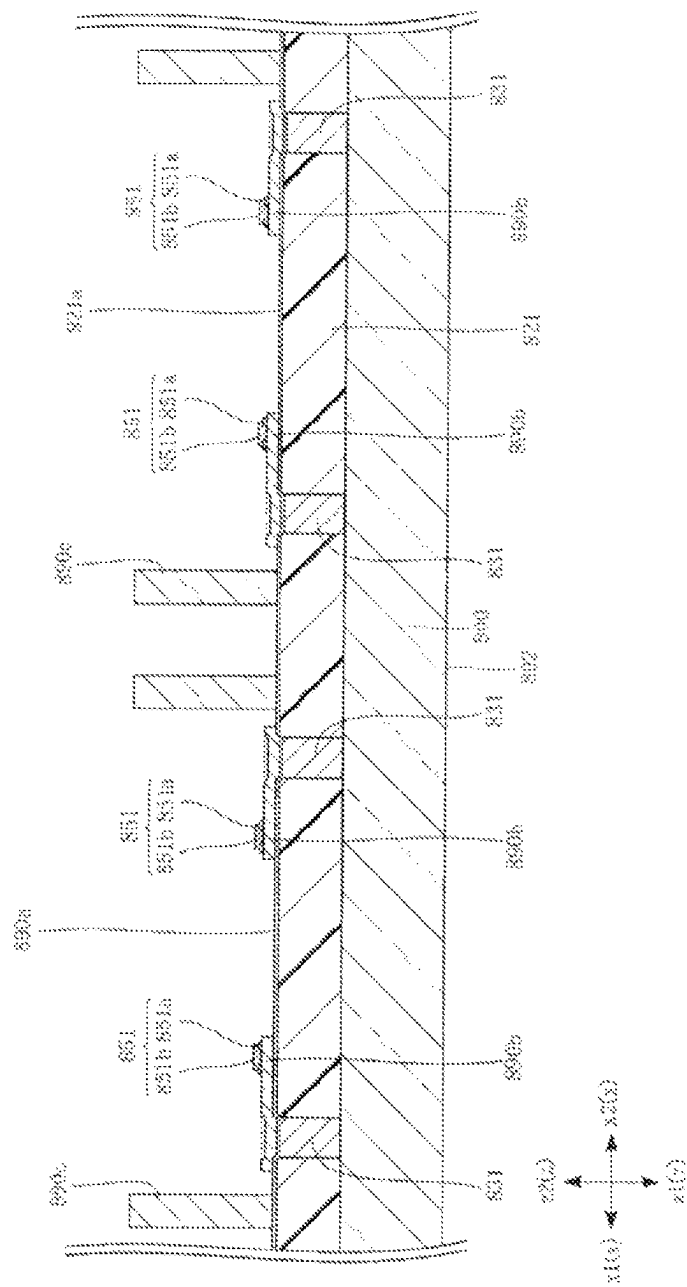


FIG. 11

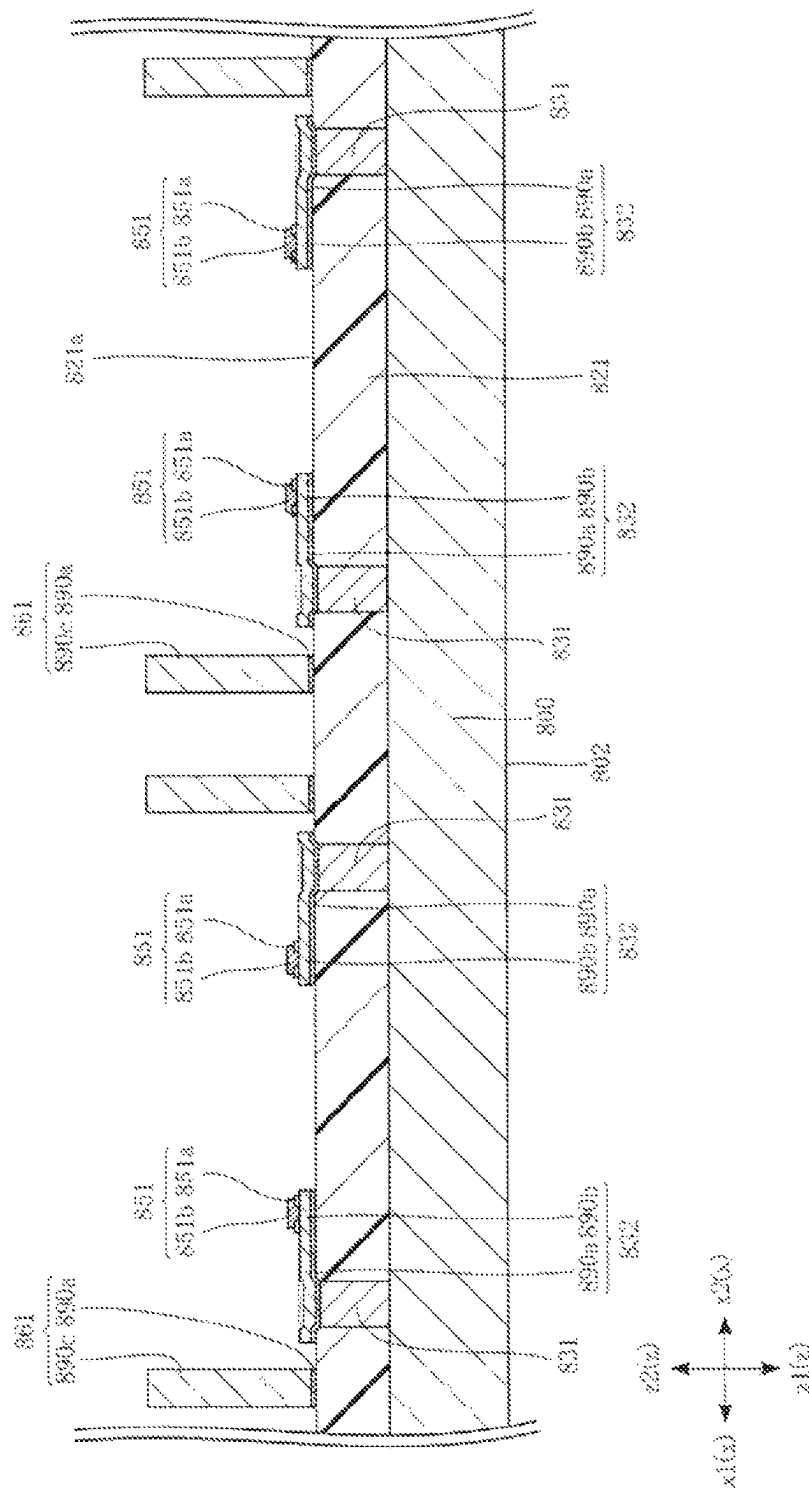


FIG. 12

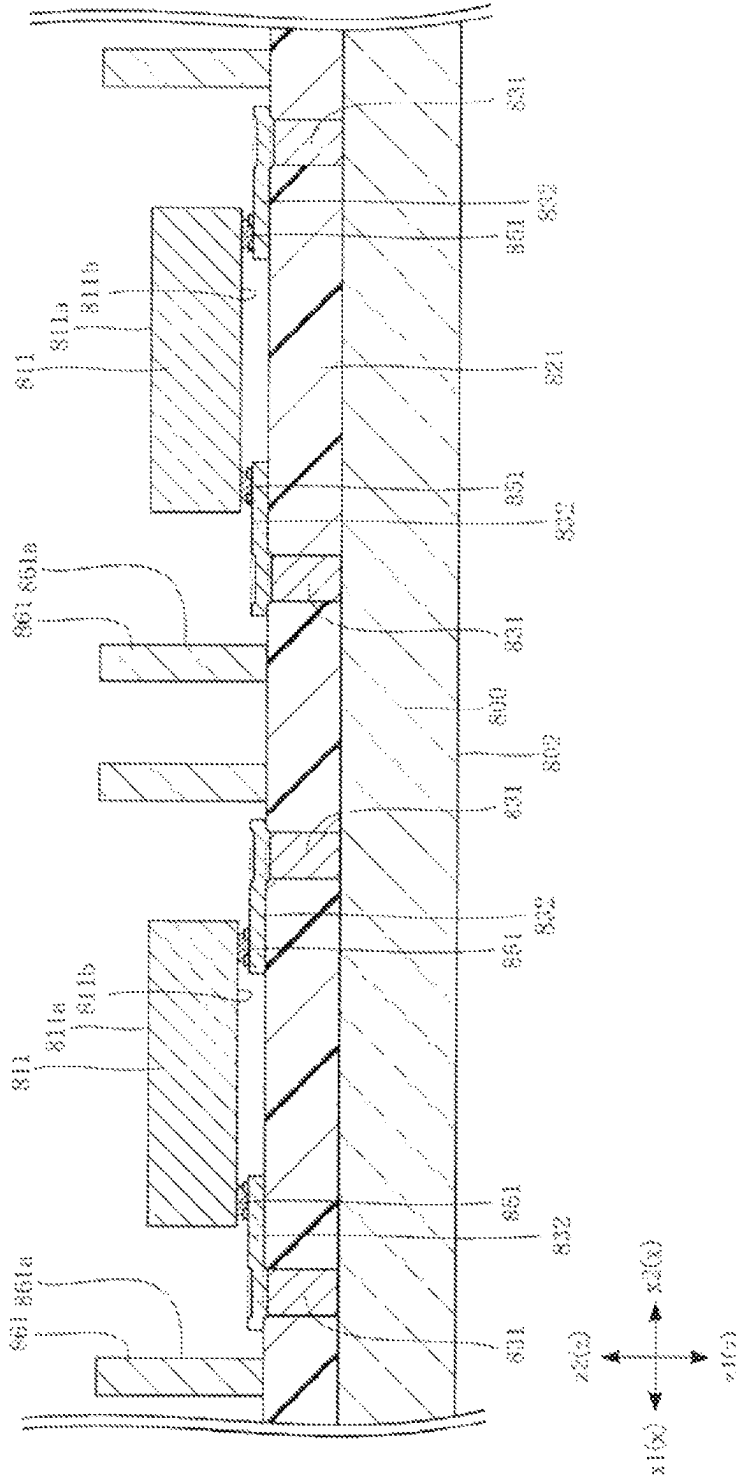


FIG.13

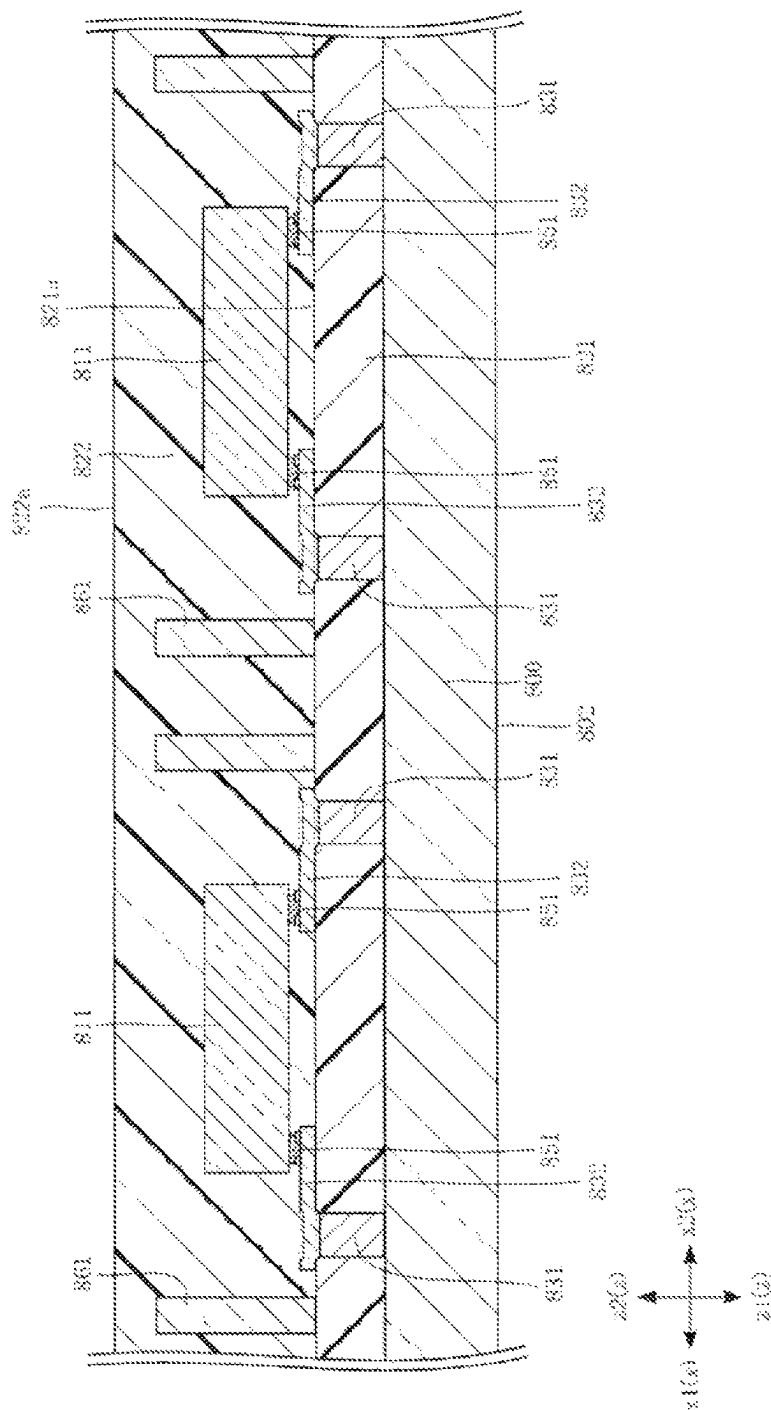


FIG. 14

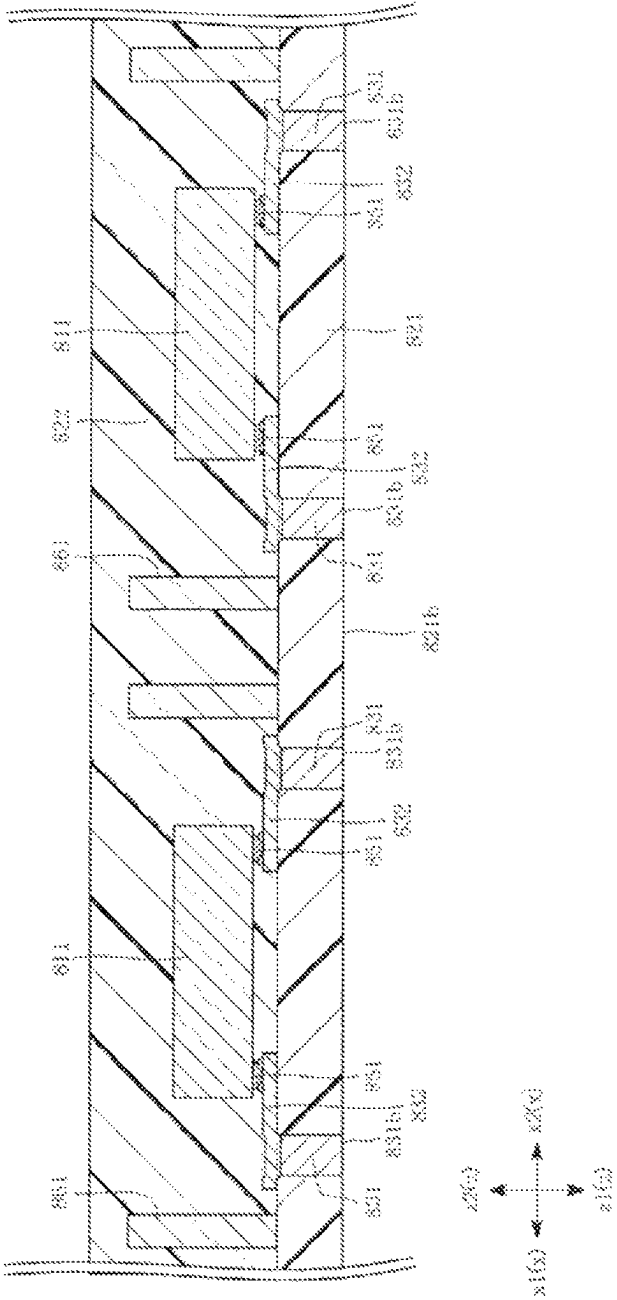


FIG. 15

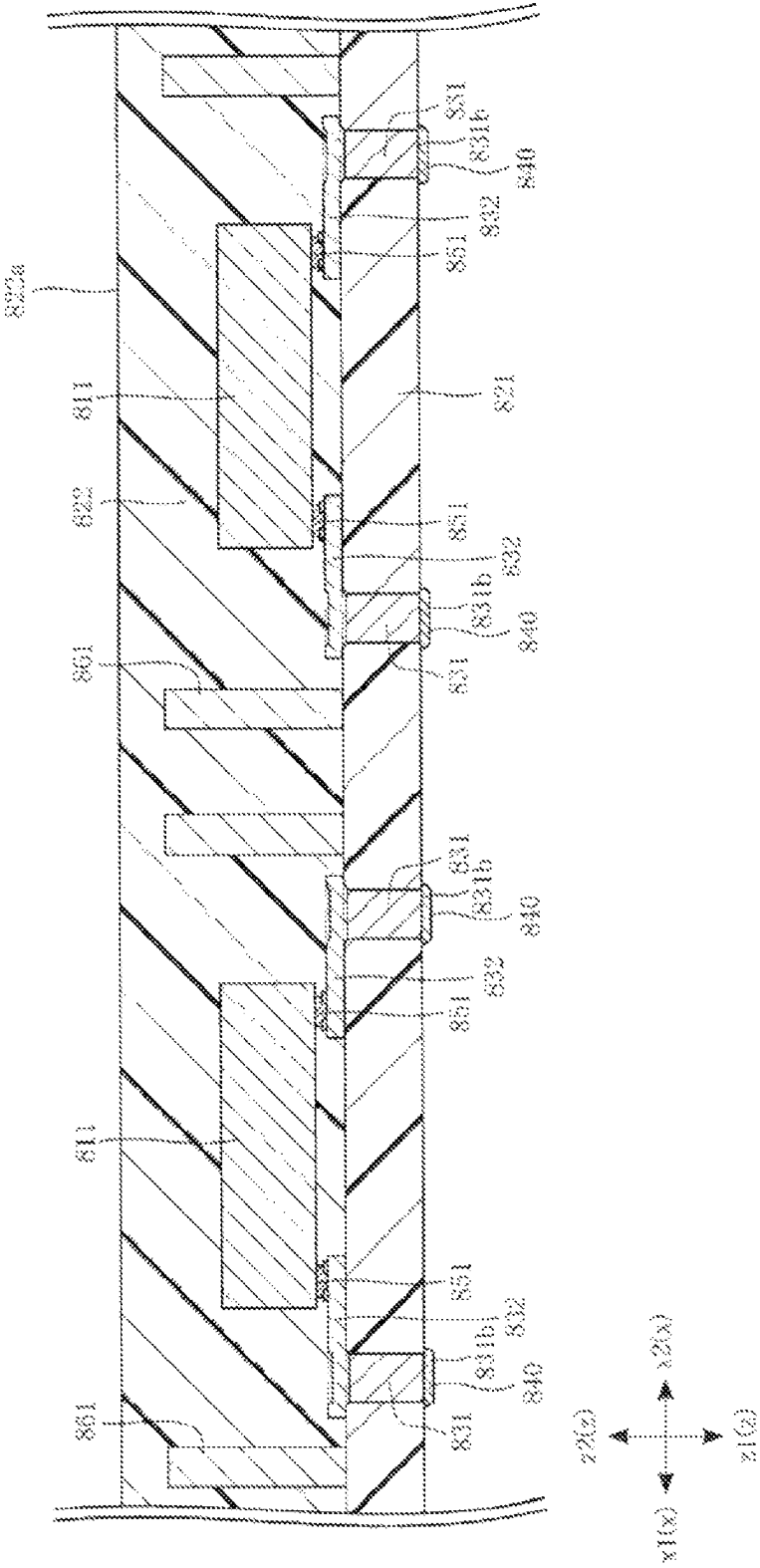


FIG.16

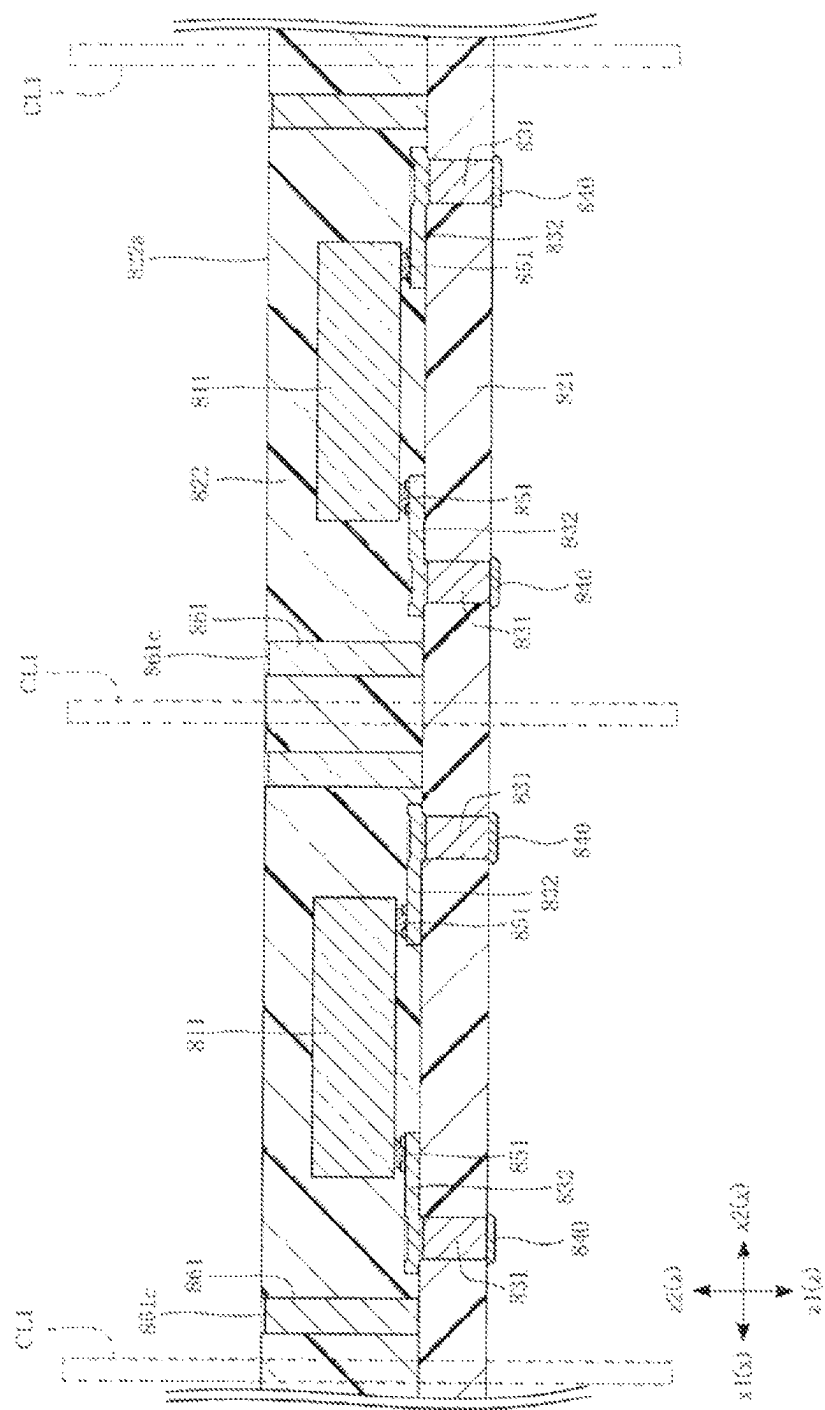


FIG.17

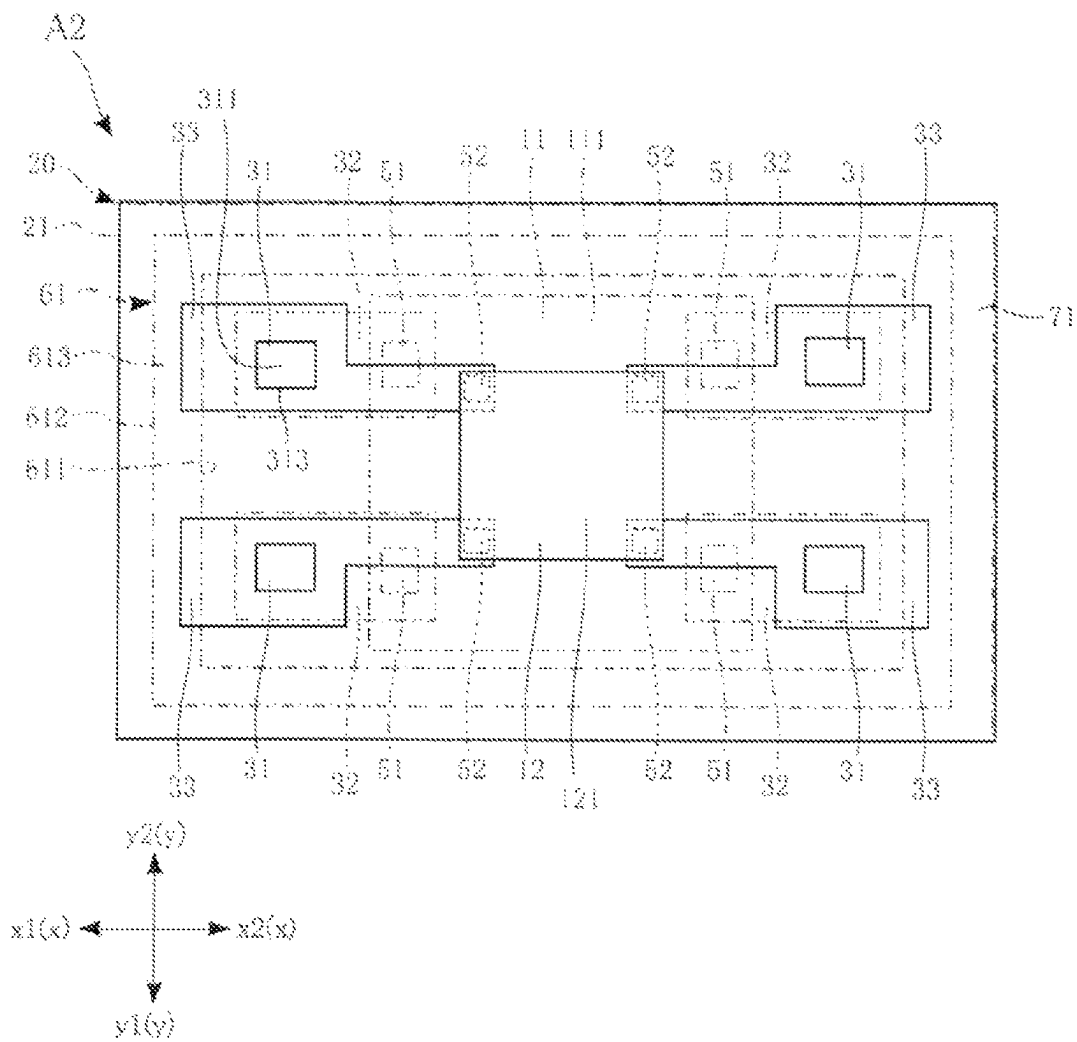


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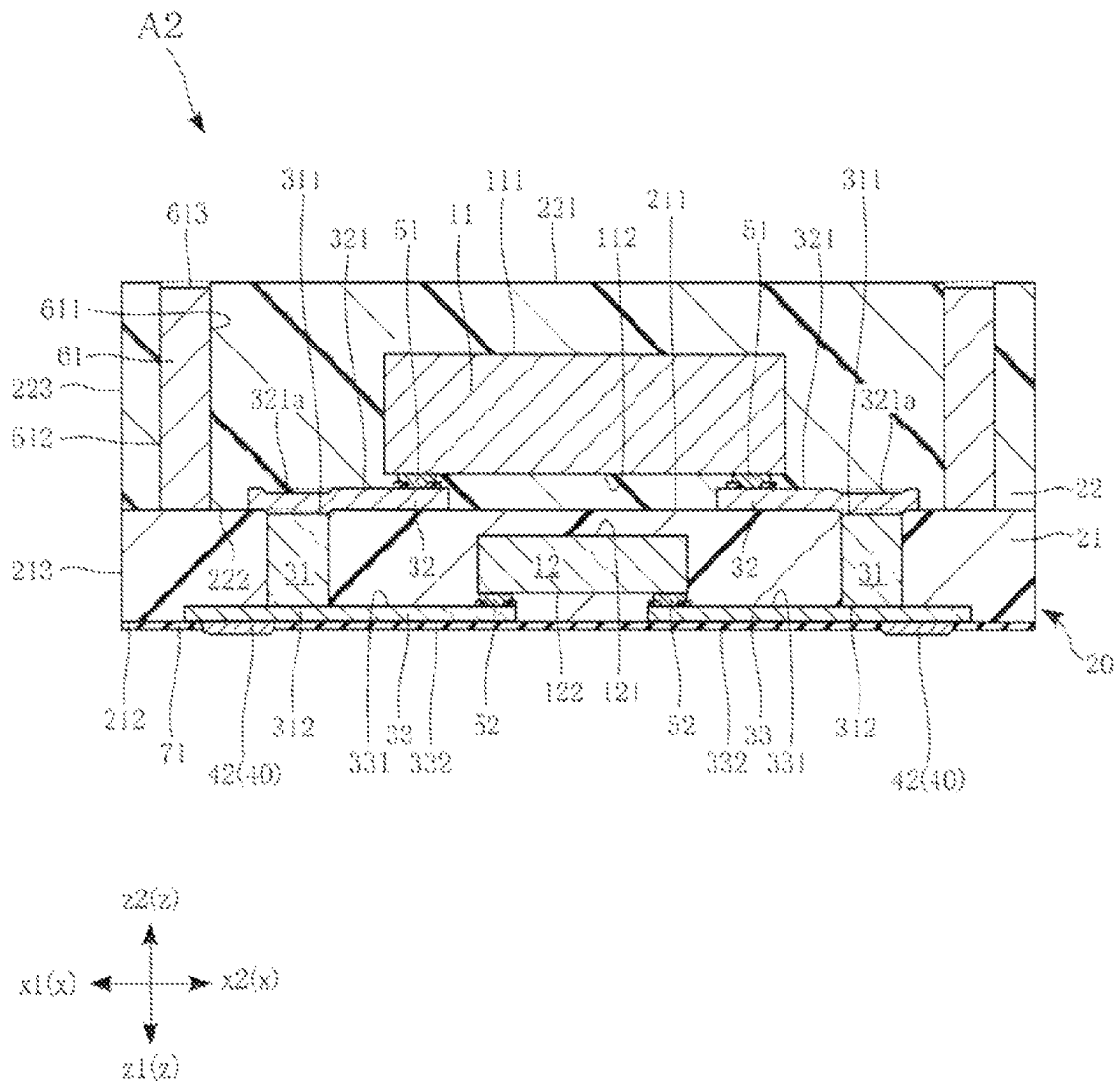


FIG. 20

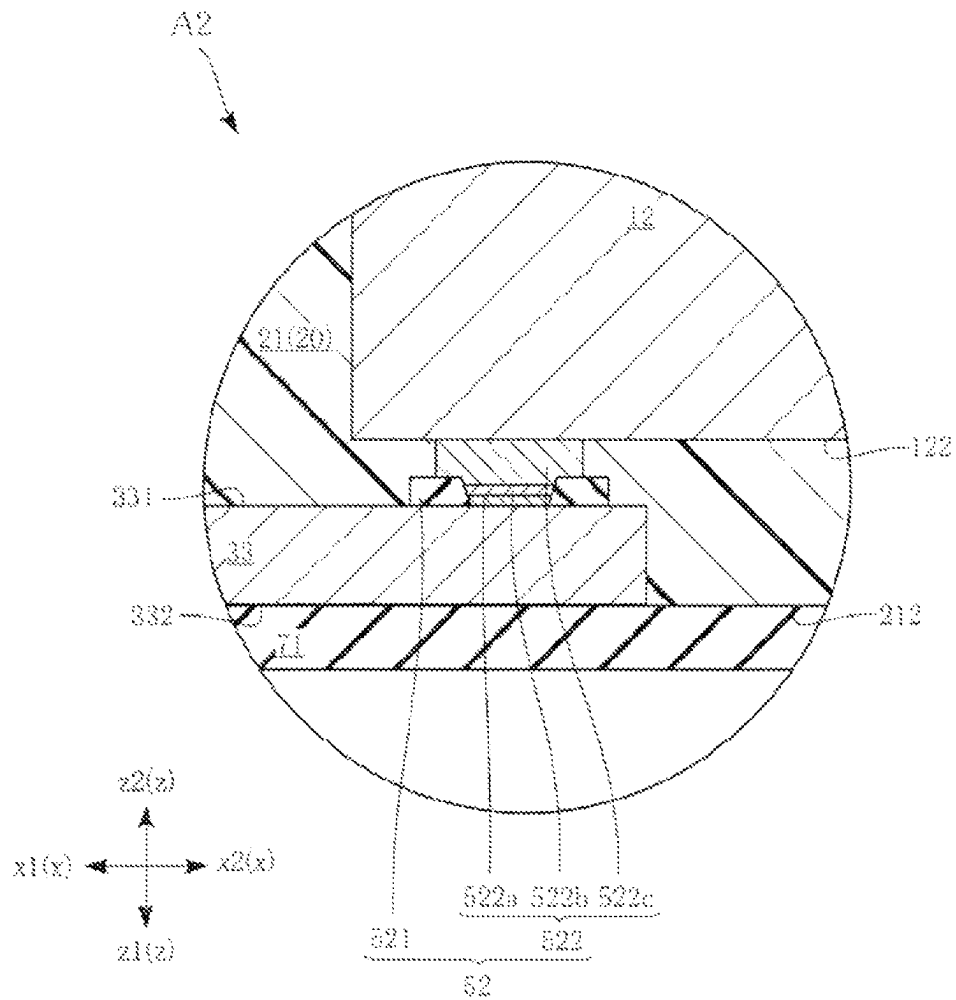


FIG. 21

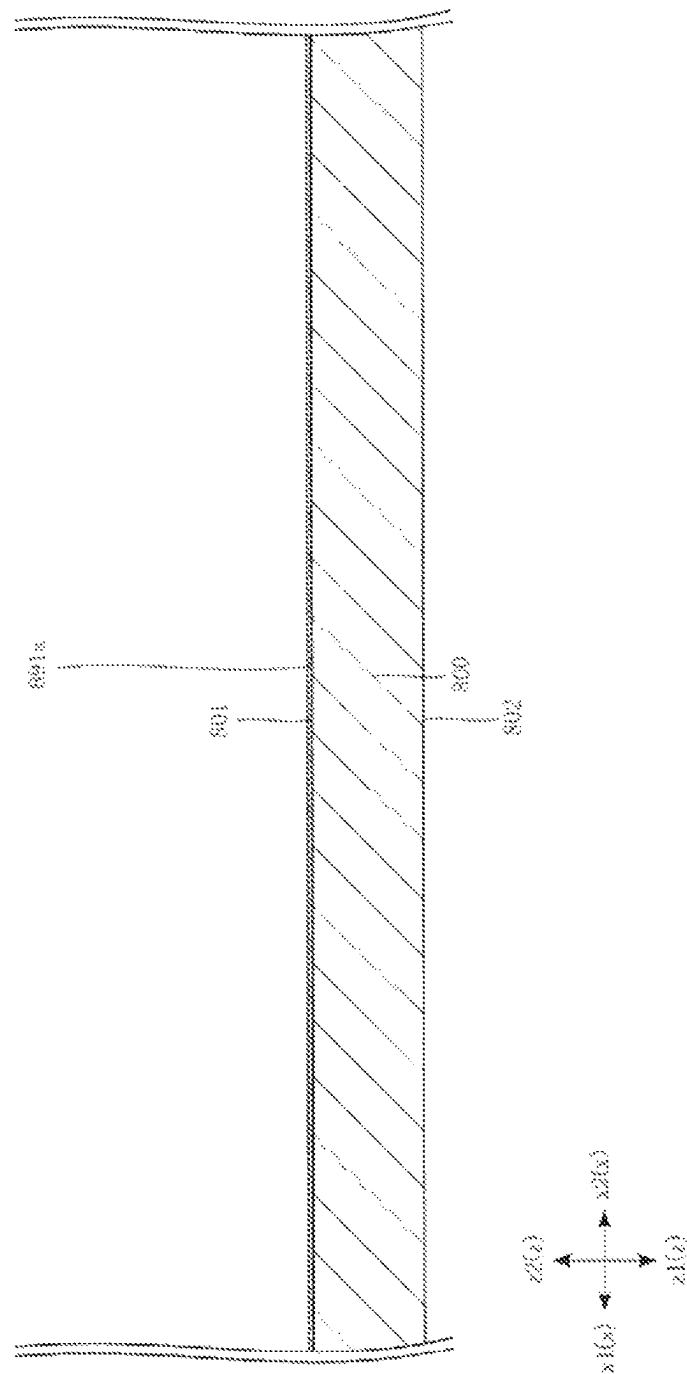


FIG.22

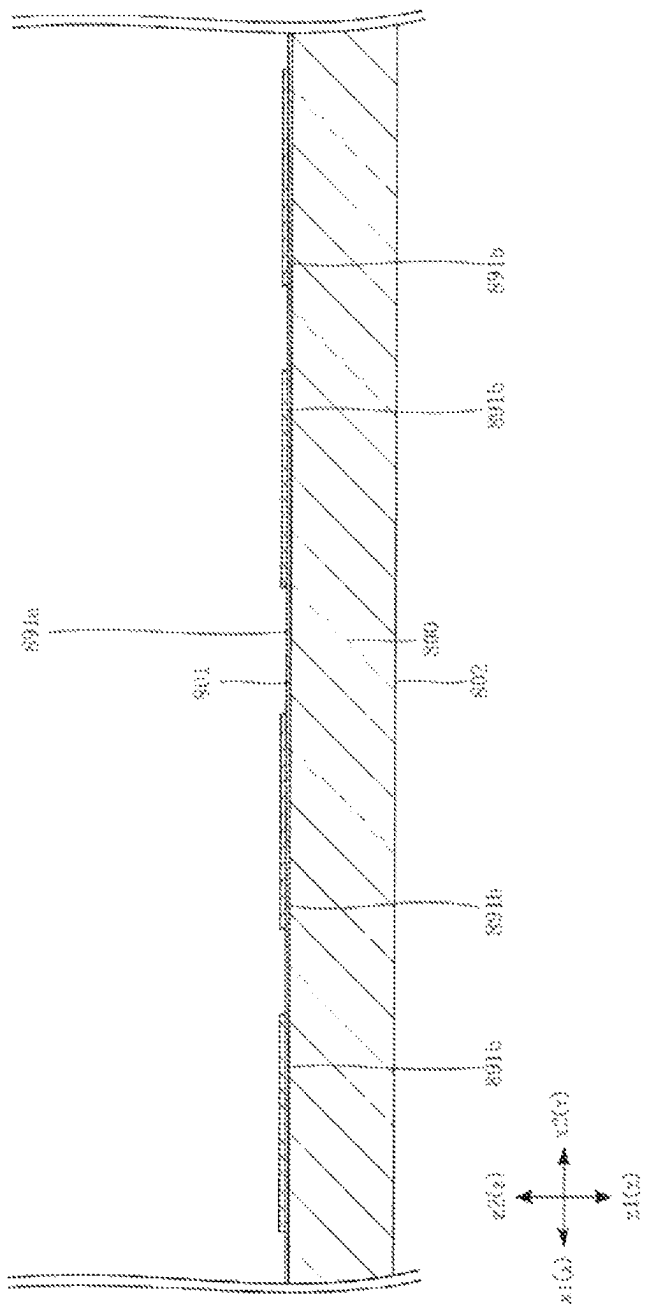


FIG. 23

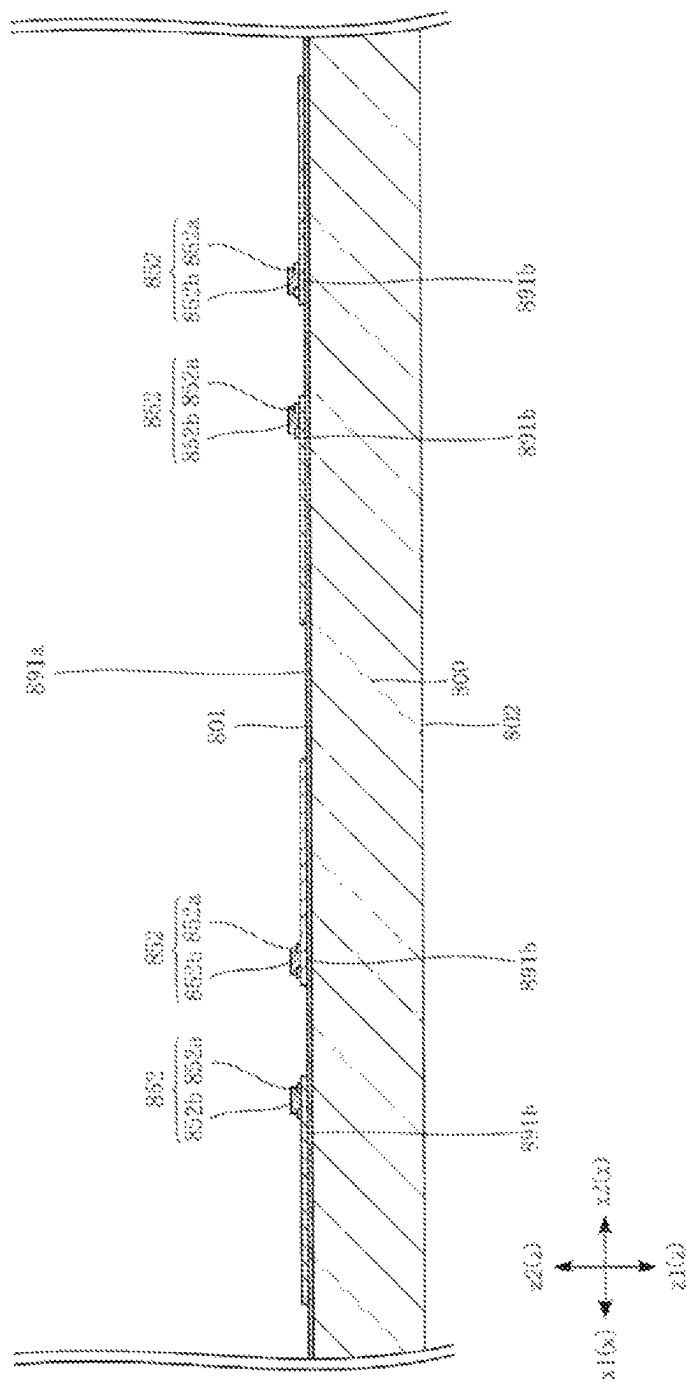


FIG. 24

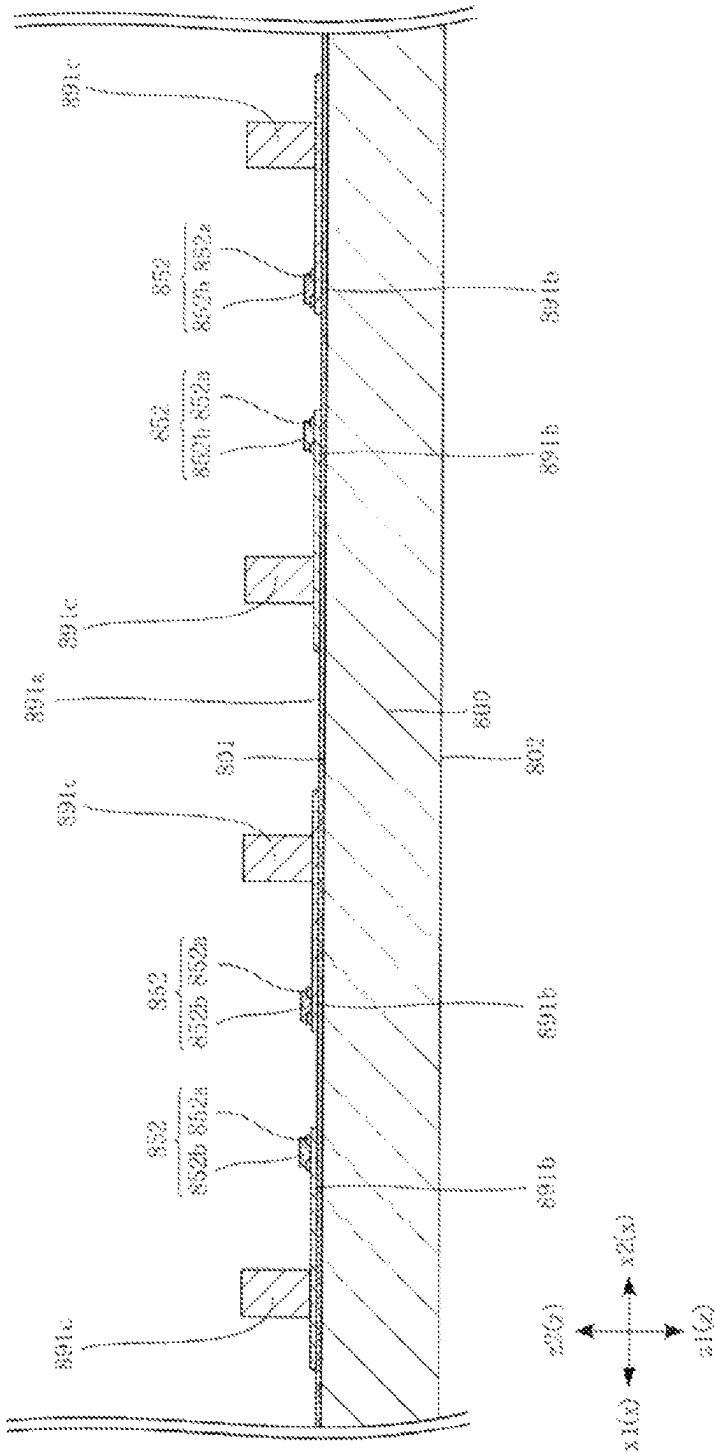


FIG.25

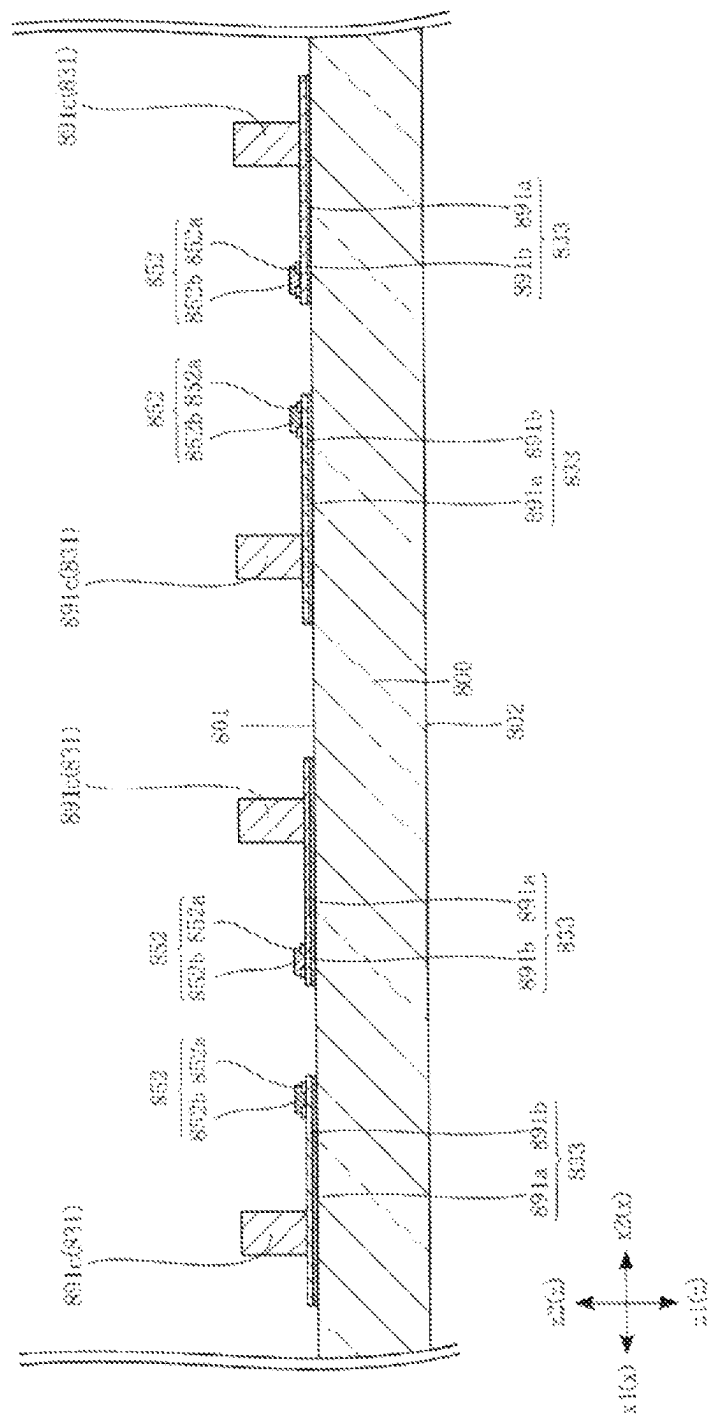


FIG. 26

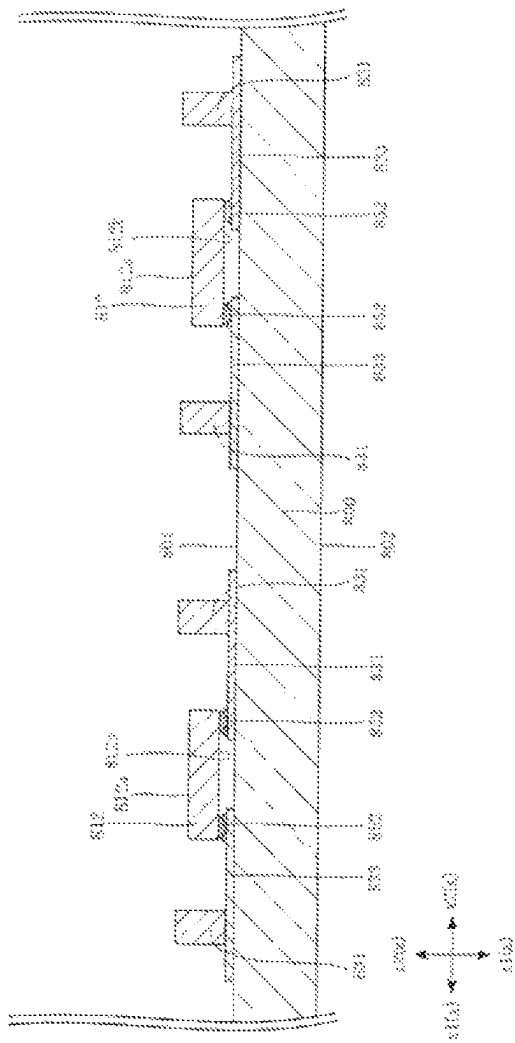


FIG. 27

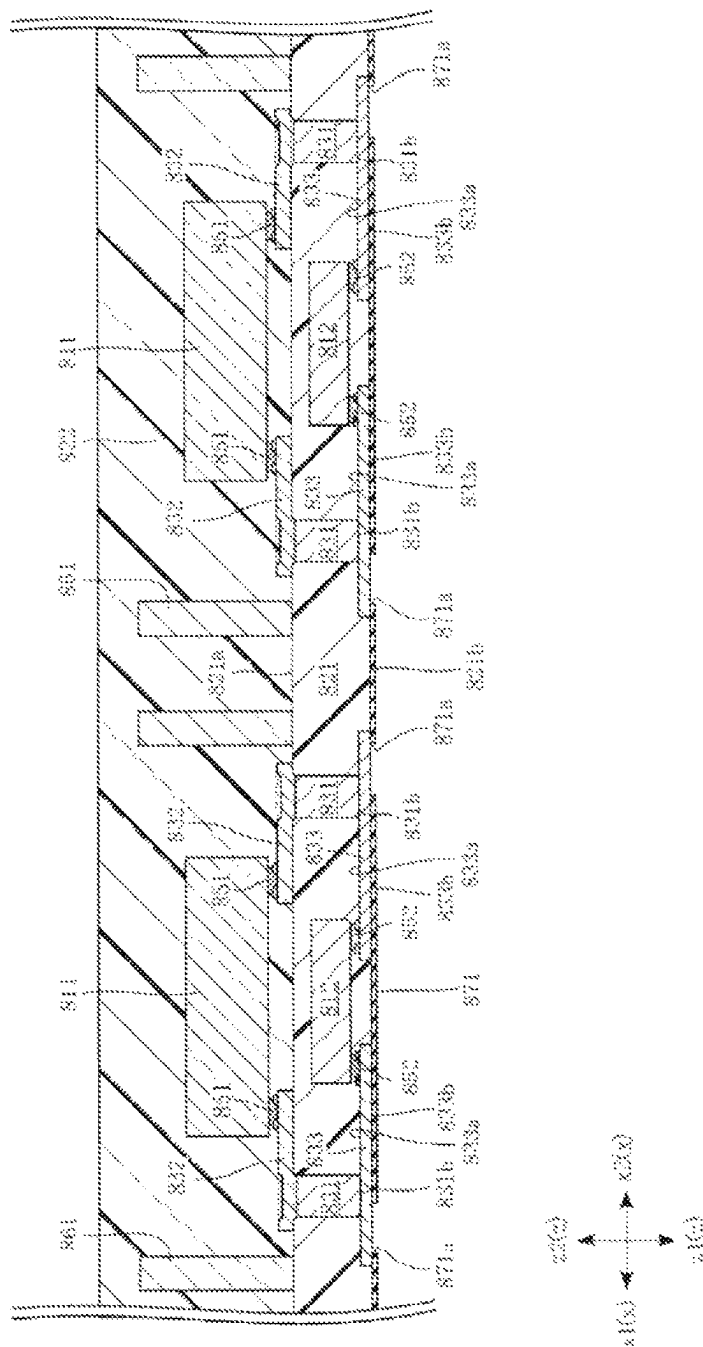


FIG.28

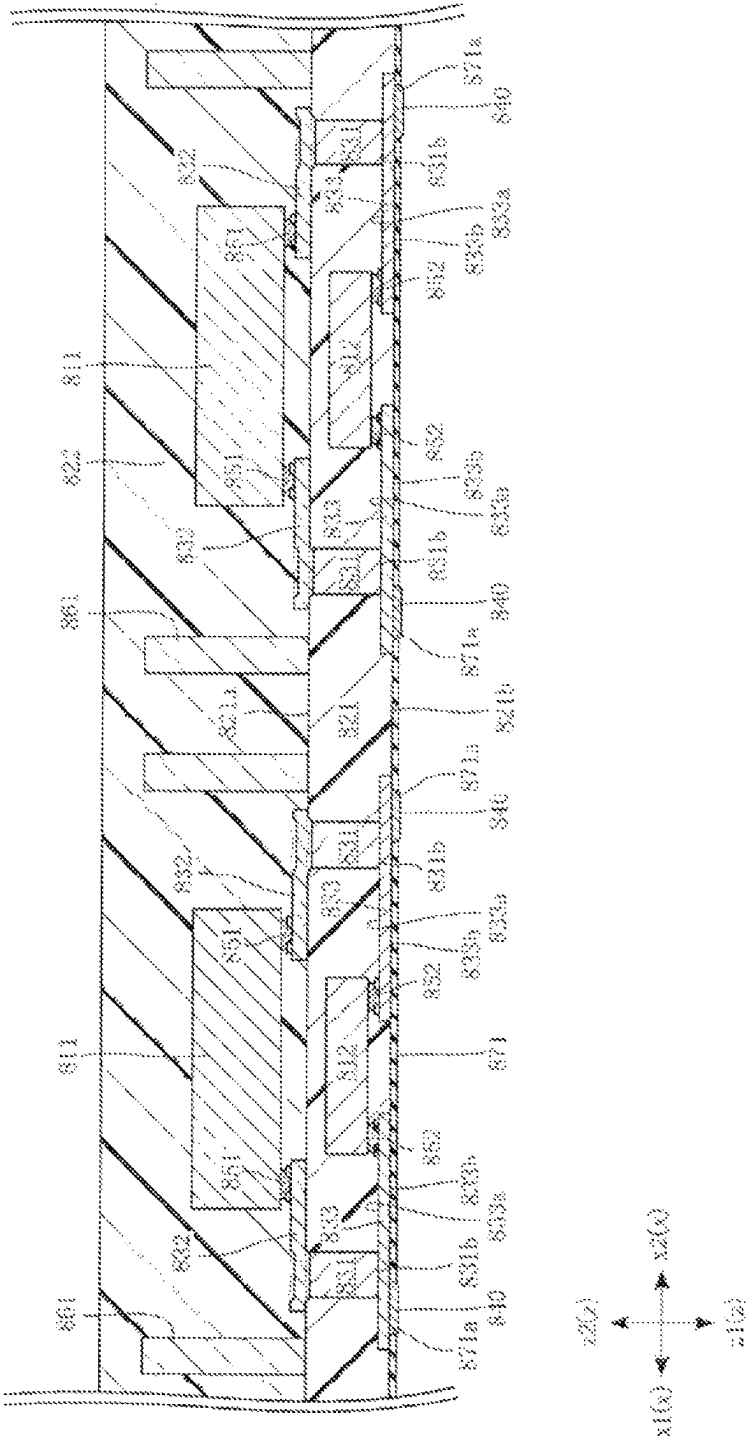


FIG. 29

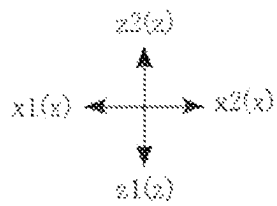
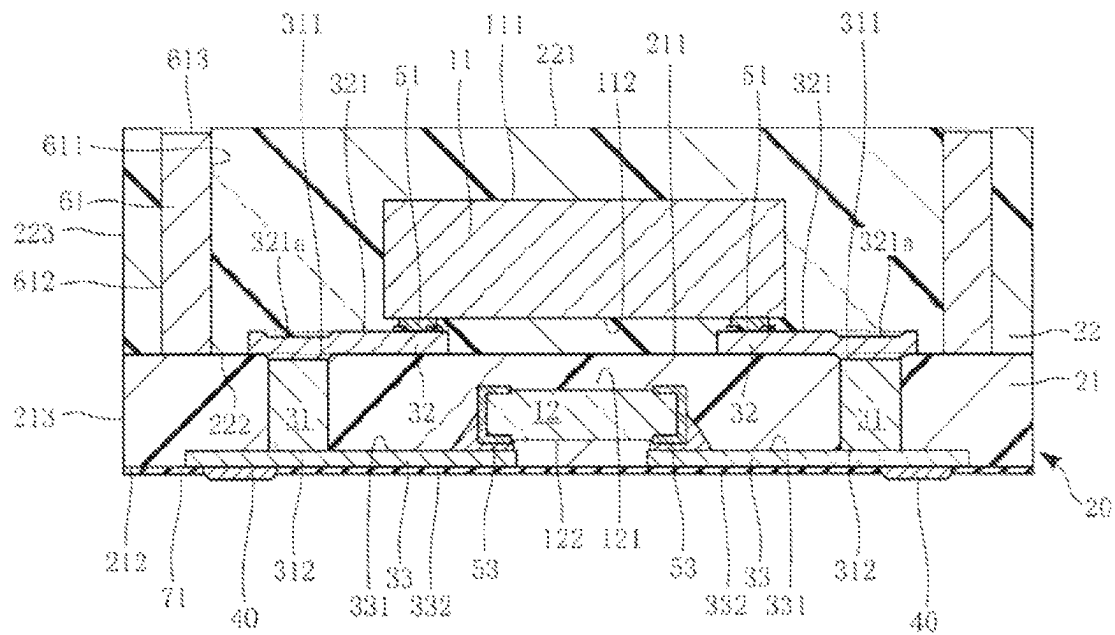


FIG.30

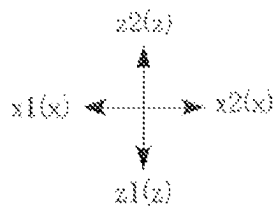
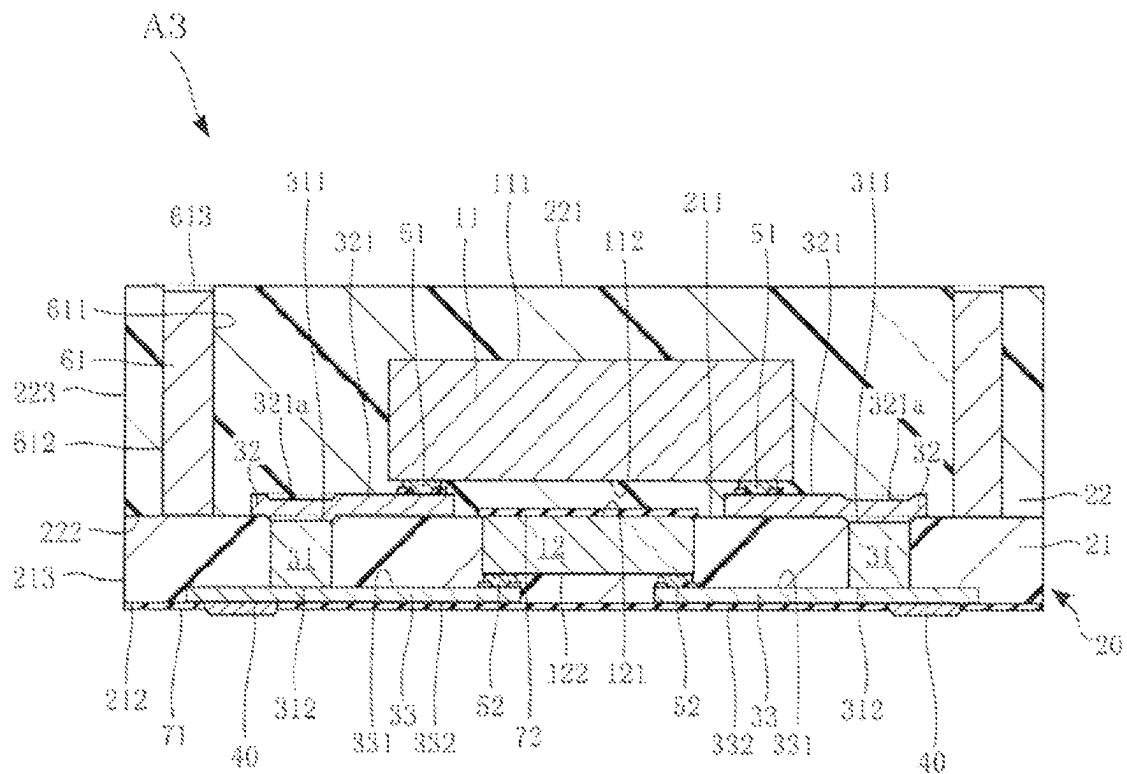


FIG.31

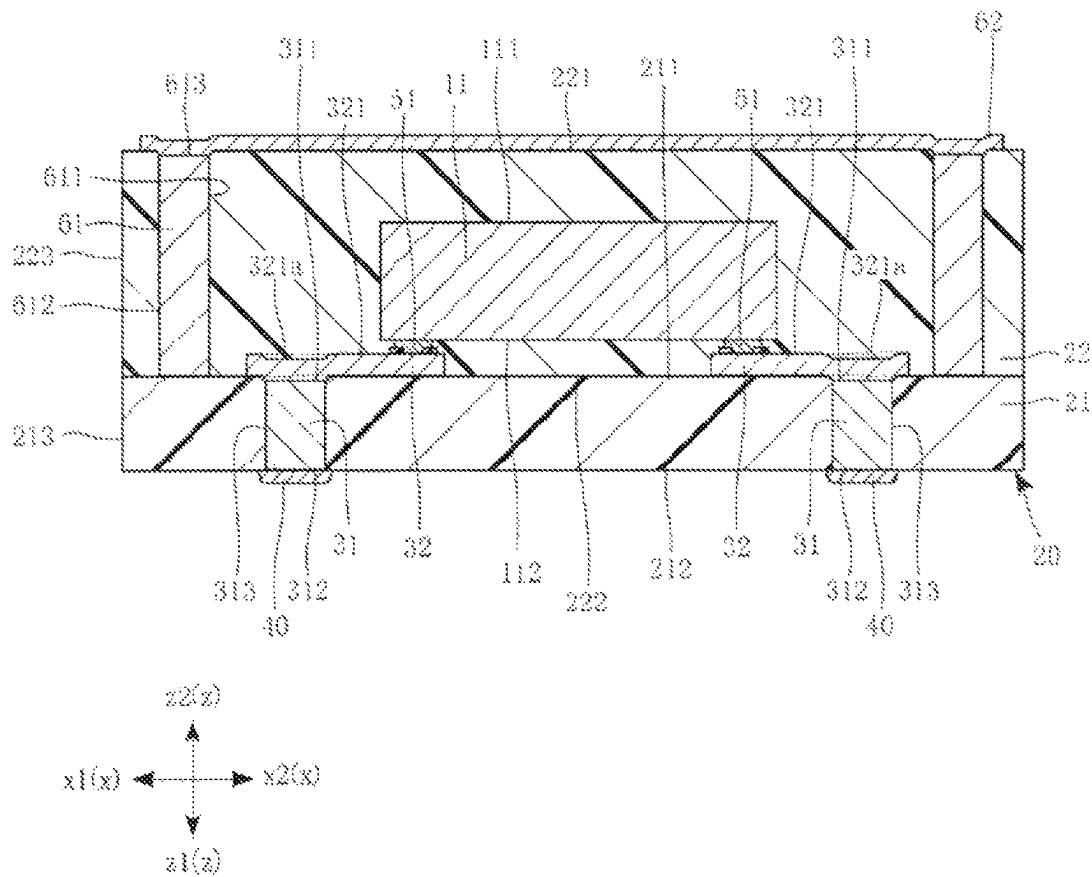


FIG.32

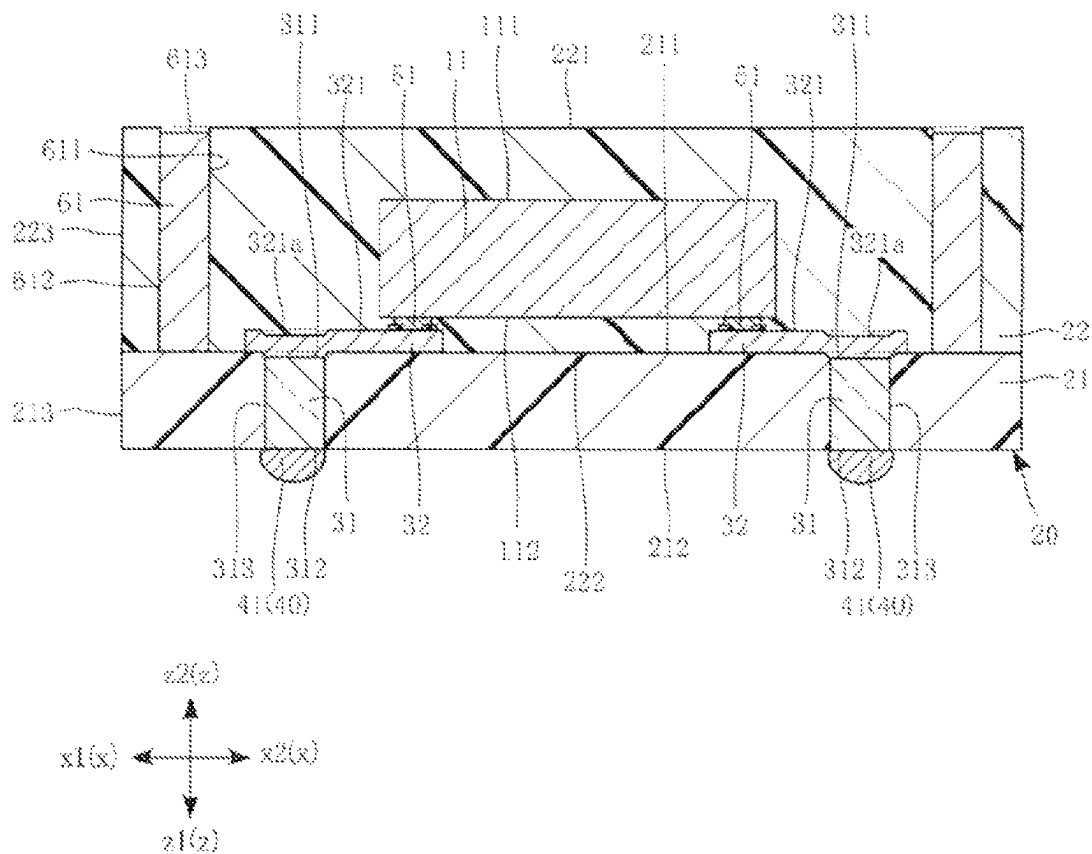


FIG. 33

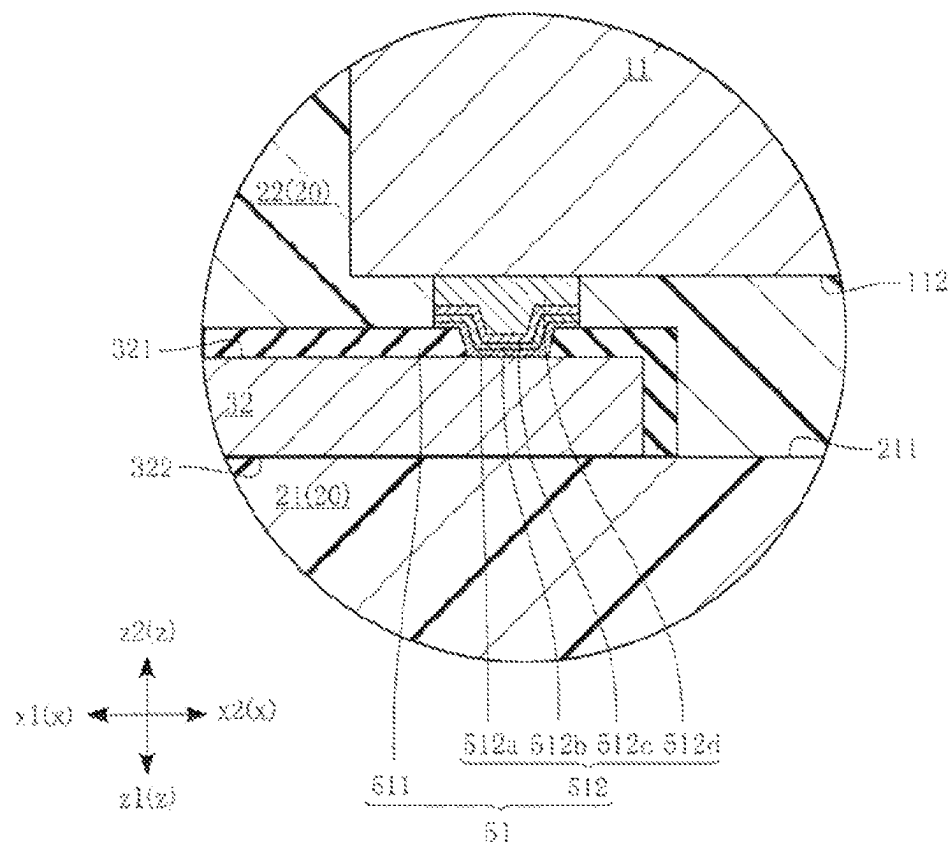


FIG.34

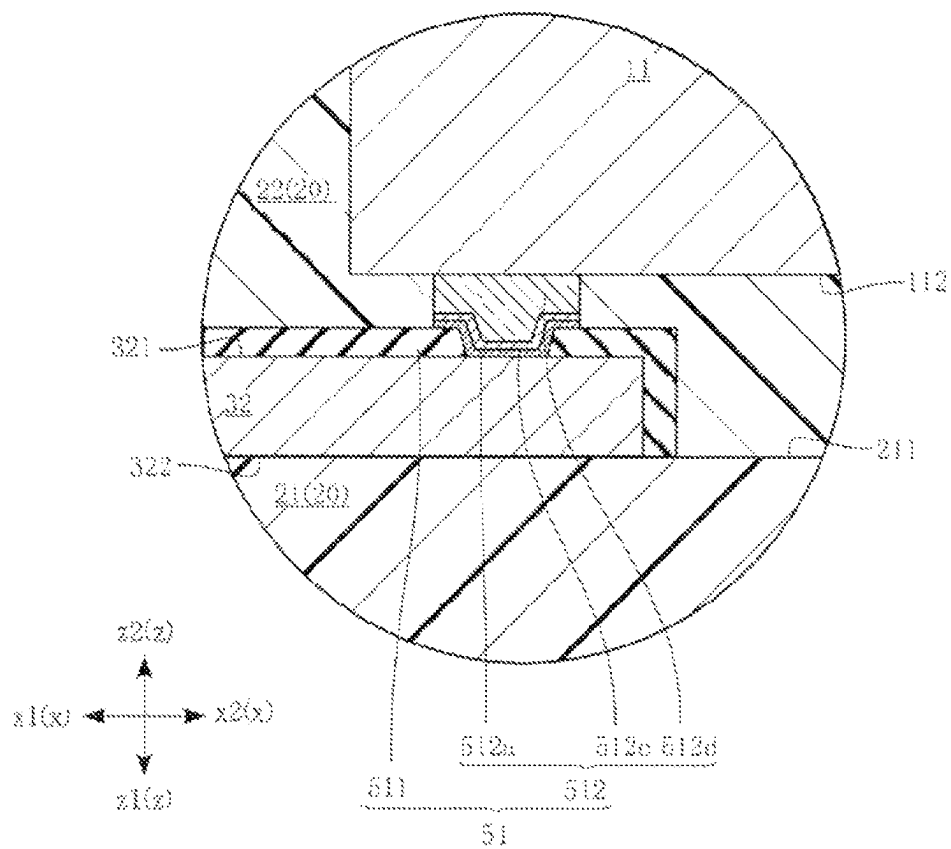


FIG. 35

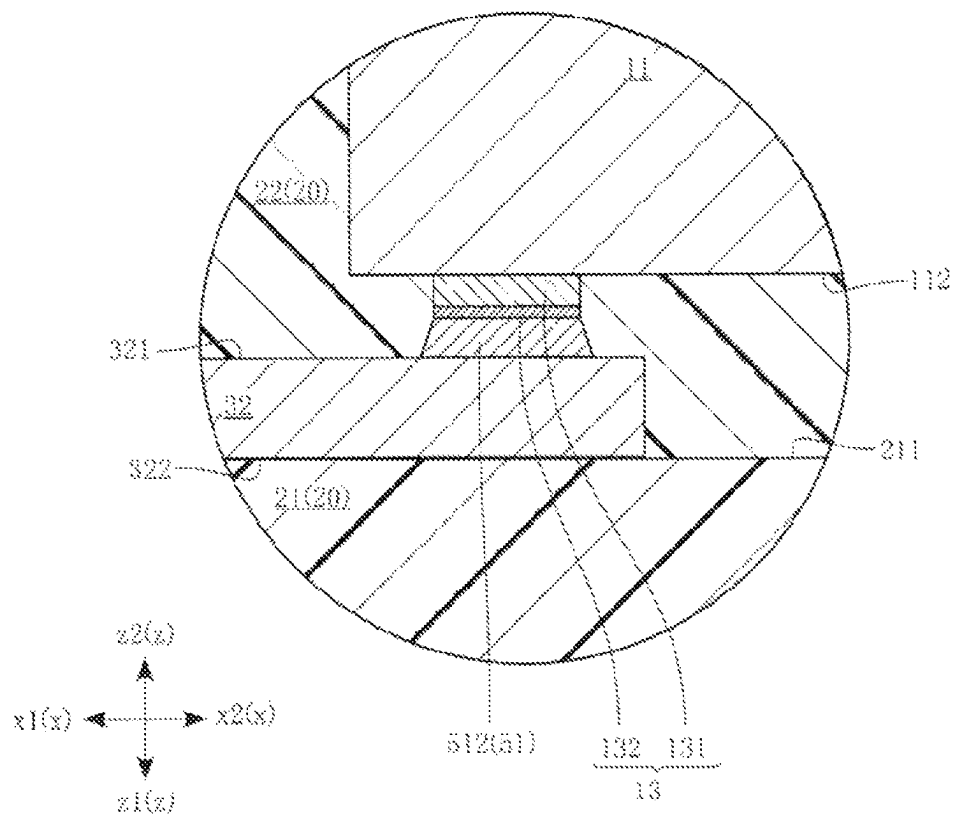


FIG.36

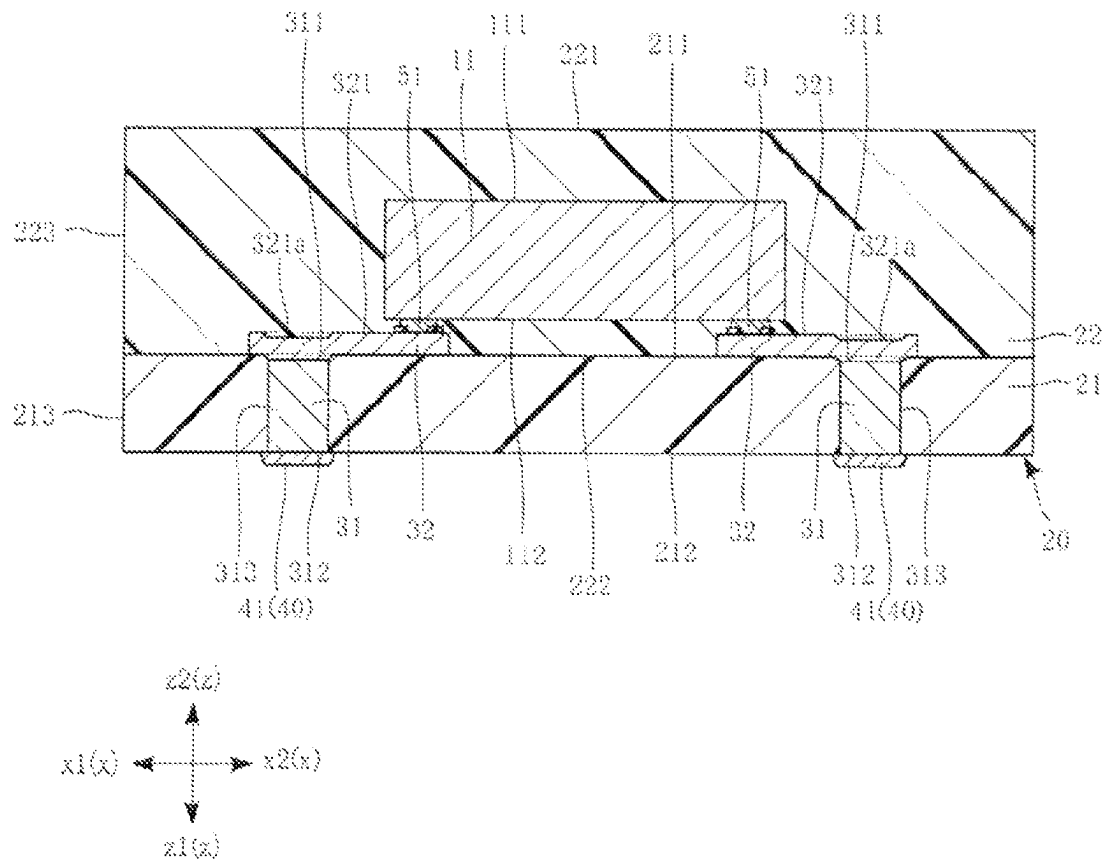


FIG.37

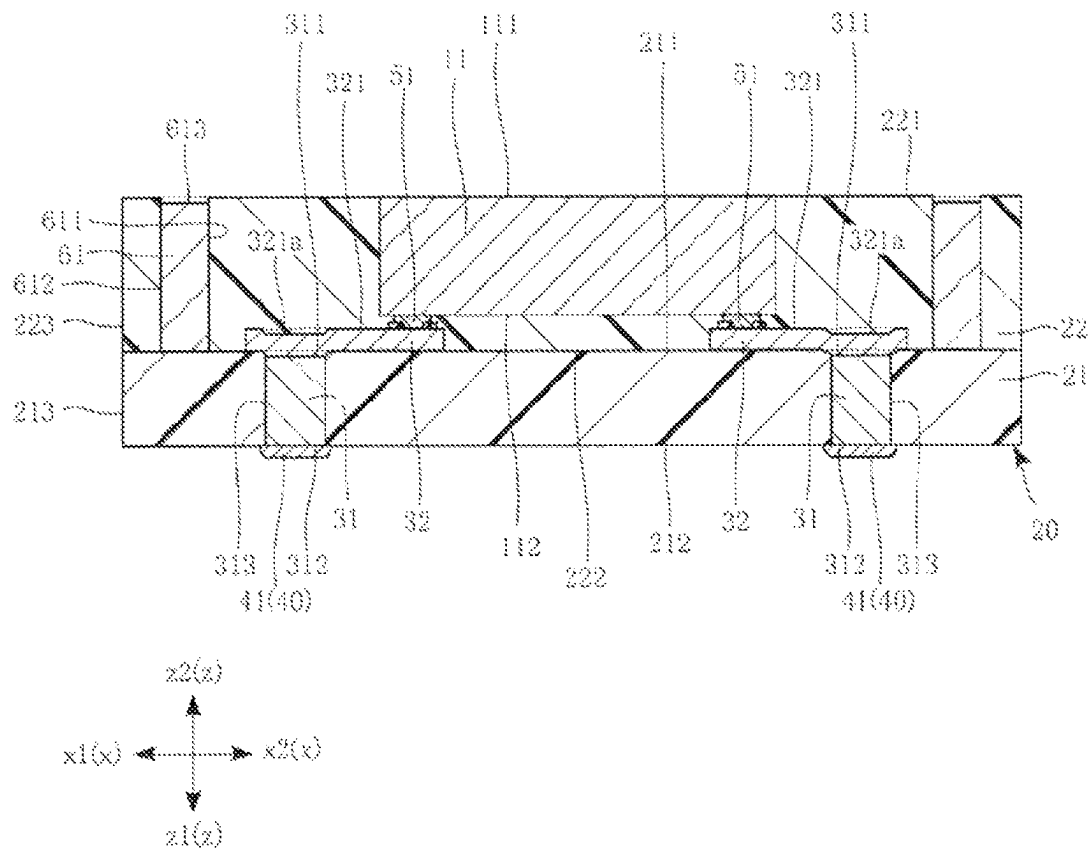


FIG.38

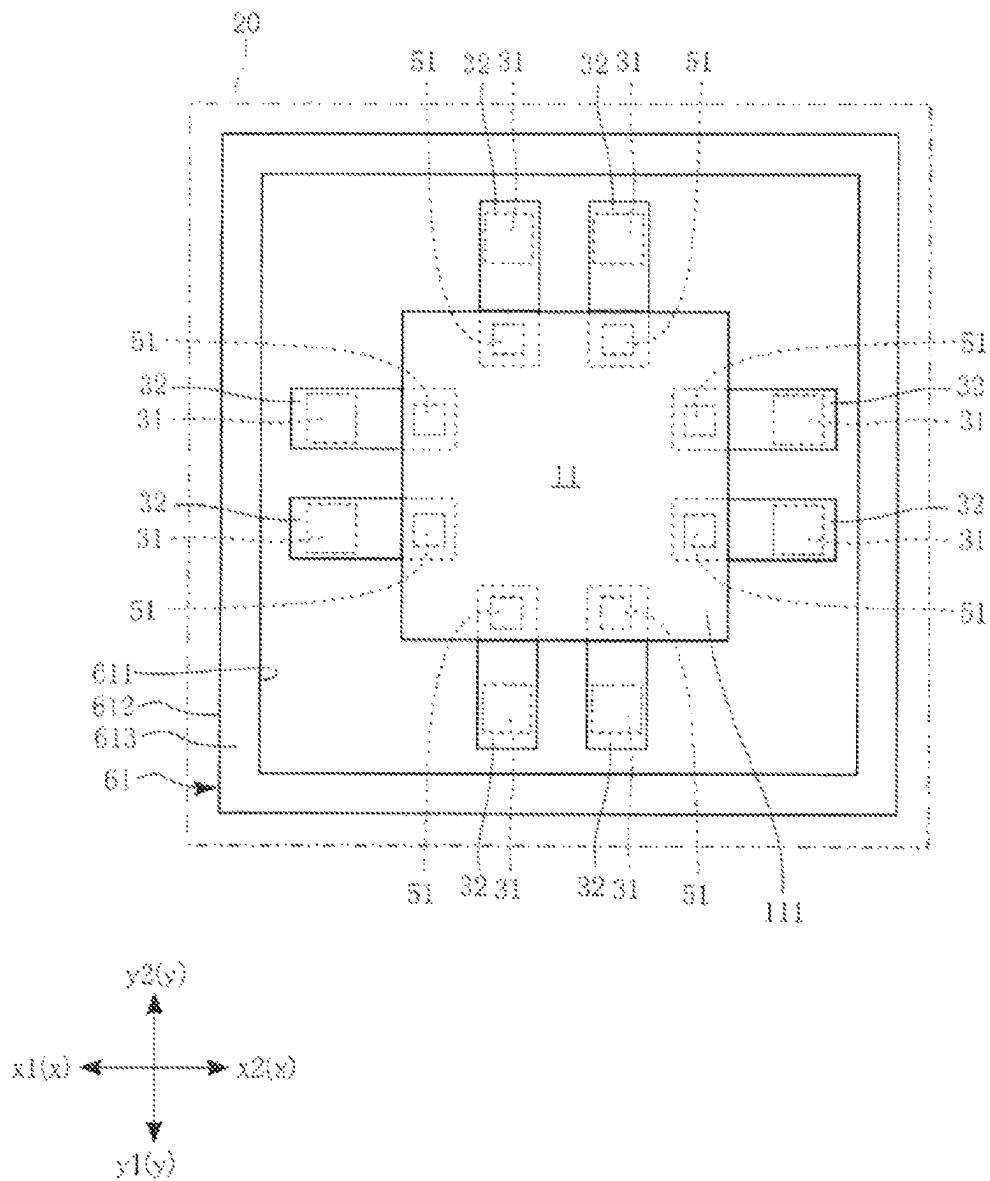


FIG.39

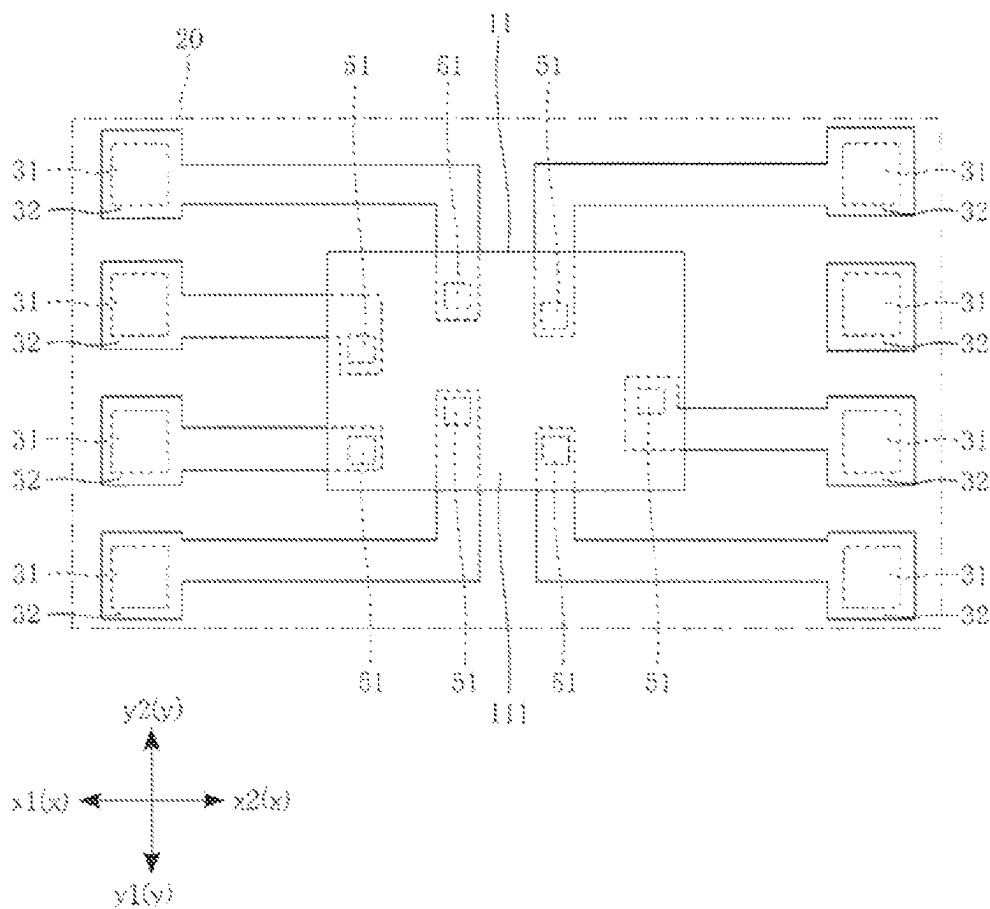


FIG. 40

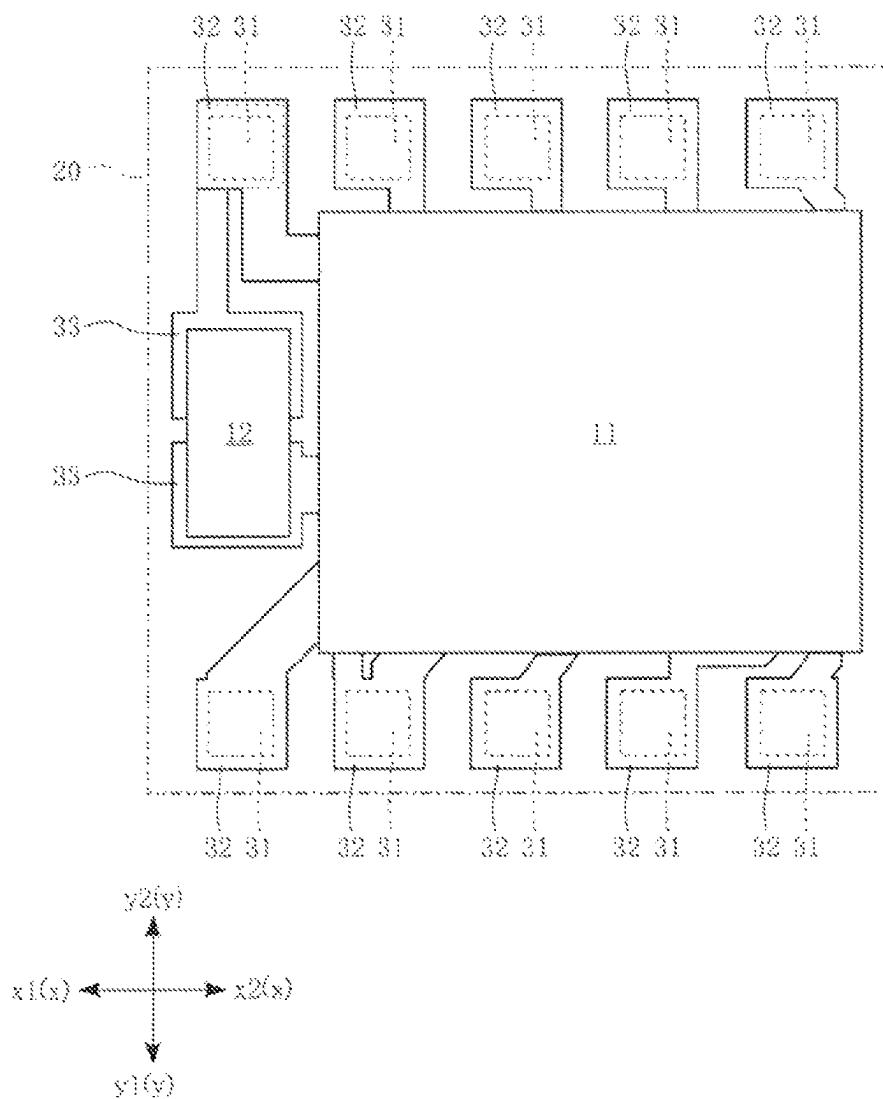


FIG. 41

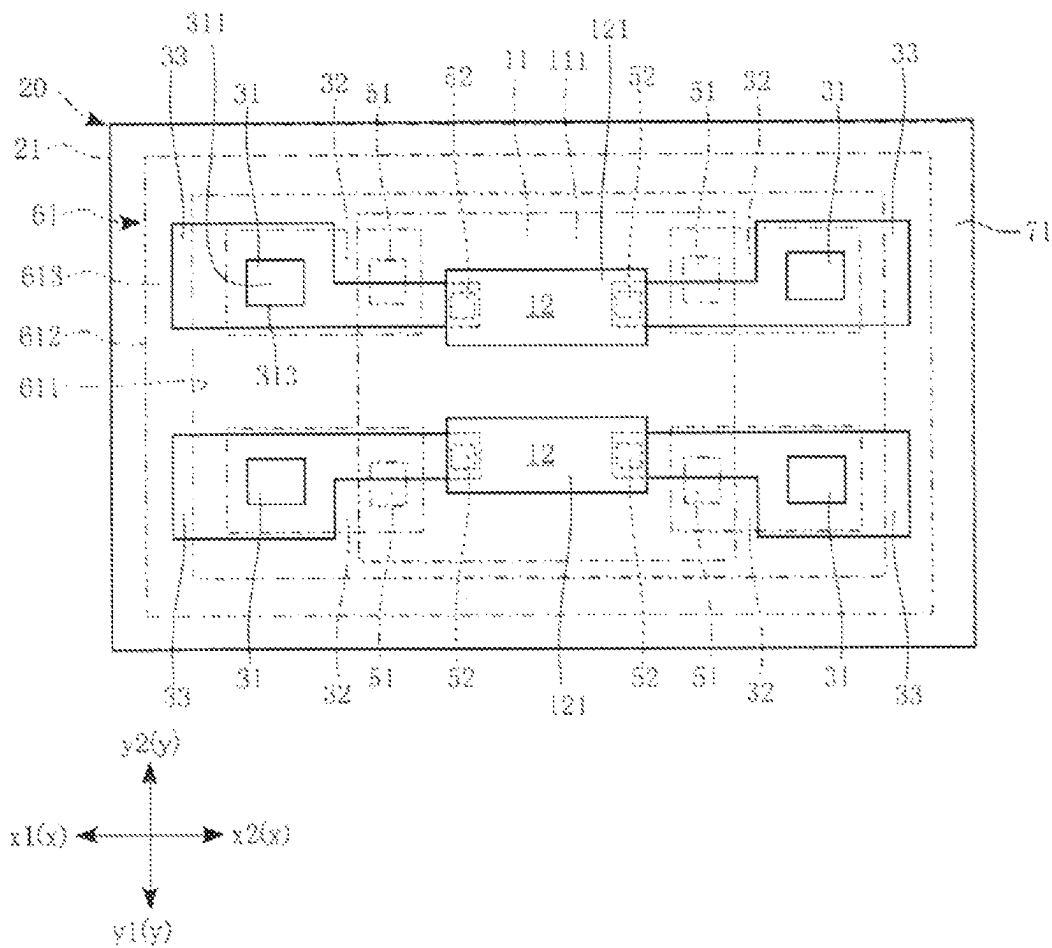


FIG. 42

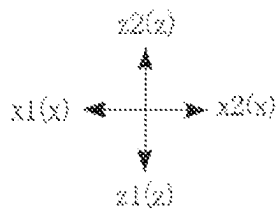
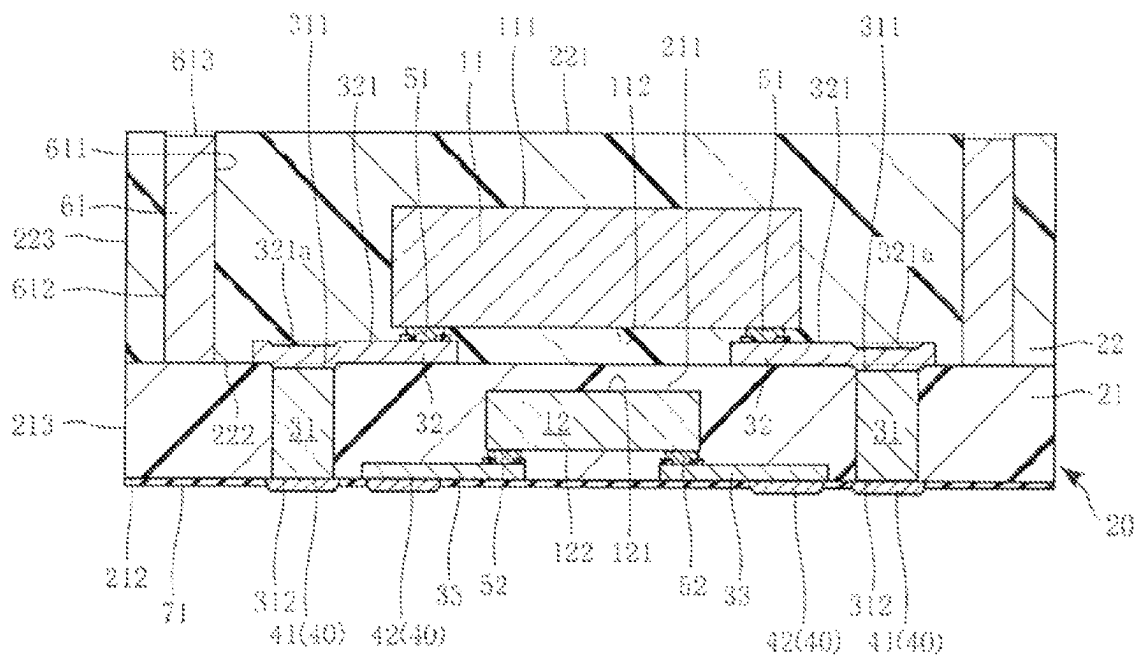


FIG.43

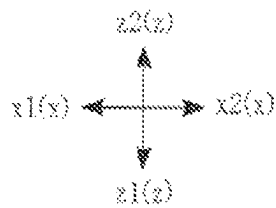
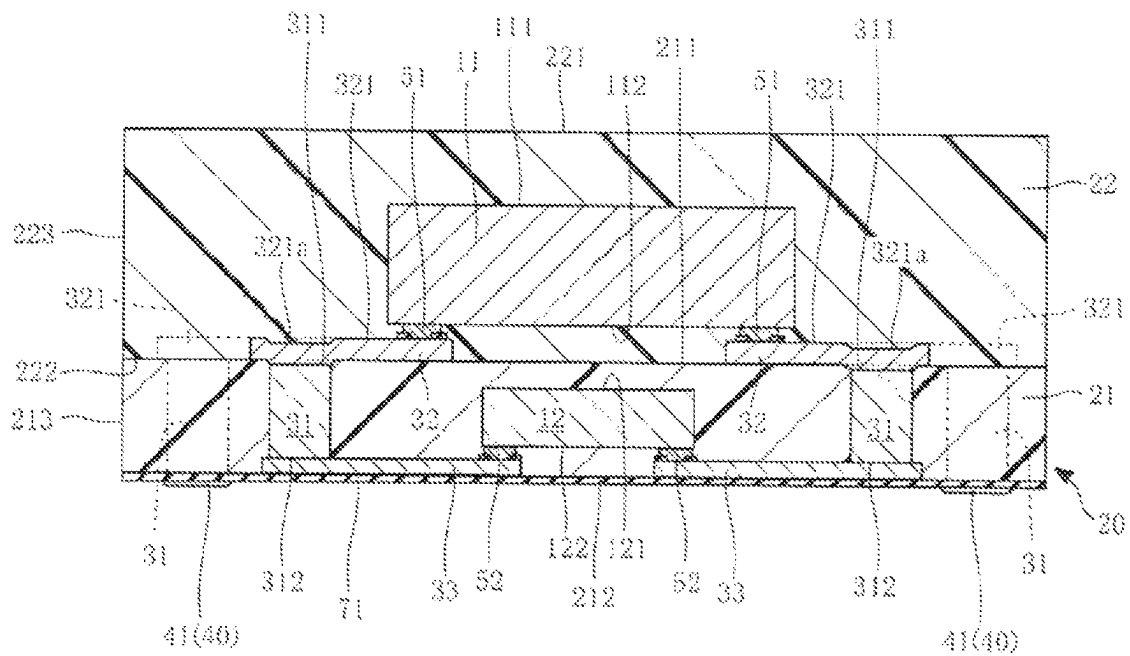


FIG.44

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ELECTRONIC DEVICE WITH MULTIPLE RESIN LAYERS REDUCING SUPPRESSION IN RELIABILITY

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation application of U.S. patent application Ser. No. U.S. Ser. No. 16/704,961 filed Dec. 5, 2019, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to an electronic device comprising electronic components and manufacturing method thereof.

BACKGROUND

Citation document 1 discloses a conventional electronic device which comprises electronic components. The electronic device of the citation document 1 comprises a semiconductor substrate, electronic components (microscopic electronic component chips) and hermetic seal resin (insulative hermetic seal resin). The semiconductor substrate is, for example, Si substrate. The electronic components are mounted on one side of the semiconductor substrate and underpinned by the semiconductor substrate. Therefore, the semiconductor substrate is a supporting part for supporting the electronic components. The hermetic seal resin is, for example, insulative epoxy resin. The hermetic seal resin is formed on the one side of the semiconductor substrate to cover the electronic components. The hermetic seal resin is a protective part for protecting the electronic components against environmental influence, such as light, heat and moisture.

BACKGROUND TECHNOLOGY DOCUMENT

Citation Document

[Citation Document 1] Japan published patent application 2009-94409.

SUMMARY OF THE INVENTION

Problem to be Solved by Present Disclosure

Electronic components of an electronic device generate heat whenever the electronic device is supplied with power. At this point, thermal stress is exerted on the interface between a semiconductor substrate (supporting part) and a hermetic seal resin (protective part) because of the difference in thermal expansion coefficient between the semiconductor substrate and the hermetic seal resin. The thermal stress is likely to cause the hermetic seal resin to be peeled from the semiconductor substrate, i.e., the protective part is peeled from the supporting part. This is the main reason why the reliability of the electronic device decreases.

To address the aforesaid issue, it is an objective of the present disclosure to provide an electronic device and manufacturing method thereof that can suppress reduction in reliability.

Technical Means to Solve Problem

The electronic device provided in the first aspect of the present disclosure comprises: a first resin layer having a first

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resin layer main surface and a first resin layer inner surface, the first resin layer main surface and the first resin layer inner surface face opposite sides in a first direction; a first conductor having a first conductor main surface and a first conductor inner surface, the first conductor main surface and the first conductor inner surface face opposite sides in the first direction, and the first conductor penetrates the first resin layer in the first direction; a first wiring layer straddling the first resin layer main surface and the first conductor main surface; a first electronic component in the first direction having a first component main surface facing the same side as the first resin layer main surface and a first component inner surface facing the same side as the first resin layer inner surface, and electrically connected with the first wiring layer; a second resin layer having a second resin layer main surface facing the same direction as the first resin layer main surface and a second resin layer inner surface being in contact with the first resin layer main surface, and covering the first wiring layer and the first electronic component; and an external electrode, disposed closer to the side where the first resin layer inner surface faces than the first resin layer and electrically connected to the first conductor.

The manufacturing method of the electronic device provided in the second aspect of the present disclosure comprises: a supporting substrate preparing step, preparing a supporting substrate having a substrate main surface and a substrate inner surface, the substrate main surface and the substrate inner surface face opposite sides in a first direction; a first conductor forming step, for forming a first conductor on the substrate main surface; a first resin layer forming step, for forming a first resin layer for covering the first conductor; a first resin layer grinding step, grinding the first resin layer in the first direction from a side which the substrate main surface faces to a side which the substrate inner surface faces such that a portion of the first conductor is exposed from the first resin layer, so as to respectively form a first conductor main surface and a first resin layer main surface, the first conductor main surface and the first resin layer main surface face the same side as the substrate main surface in the first direction; a first wiring layer forming step, forming a first wiring layer straddling the first resin layer main surface and the first conductor main surface; a first electronic component mounting step, electrically connecting a first electronic component on the first wiring layer; a second resin layer forming step, forming a second resin layer for covering the first wiring layer and the first electronic component; a supporting substrate removing step, removing the supporting substrate to expose a first resin layer inner surface facing opposite side with the first resin layer main surface in the first direction; and an external electrode forming step, forming an external electrode, the external electrode is disposed closer to the side where the first resin layer inner surface faces than the first resin layer, and the external electrode is electrically connected to the first conductor.

Effect of Present Disclosure

The present disclosure provides an electronic device capable of suppressing reduction in reliability and a manufacturing method of the electronic device.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an electronic device according to the first embodiment of the present disclosure.

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FIG. 2 is a top view of the electronic device according to the first embodiment of the present disclosure.

FIG. 3 is a cross-sectional view taken along line III-III of FIG. 2.

FIG. 4 is a partial enlarged cross-sectional view of a part of FIG. 3.

FIG. 5 is a cross-sectional view illustrative of a step of a manufacturing method of an electronic device according to the first embodiment of the present disclosure.

FIG. 6 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 7 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 8 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 9 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 10 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 11 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 12 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 13 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 14 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 15 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 16 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 17 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the first embodiment of the present disclosure.

FIG. 18 is a top view of the electronic device according to the second embodiment of the present disclosure.

FIG. 19 is a top view of the electronic device according to the second embodiment of the present disclosure.

FIG. 20 is a cross-sectional view taken along line XX-XX of FIG. 18.

FIG. 21 is a partial enlarged cross-sectional view of a part of FIG. 20.

FIG. 22 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the second embodiment of the present disclosure.

FIG. 23 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the second embodiment of the present disclosure.

FIG. 24 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the second embodiment of the present disclosure.

FIG. 25 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the second embodiment of the present disclosure.

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FIG. 26 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the second embodiment of the present disclosure.

FIG. 27 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the second embodiment of the present disclosure.

FIG. 28 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the second embodiment of the present disclosure.

FIG. 29 is a cross-sectional view illustrative of a step of the manufacturing method of the electronic device according to the second embodiment of the present disclosure.

FIG. 30 is a cross-sectional view of the electronic device according to a variant example of the second embodiment of the present disclosure.

FIG. 31 is a cross-sectional view of the electronic device according to the third embodiment of the present disclosure.

FIG. 32 is a cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 33 is a cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 34 is a partial enlarged cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 35 is a partial enlarged cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 36 is a partial enlarged cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 37 is a cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 38 is a cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 39 is a top view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 40 is a top view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 41 is a top view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 42 is a top view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 43 is a cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

FIG. 44 is a cross-sectional view of the electronic device according to a variant embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

Preferred embodiments of a manufacturing method of an electronic device of the present disclosure are illustrated by accompanying drawings and described hereunder.

Ordinal numbers, such as “first”, “second”, “third” and the like, used hereunder are intended to distinguish or correlate identical or similar components or structures and do not necessarily imply what order the components or structures are in in terms of space or time.

Unless otherwise specified hereunder, “object A is formed at object B” and “object A is formed on object B” include “object A is directly formed at object B” and “object A is formed at object B in the presence of another object disposed between object A and object B.” Likewise, unless otherwise specified, “object A is disposed at object B” and “object A is disposed on object B” include “object A is directly disposed at object B” and “object A is disposed at object B in the presence of another object disposed between object A

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and object B.” Likewise, unless otherwise specified, “object A is on object B” includes “object A is on object B while object A is in contact with object B” and “object A is on object B in the presence of another object disposed between object A and object B.” Likewise, unless otherwise specified, “object A is laminated to object B” and “object A is laminated onto object B” include “object A is directly laminated to object B” and “object A is laminated to object B in the presence of another object disposed between object A and object B.” Unless otherwise specified, “object A and object B overlap when viewed in a specific direction” includes “object A and object B have identical outlines and cover each other” and “object A and object B overlap.”

First Embodiment

FIG. 1–FIG. 4 show an electronic device in the first embodiment of the present disclosure. An electronic device A1 in the first embodiment comprises an electronic component 11, a hermetic seal resin 20, an internal electrode 30, a plurality of external electrodes 40, a plurality of connecting portions 51 and a frame-shaped conductor 61. In this embodiment, the internal electrode 30 comprises a plurality of columnar conductors 31 and a plurality of wiring layers 32.

FIG. 1 is a perspective view of the electronic device A1 when viewed from below. FIG. 2 is a top view of the electronic device A1 and depicts the hermetic seal resin 20 with an imaginary line (a double-dot and dash line). FIG. 3 is a cross-sectional view taken along line III-III of FIG. 2. FIG. 4 is a partial enlarged cross-sectional view of a part of FIG. 3.

For the sake of illustration, three directions which are perpendicular to each other are defined as direction x, direction y, and direction z, respectively. Direction z is the thickness direction of the electronic device A1. Direction x is the lateral direction in the top view of (see FIG. 2) of the electronic device A1. Direction y is the vertical direction in the top view of (see FIG. 2) of the electronic device A1. The two opposite directions of direction x are defined as direction x1 and direction x2, respectively. The two opposite directions of direction y are defined as direction y1 and direction y2, respectively. The two opposite directions of direction z are defined as direction z1 and direction z2, respectively. In this regard, direction z1 is also referred to as “downward” and direction z2 as “upward”. Direction z is equivalent to “a first direction” recited in the claims.

The electronic component 11 is crucial to the functioning of the electronic device A1. In this embodiment, the electronic component 11 is a semiconductor component which comprises a semiconductor. The electronic component 11 is an active component, for example, a component for use in voltage control, such as LSI (Large Scale Integration), IC (integrated circuit), and LDO (Low Drop Out), a component for use in amplification, such as an operational amplifier, or a discrete component, such as a transistor and a diode. The electronic component 11 comprises a semiconductor. The component is a passive component, for example, a resistor, an inductor, and a capacitor. The electronic component 11 can be surface mounted. When viewed from above, the electronic component 11 is rectangular in shape, but the present disclosure is not limited thereto. The electronic component 11 is electrically connected and joined to the wiring layers 32 by the connecting portions 51. The electronic component 11 is equivalent to “a first electronic component” recited in the claims. As shown in FIG. 3, the

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electronic component 11 has a component main surface 111 and a component inner surface 112.

The component main surface 111 and the component inner surface 112 are spaced apart from each other in direction z and face opposite sides. The component main surface 111 faces direction z2. The component inner surface 112 faces direction z1. A plurality of electrode pads (not shown) is formed on the component inner surface 112. The plurality of electrode pads respectively comprises, for example, aluminum (Al). The electrode pads are terminals in the electronic component 11. FIG. 2 is not restrictive of the quantity and position of the plurality of electrode pads. The component main surface 111 and the component inner surface 112 are equivalent to “a first component main surface” and “a first component inner surface” recited in the claims, respectively.

The hermetic seal resin 20, for example, is synthetic resin which uses black epoxy resin as a base (or the hermetic seal resin 20 is any resin, as long as the resin is capable of electrical insulation). As shown in FIG. 3, the hermetic seal resin 20 covers the electronic component 11, the internal electrode 30 and the plurality of connecting portions 51. As shown in FIG. 2, when viewed from above, the hermetic seal resin 20 is rectangular in shape. The hermetic seal resin 20 comprises a first resin layer 21 and a second resin layer 22.

The first resin layer 21 covers a part (a columnar conductor lateral surface 313 to be described later) of each columnar conductor 31. The first resin layer 21 spaces apart the wiring layers 32 for supporting the electronic component 11. The first resin layer 21 is a supporting part for supporting the electronic component 11 in the electronic device A1. The first resin layer 21 has a first resin layer main surface 211, a first resin layer inner surface 212 and a first resin layer lateral surface 213.

The first resin layer main surface 211 and the first resin layer inner surface 212 are spaced apart from each other in direction z and face opposite sides. The first resin layer main surface 211 faces direction z2, and the first resin layer inner surface 212 faces direction z1. A grinding mark is formed on the first resin layer main surface 211 by the first resin layer grinding step to be described later. In this embodiment, a part of each columnar conductor 31 is exposed from the first resin layer inner surface 212. The first resin layer lateral surface 213 connects to the first resin layer main surface 211 and the first resin layer inner surface 212. In this embodiment, the first resin layer lateral surface 213 is perpendicular to the first resin layer main surface 211 and the first resin layer inner surface 212. The first resin layer lateral surface 213 has two opposing surfaces spaced apart in direction x and two opposing surfaces spaced apart in direction y.

The second resin layer 22 covers the electronic component 11, the plurality of wiring layers 32, and a part of the frame-shaped conductor 61. The second resin layer 22 is a protective part disposed in the electronic device A1 and adapted to protect the electronic component 11. The second resin layer 22 has a second resin layer main surface 221, a second resin layer inner surface 222 and a second resin layer lateral surface 223.

The second resin layer main surface 221 and the second resin layer inner surface 222 are spaced apart from each other in direction z and face opposite sides. The second resin layer main surface 221 faces direction z2, and the second resin layer inner surface 222 faces direction z1. A grinding mark is formed on the second resin layer main surface 221 by a second resin layer grinding step to be described later. In this embodiment, a part of the frame-shaped conductor 61 is exposed from the second resin layer main surface 221. The

second resin layer lateral surface **223** connects to the second resin layer main surface **221** and the second resin layer inner surface **222**. In this embodiment, the second resin layer lateral surface **223** is perpendicular to the second resin layer main surface **221** and the second resin layer inner surface **222**. The second resin layer lateral surface **223** has two opposing surfaces spaced apart in direction x and two opposing surfaces spaced apart in direction y.

In the hermetic seal resin **20**, the first resin layer **21** and the second resin layer **22** are laminated to each other in direction z, whereas the first resin layer main surface **211** is in contact with the second resin layer inner surface **222**. In the hermetic seal resin **20**, the first resin layer lateral surface **213** and the second resin layer lateral surface **223** are coplanar.

An electrical connection path of the electronic component **11** and the plurality of external electrodes **40** is formed by the internal electrode **30** in the hermetic seal resin **20**. As mentioned above, the internal electrode **30** comprises a plurality of columnar conductors **31** and a plurality of wiring layers **32**.

Each columnar conductor **31** is formed in direction z between a corresponding one of the wiring layers **32** and a corresponding one of the external electrodes **40**, so as to electrically connect the wiring layer **32** and the external electrode **40**. Each columnar conductor **31** penetrates the first resin layer **21** in direction z. In this embodiment, each columnar conductor **31** is columnar and has a substantially rectangular cross section perpendicular to direction z. The cross section is not necessarily rectangular and thus can also be circular, elliptical, or polygonal. For example, each columnar conductor **31** is made of Cu. For example, each columnar conductor **31** comprises a basal layer and a plated layer which are laminated to each other. The basal layer comprises a Ti layer and a Cu layer which are laminated to each other and the thickness is approximately 200~800 nm. The plated layer, for example, comprises Cu and is configured to be thicker than the basal layer. The plurality of columnar conductors **31** is, for example, formed by electroplating. The above description is not restrictive of what material each columnar conductor **31** is made of and how each columnar conductor **31** is formed. The columnar conductors **31** are spaced apart from each other. Each columnar conductor **31** is equivalent to “a first conductor” of the claims. Each columnar conductor **31** has a columnar conductor main surface **311**, a columnar conductor inner surface **312** and a columnar conductor lateral surface **313**.

The columnar conductor main surface **311** and the columnar conductor inner surface **312** are spaced apart from each other in direction z and face opposite sides. The columnar conductor main surface **311** is exposed from the first resin layer main surface **211**. In this embodiment, the columnar conductor main surface **311** dents relative to the first resin layer main surface **211**. The depth (in direction z) of the dent is 1 μm approximately. The columnar conductor main surface **311** and the first resin layer main surface **211** are coplanar. The columnar conductor inner surface **312** is exposed from the first resin layer inner surface **212**. The columnar conductor inner surface **312** and the first resin layer inner surface **212** are coplanar. The columnar conductor main surface **311** is in contact with the wiring layer **32**. The columnar conductor **31** is electrically connected to the wiring layer **32**. The columnar conductor inner surface **312** is in contact with the external electrode **40**. The columnar conductor **31** is electrically connected to the external electrode **40**. The columnar conductor lateral surface **313** is connected to the columnar conductor main surface **311** and

the columnar conductor inner surface **312**. The columnar conductor lateral surface **313** is perpendicular to the columnar conductor main surface **311** and the columnar conductor inner surface **312**. The columnar conductor lateral surface **313** is in contact with the first resin layer **21**. In this embodiment, the columnar conductor lateral surface **313** has two opposing surfaces spaced apart in direction x and two opposing surfaces spaced apart in direction y. The columnar conductor main surface **311** and the columnar conductor inner surface **312** are equivalent to “a first conductor main surface” and “a first conductor inner surface” recited in the claims, respectively.

Each wiring layer **32** connects a corresponding one of the columnar conductor main surfaces **311** and a corresponding one of the first resin layer main surfaces **211**. In this embodiment, each wiring layer **32** covers the whole of the columnar conductor main surface **311** and a part of the first resin layer main surface **211** of a corresponding one of the columnar conductors **31**. The wiring layers **32** are spaced apart from each other. Each wiring layer **32** comprises a basal layer and a plated layer which are laminated to each other. The basal layer comprises a Ti layer and a Cu layer which are laminated to each other and the thickness is approximately 200~800 nm. The basal layer, for example, is formed by sputtering. For example, the plated layer comprises Cu and is configured to be thicker than the basal layer. For example, the plated layer is formed by electroplating. The above description is not restrictive of what material the wiring layers **32** are made of and how the wiring layers **32** are formed. For example, a Ni layer is formed between the basal layer and the plated layer. For example, the Ni layer is formed by electroplating. FIG. 2 is not restrictive of whatever related to the formation of the wiring layers **32**.

The wiring layers **32** each have a wiring layer main surface **321** and a wiring layer inner surface **322**. The wiring layer main surface **321** and the wiring layer inner surface **322** are spaced apart and face opposite sides in direction z. The wiring layer main surface **321** faces direction z2, and the wiring layer inner surface **322** faces direction z1. The wiring layer main surface **321** is in contact with the second resin layer **22**. The wiring layer inner surface **322** is in contact with the first resin layer **21**. Each wiring layer **32** has an end surface which faces direction x or direction y and is covered by the second resin layer **22**.

Each wiring layer **32** comprises a dent portion **321a** which dents, in direction z, relative to the wiring layer main surface **321** of the wiring layer **32**. When viewed from above, the dent portion **321a** overlaps the columnar conductor **31**. The dent portion **321a** is not formed when the columnar conductor main surface **311** and the first resin layer main surface **211** are coplanar.

The external electrodes **40** are electrically connected to the internal electrodes **30**, respectively, and external conductors of the electronic device **A1** are exposed. The external electrodes **40** function as the terminals for use in mounting the electronic device **A1** on the circuit substrates of an electronic machine. The plurality of external electrodes **40** is formed by electroless plating. In this embodiment, each external electrode **40** comprises a Ni layer, a Pd layer and a Au layer which are laminated to each other. The direction-z dimension of each external electrode **40** is, for example, 3~10 μm approximately, but the present disclosure is not limited thereto. The aforesaid description is not restrictive of the direction-z dimension of the external electrodes **40**, what material the external electrodes **40** are made, and how the external electrodes **40** are formed. For example, each exter-

nal electrode **40** comprises a Ni layer and a Au layer which are laminated to each other or is made of Sn.

The external electrodes **40** are exposed from the hermetic seal resin **20**. Each external electrode **40** is closer to the outside than the first resin layer **21** in direction **z1**. Therefore, each external electrode **40** is disposed on the bottom side of the electronic device **A1**. In this embodiment, the external electrodes **40** are electrically connected to the columnar conductors **31**, respectively. The external electrodes **40** each comprise a columnar conductor covering portion **41**.

The columnar conductor covering portions **41** cover the columnar conductor inner surfaces **312**, respectively. The columnar conductor covering portions **41** are in contact with the columnar conductor inner surfaces **312**, respectively. In this embodiment, the electronic components **11** are electrically connected to the columnar conductor covering portions **41** by the connecting portions **51**, the wiring layers **32** and the columnar conductors **31**, respectively. Therefore, the columnar conductor covering portions **41** are the terminals of the electronic device **A1** and electrically connect to the electronic component **11**. The columnar conductor covering portion **41** is equivalent to "a first conductor covering portion" recited in the claims.

The plurality of connecting portions **51** is each a conductive connecting element formed between the electronic component **11** (i.e., the electrode pad) and a corresponding one of the wiring layers **32**. The electronic component **11** is configured in such a manner as to not only allow the plurality of connecting portions **51** to be fixed to the plurality of wiring layers **32** and thereby mounted on the wiring layers **32**, but also use the plurality of connecting portions **51** to ensure that the electronic component **11** is electrically connected to the plurality of wiring layers **32**. In this embodiment, as shown in FIG. 4, the connecting portions **51** each comprise an insulating layer **511** and a connecting layer **512**.

Referring to FIG. 4, the insulating layers **511** are formed on the wiring layers **32**, respectively. When viewed from above, each insulating layer **511** is centrally-opened and frame-shaped. When viewed from above, the insulating layers **511** surrounds the connecting layers **512**, respectively. In this embodiment, when viewed from above, each insulating layer **511** has the shape of a rectangular frame. When viewed from above, each insulating layer **511** does not necessarily have the shape of a rectangular frame but can also have the shape of a circular frame, an elliptical frame or a polygonal frame. The insulating layers **511** are, for example, made of polyimide resin, but the present disclosure is not limited thereto.

The connecting layers **512** electrically connect the electronic component **11** to the wiring layers **32**, respectively. The connecting layers **512** are formed on the wiring layers **32** (wiring layer main surfaces **321**), respectively. The connecting layers **512** cover surfaces of the opening portions of the insulating layers **511**, respectively. The opening portions of the insulating layers **511** are filled with parts of the connecting layers **512**, respectively. In this embodiment, as shown in FIG. 4, the connecting layers **512** each comprise a first layer **512a**, a second layer **512b** and a third layer **512c** which are laminated to each other.

The first layers **512a** are formed on the wiring layers **32** (wiring layer main surfaces **321**), respectively, and are in contact with the wiring layer main surfaces **321**, respectively. The first layer **512a** is made of metal, such as Cu. The second layer **512b** is formed on the first layer **512a** and is in contact with the first layer **512a**. The second layer **512b** is made of metal, such as Ni. The third layer **512c** is formed on

the second layer **512b** and is in contact with the second layer **512b**. The third layer **512c** is in contact with the electronic component **11** (electrode pads). The third layer **512c** is made of metal, such as Sn. Examples of the alloy include Sn—Sb based alloy and Sn—Ag based alloy which are typical of lead-free solder. The connecting layers **512** are each equivalent to "a conductive connecting layer" recited in the claims.

When viewed from above, the frame-shaped conductor **61** surrounds the electronic component **11**. In this embodiment, when viewed from above, the frame-shaped conductor **61** surrounds the electronic component **11**. When viewed from above, the frame-shaped conductor **61** has the shape of a rectangular frame. When viewed from above, the frame-shaped conductor **61** does not necessarily have the shape of a rectangular frame but can also have the shape of a circular frame, an elliptical frame or a polygonal frame. A part of the second resin layer **22** is present between the frame-shaped conductor **61** and the electronic component **11**. The frame-shaped conductor **61** is formed on the first resin layer **21** and erected on the first resin layer main surface **211**. In this embodiment, the frame-shaped conductor **61** is spaced apart from the internal electrode **30**. The frame-shaped conductor **61** is equivalent to "a second conductor" recited in the claims.

The frame-shaped conductor **61** comprises, for example, a basal layer and a plated layer which are laminated to each other. The basal layer comprises Ti layer and Cu layer which are laminated to each other and is approximately 200–800 nm thick. The main constituent of the plated layer is Cu. The plated layer is configured to be thicker than the basal layer. For example, the frame-shaped conductor **61** is formed by electroplating. The above description is not restrictive of the material which the frame-shaped conductor **61** is made of and the method the frame-shaped conductor **61** is formed by.

The frame-shaped conductor **61** has an inner surface **611**, an outer surface **612** and a top surface **613**. The inner surface **611** is defined by the inner surface of the frame-shaped (when viewed from above) conductor **61**. The inner surface **611** faces the electronic component **11**. The outer surface **612** is defined by the outer surface of the frame-shaped (when viewed from above) conductor **61**. The top surface **613** faces direction **x2**. The top surface **613** is exposed from the second resin layer **22**. The top surface **613** dents relative to the second resin layer main surface **221** of the second resin layer **22**. The depth (in direction **z**) of the dent is 1 μ m approximately. The top surface **613** and the second resin layer main surface **221** are coplanar. The top surface **613** is covered by the second resin layer **22**. In this embodiment, the top surface **613** is closer to direction **z2** than the component main surface **111** in direction **z**. The top surface **613** is equivalent to "a second conductor main surface" recited in the claims.

FIG. 5–FIG. 17 illustrate a manufacturing method of the electronic device **A1** according to the first embodiment of the present disclosure. The manufacturing method described below is about manufacturing multiple electronic devices **A1**. FIG. 5–FIG. 17 are cross-sectional views illustrative of a step of the manufacturing method of the electronic device **A1**.

First, as shown in FIG. 5, the manufacturing method of the electronic device **A1** entails preparing a supporting substrate **800**. The supporting substrate **800** comprises a monocrystalline semiconductor. In this embodiment, the monocrystalline semiconductor is Si. A step of preparing the supporting substrate **800** (a supporting substrate preparing step), for example, entails preparing a Si wafer which functions as the supporting substrate **800**. In this embodi-

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ment, the thickness of the supporting substrate **800** is, for example, 725~775 μm approximately. The supporting substrate **800** comprises a supporting substrate main surface **801** and a supporting substrate inner surface **802** which are spaced apart and face opposite sides in direction z . The supporting substrate main surface **801** faces direction $z2$, and the supporting substrate inner surface **802** faces direction $z1$. The supporting substrate **800** thus prepared is not necessarily a Si wafer but can also be a glass substrate, for example.

Afterward, as shown in FIG. 5, the manufacturing method of the electronic device **A1** entails forming a columnar conductor **831** on the supporting substrate **800**. The columnar conductor **831** corresponds to the columnar conductor **31** of the electronic device **A1**. In a step of forming the columnar conductor **831** (a columnar conductor forming step), the basal layer in contact with the supporting substrate main surface **801** is formed. The basal layer is formed by sputtering. In this embodiment, after the Ti layer in contact with the supporting substrate main surface **801** has been formed, the Cu layer in contact with the Ti layer is formed. Therefore, the basal layer is formed of a Ti layer and a Cu layer which are laminated to each other. In this embodiment, the thickness of the Ti layer is 10~30 nm approximately, the thickness of the Cu layer is 200~800 nm approximately. The above description is not restrictive of the material which the basal layer is made of and the thickness of the basal layer. Afterward, the plated layer in contact with the basal layer is formed. A photoresist pattern is formed on the plated layer by photolithography and electroplating. Specifically speaking, a photosensitive photoresist is coated on the whole of the basal layer, and then the photosensitive photoresist undergoes exposure and development. Therefore, a patterned photoresist layer (hereinafter referred to as the "photoresist pattern") is formed. The photosensitive photoresist is, for example, coated with a spin coater, but the present disclosure is not limited thereto. At this point, a part of the basal layer is exposed from the photoresist pattern. Then, the basal layer functions as a conducting path whereby electroplating is carried out. Therefore, the plated layer is emanated from the basal layer exposed from the photoresist pattern. In this embodiment, the plated layer, for example, comprises Cu. After the plated layer has been formed, the photoresist pattern is removed. At the end of the aforesaid step, the columnar conductor **831** shown in FIG. 5 is formed. In this embodiment, the columnar conductor forming step is equivalent to "a first conductor forming step" recited in the claims.

Afterward, as shown in FIG. 6, the manufacturing method of the electronic device **A1** entails forming a first resin layer **821** for covering the columnar conductor **831**. A step of forming the first resin layer **821** (a first resin layer forming step) is, for example, carried out by die molding. In this embodiment, the first resin layer **821** capable of electrical insulation is, for example, made of synthetic resin which uses black epoxy resin as a base. Owing to the first resin layer forming step, the columnar conductor **831** is fully covered by the first resin layer **821**. Therefore, the direction- $z2$ -facing surface (a first resin layer main surface **821a**) of the first resin layer **821** is closer to the direction $z2$ than the direction $z2$ -facing surface of the columnar conductor **831**.

Afterward, as shown in FIG. 7, the manufacturing method of the electronic device **A1** entails grinding the first resin layer **821**. A step of grinding the first resin layer **821** (a first resin layer grinding step), for example, requires a mechanical grinding wheel. The grinding of the first resin layer **821** is not necessarily performed with a mechanical grinding

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wheel. In this embodiment, the first resin layer **821** is ground with a grinding stone from the first resin layer main surface **821a** toward direction $z1$. At this point, the first resin layer **821** is ground until the columnar conductor **831** is exposed. The first resin layer grinding step enables the first resin layer main surface **821a** to move in direction $z1$, and the direction- $z2$ -facing surface (columnar conductor main surface **831a**) of the columnar conductor **831** is exposed from the first resin layer **821** (first resin layer main surface **821a**). A grinding mark, i.e., a mark generated with the grinding stone, is formed on the first resin layer main surface **821a**. In this embodiment, the grinding mark extends from the first resin layer main surface **821a** to the columnar conductor main surface **831a**. In this embodiment, the grinding of the first resin layer **821** is accompanied by a smaller degree of the grinding of the columnar conductor **831**. Upon completion of the grinding process, a burr is likely to be formed on the columnar conductor main surface **831a**, because the columnar conductor **831** and the first resin layer **821** are made of different materials. Therefore, in this embodiment, the burr is removed by chemical processing. Therefore, the columnar conductor main surface **831a** dents in direction z more than the first resin layer main surface **821a**.

Afterward, as shown in FIG. 8~FIG. 12, wiring layers **832**, connecting portions **851** and frame-shaped conductors **861** are formed. The wiring layers **832**, the connecting portions **851** and the frame-shaped conductors **861** correspond to the wiring layers **32**, the connecting portions **51** and the frame-shaped conductors **61** of the electronic device **A1**, respectively. They are formed in five steps described below.

In the first step, as shown in FIG. 8, a basal layer **890a** is formed. For example, the basal layer **890a** is formed by sputtering. In the step of forming the basal layer **890a**, after the Ti layer which covers the whole of the first resin layer main surface **821a** and the whole of the columnar conductor main surface **831a** has been formed, the Cu layer in contact with the Ti layer is formed. The basal layer **890a** is formed of a Ti layer and a Cu layer which are laminated to each other.

In the second step, as shown in FIG. 9, a plated layer **890b** is formed. For example, the photoresist pattern is formed on the plated layer **890b** by photolithography and electroplating. In a step of forming the plated layer **890b**, the photosensitive photoresist is coated on the whole of the basal layer **890a** and thereby undergoes exposure and development, so as for the photoresist layer to be patterned. Therefore, the photoresist pattern is formed, and a part (which forms the plated layer **890b**) of the basal layer **890a** is exposed from the photoresist pattern. After that, the basal layer **890a** functions as a conducting path whereby electroplating is carried out, and thus the plated layer **890b** is emanated from the basal layer **890a** exposed from the photoresist pattern. In this embodiment, for example, the metal layer functioning as the plated layer **890b** and comprising Cu is emanated. At this point, the plated layer **890b** is integrally formed with the basal layer **890a**. Afterward, the photoresist pattern formed in this step is removed. Therefore, the plated layer **890b** shown in FIG. 9 is formed. As a result, the plated layer **890b** and the basal layer **890a** covered by the plated layer **890b** become the wiring layers **832**. The wiring layers **832** correspond to the wiring layers **32** of the electronic device **A1**.

In the third step, as shown in FIG. 10, the connecting portions **851** is formed. In this embodiment, the insulating layers **851a** and the connecting layers **851b** are formed to function as the connecting portions **851**. In a step of forming the insulating layer **851a**, photosensitive polyimide is coated on the whole of the plated layer **890b** and the whole of the

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basal layer **890a** exposed from the plated layer **890b**. The photosensitive polyimide is, for example, coated with a spin coater. Then, the photosensitive polyimide thus coated undergoes exposure and development to form a frame-shaped insulating layer **851a**. After that, in a step of forming the connecting layers **851b**, the photoresist pattern for forming the connecting layers **851b** is formed. The formation of the photoresist pattern entails coating the photosensitive photoresist and performing exposure and development on the coated photosensitive photoresist to pattern the photoresist layer. Therefore, the photoresist pattern is formed, and a part (which forms the connecting layers **851b**) of the plated layer **890b** is exposed from the photoresist pattern. The exposed part is located on the inner side of the frame-shaped insulating layer **851a** when viewed from above. After that, the basal layer **890a** and the plated layer **890b** function as a conducting path whereby electroplating is carried out, and thus the connecting layers **851b** is emanated from the plated layer **890b** exposed from the photoresist pattern. In this embodiment, the connecting layers **851b** is formed by sequential lamination of a Cu-containing metal layer, a Ni-containing metal layer and a Sn-containing alloy layer. The Sn-containing alloy layer is, for example, made of Sn—Sb based alloy or Sn—Ag based alloy which is typical of lead-free solder. Afterward, the photoresist pattern formed in this step is removed. Therefore, as shown in FIG. 10, the connecting portions **851** each comprising an insulating layer **851a** and a connecting layer **851b** are formed. The connecting portions **851** correspond to the connecting portions **51** of the electronic device **A1**.

In the fourth step, as shown in FIG. 11, a plated layer **890c** is formed. For example, the photoresist pattern is formed on the plated layer **890c** by photolithography and electroplating. The plated layer **890c** is formed in the same way as the plated layer **890b**. In a step of forming the plated layer **890c**, the photoresist pattern for forming the plated layer **890c** is formed. Therefore, a part (which forms the plated layer **890c**) of the basal layer **890a** is exposed from the photoresist pattern thus formed. After that, the basal layer **890a** functions as a conducting path whereby electroplating is carried out, and thus the plated layer **890c** is emanated from the basal layer **890a** exposed from the photoresist pattern. In this embodiment, for example, the metal layer comprising Cu is emanated to function as the plated layer **890c**. The plated layer **890c** is integrally formed with the basal layer **890a**. Afterward, the photoresist pattern formed in this step is removed. Therefore, the plated layer **890c** shown in FIG. 11 is formed. In this embodiment, the plated layer **890c** and the basal layer **890a** covered by the plated layer **890c** function as the frame-shaped conductor **861**. The frame-shaped conductor **861** corresponds to the frame-shaped conductor **61** of the electronic device **A1**.

In the fifth step, as shown in FIG. 12, the basal layer **890a** which is useless is removed. In this embodiment, the basal layer **890a** not covered by any one of the plated layer **890b** and the plated layer **890c** is regarded as the useless basal layer **890a** and thus removed. The useless basal layer **890a** is, for example, removed by wet etching, using a mixture solution of sulfuric acid (H_2SO_4) and hydrogen peroxide (H_2O_2). In the step of removing the useless basal layer **890a**, as shown in FIG. 12, the basal layer **890a** formed in the first step is divided into the basal layer **890a** covered by the plated layer **890b** and the basal layer **890a** covered by the plated layer **890c**. Therefore, as shown in FIG. 12, the plated layer **890b** and the basal layer **890a** covered by the plated layer **890b** are used to form the wiring layers **832**, whereas the plated layer **890c** and the basal layer **890a** covered by the

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plated layer **890c** are used to form the frame-shaped conductor **861**. Referring to FIGS. 13~17, the plated layer **890b** and the basal layer **890a** covered by the plated layer **890b** are integrally expressed as the wiring layers **832**, whereas the plated layer **890c** and the basal layer **890a** covered by the plated layer **890c** are integrally expressed as the frame-shaped conductor **861**.

Referring to FIG. 12, the aforesaid five steps are carried out to form the wiring layers **832**, the connecting portions **851** and the frame-shaped conductor **861**. In this embodiment, the same basal layer **890a** is used to form the wiring layers **832** and the frame-shaped conductor **861**; however, in a variant embodiment, formation of the wiring layers **832** and formation of the frame-shaped conductor **861** requires formation of different basal layers, respectively. In this embodiment, a step of forming the basal layer **890a**, a step of forming the plated layer **890b**, and a step of removing the useless basal layer **890a** are combined to become a step combo equivalent to “a first wiring layer forming step” recited in the claims, whereas a step of forming the basal layer **890a**, a step of forming the plated layer **890c**, and a step of removing the useless basal layer **890a** are combined to become a step combo equivalent to “a second conductor forming step” recited in the claims.

Afterward, as shown in FIG. 13, the manufacturing method of the electronic device **A1** entails mounting an electronic component **811**. The electronic component **811** corresponds to the electronic component **11** of the electronic device **A1**. The electronic component **811** has a component main surface **811a** facing direction **z2** and a component inner surface **811b** facing direction **z1**, with electrode pads (not shown) formed on the component inner surface **811b**. A step of mounting the electronic component **811** (a first electronic component mounting step) is carried out by flip-chip bonding. After flux has been coated on the component inner surfaces **811b** of the electronic components **811**, for example, the electronic component **811** is temporarily mounted on the connecting portions **851** with a flip-chip bonder. At this point, the component inner surfaces **811b** face the wiring layers **832**. Each connecting portion **851** is an electrode pad (not shown) formed between the wiring layer **832** and the component inner surface **811b** formed on the electronic component **811**. Afterward, the connecting layers **851b** of the connecting portions **851** are melted by reflow soldering and thus coupled to the electrode pads. Then, the connecting layers **851b** of the connecting portions **851** are cooled and solidified. Therefore, the electronic component **811** is mounted on the wiring layers **832** such that the electrode pads of the electronic component **811** are electrically connected to the wiring layers **832** by the connecting portions **851**.

Afterward, as shown in FIG. 14, the manufacturing method of the electronic device **A1** entails forming a second resin layer **822**. A step of forming the second resin layer **822** (a second resin layer forming step) is, for example, carried out by die molding. Both the second resin layer **822** and the first resin layer **821** are capable of electrical insulation and are, for example, made of synthetic resin which uses black epoxy resin as a base. In this embodiment, the second resin layer **822** for covering the electronic component **811** and the frame-shaped conductor **861** is formed on the first resin layer **821**. The second resin layer **822** formed in the second resin layer forming step fully covers the electronic component **811** and the frame-shaped conductor **861**. Therefore, the direction-**z2**-facing surface (a second resin layer main surface **822a**) of the second resin layer **822** is closer to direction **z2** than any one of the direction-**z2**-facing surface

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of the frame-shaped conductor **861** and the component main surface **811a**. Prior to performing die molding, the second resin layer forming step involves applying, for example, underfill, which uses epoxy resin as a base, to the electronic component **811** from below (between the electronic component **811** and the first resin layer main surface **821a**).

Afterward, as shown in FIG. 15, the manufacturing method of the electronic device **A1** entails removing the supporting substrate **800**. In a step of removing the supporting substrate **800** (a supporting substrate removing step), grinding is performed with a mechanical grinding wheel. However, the grinding is not necessarily performed with a mechanical grinding wheel. In this embodiment, the supporting substrate **800** is ground from the supporting substrate inner surface **802** toward direction **z2** until the supporting substrate **800** is fully removed. In this embodiment, not only is the supporting substrate **800** ground and fully removed, but the basal layer of the columnar conductor **831** is also ground. Therefore, the plated layer, which is a Cu-containing metal layer, functions as the columnar conductor **831**. When the basal layer of the columnar conductor **831** is kept intact while the supporting substrate **800** is being ground, the columnar conductor **831** comprises a basal layer and a plated layer. Owing to the step of removing the supporting substrate, both the direction-**z1**-facing surface (first resin layer inner surface **821b**) of the first resin layer **821** and the direction-**z1**-facing surface (columnar conductor inner surface **831b**) of the columnar conductor **831** are exposed to the outside. When the supporting substrate **800** is a glass substrate, the glass substrate is stripped away by chemical processing or laser irradiation, thereby removing the supporting substrate **800**.

Afterward, as shown in FIG. 16, the manufacturing method of the electronic device **A1** entails forming an external electrode **840**. A step of forming the external electrode **840** (an external electrode forming step) requires electroless plating. In this embodiment, by electroless plating, the Ni layer, Pd layer and Au layer are emanated sequentially. At this point, the Ni layer which is in contact with the columnar conductor inner surface **831b** and covers the columnar conductor inner surface **831b** is formed, the Pd layer is formed on the Ni layer, and the Au layer is formed on the Pd layer. Therefore, the external electrode **840** shown in FIG. 16 is formed. The above description is not restrictive of how the external electrode **840** is formed; hence, the present disclosure allows the Ni layer and Au layer to be emanated sequentially, allows the sole presence of the Au layer, or allows the sole presence of Sn.

Afterward, as shown in FIG. 17, the manufacturing method of the electronic device **A1** entails grinding a second resin layer **822**. A step of grinding the second resin layer **822** (a second resin layer grinding step) entails, for example, grinding the second resin layer **822** with a mechanical grinding wheel, such as a grinding stone. The present disclosure is not restrictive of how the second resin layer **822** is ground. In this embodiment, the second resin layer **822** is ground from the second resin layer main surface **822a** toward direction **z1** until the frame-shaped conductor **861** is exposed. Therefore, the second resin layer main surface **822a** moves in direction **z1**, and the direction-**z**-facing surface (a top surface **861c**) of the frame-shaped conductor **861** is exposed from the second resin layer **822** (the second resin layer main surface **822a**). In this embodiment, the grinding of the second resin layer **822** is accompanied by a smaller degree of the grinding of the frame-shaped conductor **861**. Upon completion of the grinding process, a burr is likely to be formed on the top surface **861c**, because the

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frame-shaped conductor **861** and the second resin layer **822** are made of different materials. Therefore, the burr is removed by chemical processing. Therefore, the top surface **861c** of the frame-shaped conductor **861** dents in direction **z** more than the second resin layer main surface **822a**.

Afterward, the manufacturing method of the electronic device **A1** entails performing singulation to attain each electronic component **811**. In a singulation step (a singulating step), for example, the first resin layer **821** and the second resin layer **822** are cut with a knife blade. At this point, the first resin layer **821** and the second resin layer **822** are cut along a cutting line **CL1** shown in FIG. 17 and depicted in a rectangular shape because of the thickness of the knife blade. The cutting process is not necessarily carried out with a knife blade. In a variant embodiment, the cutting process is carried out by laser cutting or plasma cutting. At the end of the singulation step, the first resin layer **821** and the second resin layer **822** are cut and turned into the electronic devices **A1**, one of which is shown in FIG. 1~FIG. 4.

The aforesaid steps are carried out to manufacture the electronic devices **A1**, one of which is shown in FIG. 1~FIG. 4. The above description of the manufacturing method of the electronic device **A1** merely serves exemplary purposes, and thus the present disclosure is not limited thereto. For example, in a variant embodiment, the second resin layer grinding step precedes the supporting substrate removing step and the external electrode forming step. Under the aforesaid condition, to prevent the external electrode **840** from being formed, by electroless plating, on the top surface **861c** of the frame-shaped conductor **861** exposed from the second resin layer **822** in the external electrode forming step, a cutting protective tape is adhered to the second resin layer main surface **822a** of the second resin layer **822** before the external electrode forming step. In a variant embodiment, when a connecting part, such as a solder bump, is formed on electrode pads of the electronic component **811**, the connecting layers **851b** of the connecting portions **851** are formed without carrying out the step of forming the connecting layers **851b**.

The effect and advantage of the electronic device **A1** and the manufacturing method of the electronic device **A1** according to the first embodiment are described below.

The electronic device **A1** comprises the first resin layer **21** and the second resin layer **22**. The first resin layer **21** spaces apart the wiring layers **32** to underpin the electronic component **11**. The second resin layer **22** is formed on the first resin layer **21** and covers the electronic component **11**. Given the aforesaid structural features, the first resin layer **21** is a supporting part for supporting the electronic component **11**, and the second resin layer **22** is a protective part for covering the electronic component **11**. Therefore, the difference in thermal expansion coefficient between the supporting part and the protective part decreases. In this embodiment, there is hardly any difference in thermal expansion coefficient between the supporting part and the protective part, because both the first resin layer **21** and the second resin layer **22** are made of epoxy resin. Therefore, thermal stress on the interface between the supporting part (first resin layer **21**) and the protective part (second resin layer **22**) is lessened by the heat generated by the electronic component **11** while the electronic device **A1** is powered. Therefore, the protective part is less likely to be stripped away from the supporting part, thereby enhancing the reliability of the electronic device **A1**.

Regarding the electronic device **A1**, the electronic component **11** is underpinned by the first resin layer **21** formed

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by die molding. Regarding an electronic device different from the electronic device A1 of the present disclosure, for example, an electronic device disclosed in citation document 1, the electronic component 11 is underpinned by the semiconductor substrate (silicon substrate). Therefore, formation of terminals on the bottom side of the electronic device entails forming a penetrating electrode known as TSV (through-silicon via). The formation of the TSV entails, for example, forming a penetrating hole by an etching process known as Bosch process; however, the thicker the semiconductor substrate is, the more difficult the formation of the penetrating hole is. Therefore, it is impossible to form the penetrating electrode which penetrates the supporting part (the semiconductor substrate). According to this embodiment, after the columnar conductor 31 (the columnar conductor 831) has been formed by electroplating, the first resin layer 21 (first resin layer 821) is formed by die molding. Therefore, a penetrating electrode (the columnar conductor 31) which penetrates the supporting part (first resin layer 21) can be formed easily. Therefore, the manufacturing of the electronic device A1 is easier than is the case when the semiconductor substrate is used as the supporting part.

Regarding the electronic device A1, a grinding mark is formed on the first resin layer main surface 211 of the first resin layer 21. Therefore, fine grooves and ridges are formed on the first resin layer main surface 211 because of the grinding mark. Given the structural features, the strength of adhesion between the first resin layer 21 and the second resin layer 22 is augmented by the anchoring effect. Therefore, separation of the protective part (second resin layer 22) from the supporting part (first resin layer 21) is prevented, so as to enhance the reliability of the electronic device A1.

Regarding the electronic device A1, the connecting portions 51 each comprise an insulating layer 511. Given this structural feature, the connecting layers 851b is less likely to extend to any unexpected locations when the connecting layers 851b (especially parts thereof corresponding to the third layer 512c) are melted by heat generated from reflow soldering during the first electronic component mounting step. Therefore, inadvertent short circuits are less likely to occur, thereby reducing the malfunctioning of the electronic device A1.

The electronic device A1 comprises the frame-shaped conductor 61. The frame-shaped conductor 61 is made of metal and surrounds the electronic component 11 when viewed from above. Given this structural feature, electromagnetic shielding achieved by the frame-shaped conductor 61 suppresses electromagnetic waves from the lateral side of the electronic component 11. Therefore, the malfunctioning of the electronic device A1 is less likely.

More embodiments of the electronic device of the present disclosure and the manufacturing method of the electronic device of the present disclosure are described below. Identical or similar structural features are hereunder denoted by identical reference numerals and not described whenever the structural features are disclosed in the first embodiment in the same way as the embodiments to be described below.

Second Embodiment

FIG. 18~FIG. 21 illustrate an electronic device according to the second embodiment of the present disclosure. The main feature which distinguishes an electronic device A2 of the second embodiment from the electronic device A1 is that the electronic device A2 comprises an electronic component 12 which is different from the electronic component 11.

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FIG. 18 is a top view of the electronic device A2 and depicts the hermetic seal resin 20 with an imaginary line (a double-dot and dash line). FIG. 19 is a top view of the electronic device A2 and depicts the electronic component 11, the hermetic seal resin 20, the wiring layers 32, the connecting portions 51 and the frame-shaped conductor 61 with an imaginary line (a double-dot and dash line). FIG. 20 is a top view taken along line XX-XX of FIG. 18. FIG. 21 is a partial enlarged cross-sectional view of a part of FIG. 20.

Referring to FIG. 18~FIG. 21, the electronic device A2 comprises the electronic components 11, 12, the hermetic seal resin 20 (first resin layer 21 and second resin layer 22), a plurality of columnar conductors 31, a plurality of wiring layers 32, 33, a plurality of external electrodes 40, a plurality of connecting portions 51, 52, the frame-shaped conductor 61 and an external protective film 71. Therefore, as shown in FIG. 18~FIG. 21, unlike the electronic device A1, the electronic device A2 further comprises the electronic component 12, the plurality of wiring layers 33, the plurality of connecting portions 52 and the external protective film 71.

The electronic component 12 and the electronic component 11 together are crucial to the functioning of the electronic device A2. In this embodiment, both the electronic component 12 and the electronic component 11 are semiconductor components which comprise the same semiconductor. Like the electronic component 11, the electronic component 12 is, for example, a component for use in voltage control, such as LSI (Large Scale Integration), IC (integrated circuit), and LDO (Low Drop Out), a component for use in amplification, such as an operational amplifier, or a discrete component, such as a transistor and a diode. The electronic component 12 comprises a semiconductor. The component is a passive component, for example, a resistor, an inductor, and a capacitor. When viewed from above, the electronic component 12 is rectangular in shape. When viewed from above, the electronic component 12 is smaller than the electronic component 11 and falls within the outline of the electronic component 11. The electronic component 12 is closer to direction z1 than the electronic component 11 in direction z. When viewed from above, the electronic component 12 is larger than the electronic component 11 in a variant embodiment. The electronic component 12 is electrically connected and joined to the wiring layers 33 by the connecting portions 52. The electronic component 12 can be surface mounted. The electronic component 12 is covered by the first resin layer 21. The electronic component 12 is equivalent to "a second electronic component" recited in the claims. As shown in FIG. 20, the electronic component 12 has a component main surface 121 and a component inner surface 122.

The component main surface 121 and the component inner surface 122 are spaced apart from each other in direction z and face opposite sides. The component main surface 121 faces direction z2. The component inner surface 122 faces direction z1. The component main surface 121 is covered by the first resin layer 21. A plurality of electrode pads (not shown) is formed on the component inner surface 122. The plurality of electrode pads each comprise aluminum (Al), for example. The electrode pads are terminals in the electronic component 12. FIG. 18 and FIG. 19 is not restrictive of the quantity and position of the plurality of electrode pads. The component main surface 121 is equivalent to "a second component main surface" recited in the claims.

In this embodiment, the columnar conductors 31 are formed on the wiring layers 33, respectively. The columnar conductor inner surfaces 312 of the columnar conductors 31

are in contact with the wiring layers 33, respectively. In this embodiment, the columnar conductors 31 are made of Cu. Each columnar conductor 31 comprises a basal layer and a plated layer which are laminated to each other. Under the aforesaid condition, the basal layer comprises Ti layer and Cu layer, with the Ti layer formed on the wiring layers 33, and the Cu layer formed on the Ti layer. The plated layer comprises Cu and is formed on the Cu layer of the basal layer.

The wiring layers 33 electrically connect the electronic component 12 to the columnar conductors 31, respectively. The wiring layers 33 each comprise a basal layer and a plated layer which are laminated to each other. The basal layer comprises a Ti layer and a Cu layer which are laminated to each other and is approximately 200–800 nm thick. The plated layer, for example, comprises Cu and is configured to be thicker than the basal layer. The above description is not restrictive of what material the wiring layers 33 are made of. FIG. 18 and FIG. 19 are not restrictive of the size of the wiring layers 33. The wiring layers 33 are each equivalent to “a second wiring layer” recited in the claims.

The wiring layers 33 each have a wiring layer main surface 331 and a wiring layer inner surface 332. The wiring layer main surface 331 and the wiring layer inner surface 332 are spaced apart and face opposite sides in direction z. The wiring layer main surface 331 faces direction z2, and the wiring layer inner surface 332 faces direction z1. The wiring layer main surface 331 is covered by the first resin layer 21. The columnar conductors 31 and the connecting portions 52 are formed, one by one, on each wiring layer main surface 331. A part of the wiring layer main surface 331 is in contact with the columnar conductor inner surface 312. The wiring layer inner surface 332 is exposed from the first resin layer 21 (first resin layer inner surface 212). In this embodiment, the wiring layer inner surface 332 and the first resin layer inner surface 212 are coplanar. A part of the wiring layer inner surface 332 is in contact with the external electrode 40. The wiring layer main surface 331 and the wiring layer inner surface 332 are equivalent to “a second wiring layer main surface” and “a second wiring layer inner surface” recited in the claims, respectively.

In this embodiment, the plurality of external electrodes 40 does not include a plurality of columnar conductor covering portions 41 but comprises a plurality of wiring layer covering portions 42.

Each wiring layer covering portion 42 covers a part of a corresponding one of the wiring layer inner surfaces 332. The wiring layer covering portions 42 are in contact with the wiring layer inner surfaces 332, respectively. In this embodiment, the electronic component 11 is electrically connected to the wiring layer covering portions 42 by the connecting portions 51, the wiring layers 32, the columnar conductors 31 and the wiring layers 33, respectively. The electronic component 12 is electrically connected to the wiring layer covering portions 42 by the connecting portions 52 and the wiring layers 33, respectively. Therefore, the wiring layer covering portions 42 are terminals electrically connected to both the electronic component 11 and the electronic component 12. The wiring layer covering portions 42 are each equivalent to “a second wiring layer covering portion” recited in the claims.

The connecting portions 52 are conductive connecting elements formed between the electronic component 12 (electrode pads) and the wiring layers 33. The electronic component 12 is fixed to the plurality of wiring layers 33 by the plurality of connecting portions 52 and thus mounted on

the wiring layers 33. The connecting portions 52 ensure that the electronic component 12 is electrically connected to the wiring layers 33. In this embodiment, as shown in FIG. 21, the connecting portion 52 comprises an insulating layer 521 and a connecting layer 522.

Referring to FIG. 21, the insulating layers 521 are formed on the wiring layers 33, respectively. The insulating layer 521 and the insulating layer 511 are of the same structure. When viewed from above, the insulating layer 521 is centrally-opened and frame-shaped. When viewed from above, the insulating layer 521 has the shape of a rectangular frame. When viewed from above, the insulating layer 521 does not necessarily have the shape of a rectangular frame but can also have the shape of a circular frame, an elliptical frame or a polygonal frame. When viewed from above, the insulating layer 521 includes the connecting layer 522. The insulating layer 521 is, for example, made of polyimide resin, but the present disclosure is not limited thereto.

Owing to the connecting layers 522, the electronic component 12 is electrically connected and joined to the wiring layers 33. The connecting layers 522 are formed on the wiring layers 33 (wiring layer main surfaces 331), respectively. The connecting layers 522 and the connecting layers 512 are of the same structure. The connecting layers 522 cover the open parts of the insulating layers 521. In this embodiment, the open parts of the insulating layers 521 are filled with parts of the connecting layers 522, respectively. In this embodiment, as shown in FIG. 21, each connecting layer 522 comprises a first layer 522a, a second layer 522b and a third layer 522c which are laminated to each other. The first layer 522a, the second layer 522b and the third layer 522c are of the same structure as the first layer 512a, the second layer 512b and the third layer 512c of the connecting layer 512 of each connecting portion 51, respectively.

The external protective film 71 has an insulative resin film. The external protective film 71 is, for example, made of polymeric resin. Examples of the polymeric resin include polyimide resin and phenol-based resin. The present disclosure is not restrictive of what material the external protective film 71 is made of; hence, the external protective film 71 will work, provided that the material which the external protective film 71 is made of is insulative resin. The external protective film 71 at the very least covers the wiring layer inner surfaces 332 exposed from the wiring layer covering portions 42 of the external electrode 40. In this embodiment, the external protective film 71 covers the wiring layer inner surfaces 332 exposed from the wiring layer covering portions 42 of the external electrode 40 and the whole of the first resin layer inner surface 212. The external protective film 71 is equivalent to “a protective film” recited in the claims.

Referring to FIG. 22~FIG. 29, an example of the manufacturing method of the electronic device A2 according to the second embodiment is described below. FIG. 22~FIG. 29 are cross-sectional views of a step of the manufacturing method of the electronic device A2. A step disclosed in the second embodiment is not described whenever the step is disclosed in the second embodiment in a way identical or similar to the first embodiment.

First, like the first embodiment, the second embodiment includes a supporting substrate preparing step for preparing the supporting substrate 800.

Afterward, as shown in FIG. 22~FIG. 26, the wiring layers 833, the connecting portions 852 and the columnar conductors 831 are formed. The wiring layers 833, the connecting portions 852 and the columnar conductors 831 correspond to the wiring layers 32, the connecting portions 52 and the columnar conductors 31 of the electronic device

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A2, respectively. The wiring layers **833**, the connecting portions **852** and the columnar conductors **831** are formed in five steps described below.

In the first step, as shown in FIG. 22, a basal layer **891a** is formed. The basal layer **891a** is, for example, formed by sputtering. In the basal layer **891a** forming step, after the Ti layer for covering the whole of the supporting substrate main surface **801** has been formed, the Cu layer in contact with the Ti layer is formed. The basal layer **891a** is formed of a Ti layer and a Cu layer which are laminated to each other.

In the second step, as shown in FIG. 23, a plated layer **891b** is formed. The photoresist pattern is formed on the plated layer **891b** by photolithography and electroplating. In a step of forming the plated layer **891b**, the photosensitive photoresist is coated on the whole of the basal layer **891a** and thereby undergoes exposure and development so as to pattern the photoresist layer. Therefore, the photoresist pattern is formed, and a part (which forms the plated layer **891b**) of the basal layer **891a** is exposed from the photoresist pattern. After that, the basal layer **891a** functions as a conducting path whereby electroplating is carried out, and thus the plated layer **891b** is emanated from the basal layer **891a** exposed from the photoresist pattern. In this embodiment, for example, the metal layer functioning as the plated layer **891b** and comprising Cu is emanated. At this point, the plated layer **891b** is integrally formed with the basal layer **891a**. Afterward, the photoresist layer formed in this step is entirely removed. Therefore, the plated layer **891b** shown in FIG. 23 is formed. Afterward, the plated layer **891b** and the basal layer **891a** covered by the plated layer **891b** become the wiring layers **833**. The wiring layers **833** correspond to the wiring layers **33** of the electronic device A2.

In the third step, as shown in FIG. 24, the connecting portions **852** are formed. In this embodiment, an insulating layer **852a** and the connecting layers **852b** are formed to function as the connecting portions **852**. In a step of forming the insulating layer **852a**, photosensitive polyimide is coated on the whole of the plated layer **891b** and the whole of the basal layer **891a** exposed from the plated layer **891b**. The photosensitive polyimide is, for example, coated with a spin coater. Then, the photosensitive polyimide thus coated undergoes exposure and development to form the frame-shaped insulating layer **852a**. After that, in a step of forming the connecting layers **852b**, the photoresist pattern for forming the connecting layers **852b** is formed. The formation of the photoresist pattern entails coating the photosensitive photoresist and performing exposure and development on the coated photosensitive photoresist to pattern the photoresist layer. Therefore, the photoresist pattern is formed, and a part (which forms the connecting layer **852b**) of the plated layer **891b** is exposed from the photoresist pattern. The exposed part is located on the inner side of the frame-shaped insulating layer **852a** when viewed from above. After that, the basal layer **891a** and plated layer **891b** function as a conducting path whereby electroplating is carried out, and thus the connecting layer **852b** is emanated from the plated layer **891b** exposed from the photoresist pattern. In this embodiment, the connecting layer **852b** is formed of a Cu-containing metal layer, a Ni-containing metal layer and a Sn-containing alloy layer by sequential lamination. The Sn-containing alloy layer is, for example, made of Sn—Sb based alloy or Sn—Ag based alloy which is typical of lead-free solder. Afterward, the photoresist pattern formed in this step is removed. Therefore, as shown in FIG. 24, the connecting portions **852** each comprising an insulating layer

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852a and a connecting layer **852b** are formed. The connecting portions **852** correspond to the connecting portions **52** of the electronic device A2.

In the fourth step, as shown in FIG. 25, a plated layer **891c** is formed. For example, the photoresist pattern is formed on the plated layer **891c** by photolithography and electroplating. The plated layer **891c** is formed in the same way as the plated layer **891b**. In a step of forming the plated layer **891c**, the photoresist pattern for forming the plated layer **891c** is formed. Therefore, a part (which forms the plated layer **891c**) of the plated layer **891b** is exposed from the photoresist pattern thus formed. After that, the basal layer **891a** and the plated layer **891b** function as a conducting path whereby electroplating is carried out, and thus the plated layer **891c** is emanated from the plated layer **891b** exposed from the photoresist pattern. In this embodiment, for example, the metal layer functioning as the plated layer **891c** and comprising Cu is emanated. Afterward, the photoresist pattern formed in this step is removed. Therefore, the plated layer **891c** shown in FIG. 25 is formed. In this embodiment, the plated layer **891c** becomes the columnar conductor **831**.

In the fifth step, as shown in FIG. 26, the useless basal layer **891a** is removed. In this embodiment, the basal layer **891a** not covered by the plated layer **891b** is regarded as the useless basal layer **891a** and thus removed. Both the removal of the useless basal layer **891a** and the removal of the useless basal layer **890a** are carried out by wet etching. Given the useless basal layer **891a** removing step, as shown in FIG. 26, the wiring layers **833** are formed, using the plated layer **891b** and the basal layer **891a** covered by the plated layer **891b**. As shown in FIG. 27~FIG. 29, the plated layer **891b** and the basal layer **891a** covered by the plated layer **891b** are integrally expressed as the wiring layers **833**, and the plated layer **891c** is expressed as the columnar conductor **831**.

As shown in FIG. 26, the wiring layers **833**, the connecting portions **852**, and the columnar conductors **831** are formed by the aforesaid five steps. In this embodiment, the step of forming the basal layer **891a**, the step of forming the plated layer **891c**, and the step of removing the useless basal layer **891a** are combined to become a step combo equivalent to “a second wiring layer forming step” recited in the claims. The step of forming the basal layer **891a**, the step of forming the plated layer **891c**, and the step of removing the useless basal layer **891a** are combined to become a step combo equivalent to “a first conductor forming step” recited in the claims.

Afterward, as shown in FIG. 27, the electronic component **812** is mounted. The electronic component **812** corresponds to the electronic component **12** of the electronic device A2. The electronic component **812** has a component main surface **812a** facing direction **z2** and a component inner surface **812b** facing direction **z1**, with electrode pads (not shown) formed on the component inner surface **812b**. A step of mounting the electronic component **812** (a second electronic component mounting step) is carried out by flip-chip bonding. After flux has been coated on the electronic component **812**, for example, the electronic component **812** is temporarily mounted on the connecting portions **852** with a flip-chip bonder. At this point, the connecting portions **852** are formed between the wiring layers **833** and formed between electrode pads (not shown) on the component inner surface **812b** of the electronic component **812**. Afterward, the connecting layers **852b** of the connecting portions **852** are melted by reflow soldering and thus coupled to the electrode pads. Then, the connecting layers **852b** of the connecting portions **852** are cooled and solidified. Therefore, the electronic component **812** is mounted on the wiring

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layers **833** such that the wiring layers **833** and the electrode pads of the electronic component **812** are electrically connected by the connecting portions **852**, respectively.

Afterward, like the manufacturing method of the electronic device **A1**, the manufacturing method of the electronic device **A2** entails carrying out the first resin layer forming step, the first resin layer grinding step, the wiring layers **832** forming step, the connecting portions **851** forming step, the frame-shaped conductor **861** forming step, the first electronic component mounting step, the second resin layer forming step, and the supporting substrate removing step (see FIG. 6–FIG. 15). This embodiment dispenses with the columnar conductor forming step.

Afterward, as shown in FIG. 28, an external protective film **871** is formed. In a step of forming the external protective film **871** (an external protective film forming step), the polymeric resin which connects the wiring layer inner surface **833b** and the first resin layer inner surface **821b** is formed, except for a part (reserved for lateral formation of the external electrode **840**) of the wiring layer inner surface **833b**. In this embodiment, examples of the polymeric resin include polyimide resin and phenol-based resin. The external protective film **871** thus formed has an opening portion **871a** which a part of each wiring layer inner surface **833b** is exposed from.

Afterward, as shown in FIG. 29, the external electrode **840** is formed. Like the external electrode forming step in the first embodiment, the external electrode forming step in this embodiment requires electroless plating. A Ni layer, a Pd layer and a Au layer are sequentially laminated to part of each wiring layer inner surface **833b** exposed from the opening portion **871a** of the external protective film **871**. Therefore, the external electrode **840** is of a structure formed by lamination of the Ni layer, Pd layer and Au layer.

Afterward, like the first embodiment, the second embodiment entails carrying out the second resin layer grinding step and the singulation step. Therefore, the electronic device **A2** shown in FIG. 18–FIG. 21 is manufactured. The above description of the manufacturing method of the electronic device **A2** merely serves exemplary purposes, and thus the present disclosure is not limited thereto. For example, when a connecting part, such as a solder bump, is formed on electrode pads of the electronic component **812**, the connecting layers **852b** of the connecting portions **852** are formed without carrying out the step of forming the connecting layers **852b**.

The effect and advantage of the electronic device **A2** and the manufacturing method of the electronic device **A2** according to the second embodiment are described below.

Like the electronic device **A1**, the electronic device **A2** comprises the first resin layer **21** and the second resin layer **22**. The first resin layer **21** spaces apart the wiring layers **32** to underpin the electronic component **11**. The second resin layer **22** is formed on the first resin layer **21** and covers the electronic component **11**. Therefore, like the first embodiment, the second embodiment is effective in reducing the difference in thermal expansion coefficient between the supporting part (first resin layer **21**) and the protective part (second resin layer **22**). Therefore, like the first embodiment, the second embodiment is effective in reducing the thermal stress on the interface between the supporting part and the protective part, and thus the protective part is less likely to be stripped away from the supporting part. Therefore, the reliability of the electronic device **A2** is enhanced.

If the electronic devices **A1**, **A2** have an identical or similar technical feature in common, the technical feature will have the same effect on the electronic devices **A1**, **A2**.

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The electronic device **A2** comprises a plurality of electronic components **11**, **12**. The electronic components **11** are covered by the second resin layer **22**, and the electronic components **12** are covered by the first resin layer **21**. The first resin layer **21** is laminated to the second resin layer **22** in direction **z**. Therefore, the electronic components **11** and the electronic components **12** become structures capable of polyphasic mounting in direction **z**. Therefore, the electronic components **11**, **12** overlap in direction **z**, and thus the electronic device **A2** looks smaller when viewed from above. The formation of the electronic components **11**, **12** involves forming the first resin layer **21** and second resin layer **22** in a polyphasic manner. None of the electronic components **11**, **12** includes a semiconductor substrate, and thus the processing of a semiconductor substrate is not required, thereby rendering polyphasic formation of the electronic components **11**, **12** easy.

The second embodiment is not restrictive of the structure of the electronic component **12**. FIG. 30 shows the electronic device with the electronic component **12** of a variant structure. FIG. 30 is a cross-sectional view of the electronic device according to this variant embodiment and corresponds to the cross-sectional view of FIG. 20. As shown in FIG. 30, in this variant embodiment, electrodes are formed at two ends of the electronic component **12** in direction **x**. Examples of the electronic component **12** of the aforesaid structure include a chip capacitor and a chip resistor. As shown in FIG. 30, the electronic component **12** is joined to the wiring layers **833** by the connecting portions **53**. The connecting portions **53** are conductive connecting elements in the form of solder paste or silver solder paste. Fillets are formed at the connecting portions **53**.

Third Embodiment

FIG. 31 illustrates an electronic device according to the third embodiment of the present disclosure. The main feature which distinguishes an electronic device **A3** of the third embodiment from the electronic device **A2** is that the component main surface **121** is exposed from the first resin layer **21**.

FIG. 31 is a cross-sectional view of the electronic device **A3** and corresponds to the cross-sectional view of the electronic device **A2** shown in FIG. 20.

Regarding the electronic device **A3**, the component main surface **121** of the electronic component **12** is exposed from the first resin layer main surface **211** of the first resin layer **21**. In this embodiment, the component main surface **121** and the first resin layer main surface **211** are coplanar. To allow the component main surface **121** to be exposed from the first resin layer main surface **211**, for example, the first resin layer grinding step involves grinding the first resin layer **821** until the component main surface **812a** of the electronic component **812** is exposed.

The component protective film **72** is insulative film. The component protective film **72** covers the component main surface **121** of the electronic component **12**. When viewed from above, the component protective film **72** overlaps the electronic component **12**. Both the component protective film **72** and the external protective film **71** are made of polymeric resin. The above description is not restrictive of the material which the component protective film **72** is made of. The formation of the component protective film **72**, for example, occurs after the first resin layer grinding step but before the step of forming the basal layer **891a**. This timing merely serves exemplary purposes and thus is not restrictive of the formation of the component protective film **72**. In this

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embodiment, the electronic device A3 comprises the component protective film 72; however, in a variant embodiment, the electronic device A3 does not include the component protective film 72.

The effect and advantage of the electronic device A3 and the manufacturing method of the electronic device A3 according to the third embodiment are described below.

Like the electronic device A1, the electronic device A3 comprises the first resin layer 21 and the second resin layer 22. The first resin layer 21 spaces apart the wiring layers 32 to underpin the electronic component 11. The second resin layer 22 is formed on the first resin layer 21 and covers the electronic component 11. Therefore, like the first embodiment, the third embodiment is effective in reducing the difference in thermal expansion coefficient between the supporting part (first resin layer 21) and the protective part (second resin layer 22). Therefore, like the first embodiment, the third embodiment is effective in reducing the thermal stress on the interface between the supporting part and the protective part, and thus the protective part is less likely to be stripped away from the supporting part. Therefore, the reliability of the electronic device A3 is enhanced.

The electronic device A3 can have a structural feature identical or similar to that of the electronic devices A1, A2 to bring about an effect which the electronic device A3 and the electronic device A1, A2 have in common.

Regarding the electronic device A3, in the first resin layer grinding step, the first resin layer 821 is ground until the component main surface 812a of the electronic component 812 is exposed. Given the structural feature, the direction-z dimension of the electronic device A3 decreases. Therefore, the electronic device A3 is downsized.

The electronic device A3 comprises a component protective film 72 for covering the component main surface 121 of the electronic component 12. In the electronic device A3, the component main surface 121 of the electronic component 12 is exposed from the first resin layer 21, and thus a conductor is likely to be inadvertently formed on the component main surface 121 during the manufacturing process of the electronic device A3. Therefore, inadvertent short circuits are likely to occur to the electronic component 12. With the component protective film 72 being formed, the whole surface of the electronic component 12 is covered by the first resin layer 21 and component protective film 72, inadvertent short circuits are less likely to occur to the electronic component 12. Therefore, reliability of the electronic device A3 is enhanced.

Other variant embodiments of the electronic device of the present disclosure are described below. The variant embodiments described below can be appropriately combined.

Regarding the electronic device of the present disclosure, a metal film can be formed on the second resin layer main surface 221 of the second resin layer 22. FIG. 32 shows that the electronic device A1 in the first embodiment comprises the metal film (metal film 62). FIG. 32 is a cross-sectional view of the electronic device in the variant embodiment and corresponds to the cross-sectional view of FIG. 3. The metal film 62 is, for example, formed by sequential lamination of a Ti layer, a Cu layer and a stainless steel layer. The metal film 62 is, for example, formed by sputtering. The above description is not restrictive of what material the metal film 62 is made of and how the metal film 62 is formed. The metal film 62 is in contact with the top surface 613 of the frame-shaped conductor 61. Therefore, in the presence of the metal film 62, the electronic component 11 is fully covered, and thus interference electromagnetic waves from

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the outside can be blocked. The electronic devices A2, A3 each comprise the metal film 62.

Regarding the electronic device of the present disclosure, the first, second and third embodiments are not restrictive of the structure of each external electrode 40. For example, each external electrode 40 can also be a ball-shaped solder bump (solder ball). FIG. 33 shows that, in the electronic device A1 of the first embodiment, the external electrodes 40 are solder balls. FIG. 33 is a cross-sectional view of the electronic device according to the variant embodiment and corresponds to the cross-sectional view of FIG. 3. In the electronic devices A2, A3, the external electrodes 40 are also solder balls.

Regarding the electronic device of the present disclosure, the first, second and third embodiments are not restrictive of the structure of each connecting portion 51. FIG. 34~FIG. 36 shows different structures of the connecting portions 51 of the electronic device A1 according to the first embodiment. FIG. 34~FIG. 36 are partial enlarged cross-sectional views of the electronic device according to their respective variant embodiments and correspond to the partial enlarged cross-sectional view of FIG. 4. The connecting portions are not only found in the electronic device A1 but also found in the electronic devices A2, A3. The connecting portions 52 illustrated by the second embodiment and the third embodiment can also be of the same structure as the connecting portions 51 shown in FIG. 34~FIG. 36, respectively.

Referring to FIG. 34, the insulating layer 511 of the connecting portion 51 is, for example, flux which covers the wiring layers 32. A part of the insulating layer 511 above each wiring layer 32 is opened. The open part of the insulating layer 511 is filled with a part of the connecting layer 512. As shown in FIG. 34, the connecting layer 512 comprises a first layer 512a, a second layer 512b, a third layer 512c and a fourth layer 512d which are laminated to each other. The first layer 512a comprises a Ti layer and a Cu layer which are laminated to each other. The Ti layer is in contact with the wiring layers 32. The first layer 512a is, for example, formed by sputtering. The second layer 512b is made of Cu-containing metal. The third layer 512c is made of Ni-containing metal. The fourth layer 512d is, for example, made of Sn-containing metal. Examples of the alloy include Sn—Sb based alloy and Sn—Ag based alloy which are typical of lead-free solder. The second layer 512b, the third layer 512c and the fourth layer 512d are, for example, formed by electroplating, respectively.

Referring to FIG. 35, the connecting layer 512 of the connecting portion 51 comprises a first layer 512a, a third layer 512c and a fourth layer 512d. Hence, unlike FIG. 34, FIG. 35 shows that the connecting layer 512 of the connecting portion 51 does not include the second layer 512b. The insulating layer 511 shown in FIG. 35 is of the same structure as its counterpart in FIG. 34.

The connecting portion 51 shown in FIG. 36 does not include insulating layer 511 but includes the connecting layer 512. The connecting layer 512 is made of metal, such as Sn. The alloy is, for example, Sn—Sb based alloy or Sn—Ag based alloy which is typical of lead-free solder. FIG. 36 shows the electrode pad 13 of the electronic component 11. The electrode pad 13 comprises a first layer 131 and a second layer 132 which are laminated to each other. The first layer 131 is a metal layer which contains Cu, for example. The second layer 132 is a metal layer which contains Ni, for example. In a variant embodiment illustrated by FIG. 36, the connecting portion 51 includes the insulating layer 511.

It is also feasible that the electronic device of the present disclosure dispenses with the frame-shaped conductor 61. FIG. 37 shows the electronic device according to this variant embodiment. FIG. 37 is a cross-sectional view of the electronic device according to this variant embodiment and corresponds to the cross-sectional view of FIG. 3. FIG. 37 shows that the electronic device A1 in the first embodiment dispenses with the frame-shaped conductor 61, and the electronic devices A2, A3 can also be of this same structure as the electronic device A1.

Regarding the electronic device of the present disclosure, the component main surface 111 of the electronic component 11 is exposed from the second resin layer main surface 221 of the second resin layer 22. FIG. 38 shows the electronic device in this variant embodiment. FIG. 38 is a cross-sectional view of the electronic device in this variant embodiment, and it corresponds to the cross section shown in FIG. 3. FIG. 38 shows that, in the electronic device A1 of the first embodiment, the component main surface 111 is exposed from the second resin layer main surface 221; this feature also occurs to the electronic devices A2, A3. For example, in the second resin layer grinding step, the second resin layer 822 is ground until the component main surface 811a of the electronic component 811 is exposed. In this variant embodiment, the component main surface 111 of the electronic component 11 is exposed from the electronic device, and thus a protective film for covering the component main surface 111 is formed in advance. In this variant embodiment, the second resin layer 22 is ground until the component main surface 111 of the electronic component 11 is exposed, and thus the thickness (in direction z) of the second resin layer 22 is reduced. With its thickness (in direction z) being reduced, the electronic device is downsized. Furthermore, the electronic component 11 can dissipate heat better, because the component main surface 111 of the electronic component 11 is exposed from the second resin layer 22.

Regarding the electronic device of the present disclosure, the first, second and third embodiments are not restrictive of the size of the wiring layers 32. For example, the size of the wiring layers 32 is subject to changes according to the quantity and position of electrode pads formed on the component inner surface 112 of the electronic component 11 and the quantity and position of terminals (external electrodes 40) of the electronic device. FIG. 39 and FIG. 40 show that the size of the wiring layers 32 of the electronic device A1 in the first embodiment differs. FIG. 39 and FIG. 40 are top views of the electronic device in this variant embodiment and merely serve exemplary purposes. As shown in FIG. 39 and FIG. 40, the electronic component 11 has eight electrode pads, the size of the wiring layers 32 is subject to changes according to the quantity of the electrode pads. Likewise, the size of the wiring layers 32 of the electronic devices A2, A3 is subject to changes.

Regarding the electronic device of the present disclosure, the second and third embodiments are not restrictive of the size of the wiring layers 33. The size of the wiring layers 33 and the size of the wiring layers 32 is subject to changes, for example, according to the quantity and position of the electrode pads of the electronic component 11, the quantity and position of the electrode pads of the electronic component 12, an electrical connection path of the electronic component 11 and the electronic component 12, and the quantity and position of terminals (external electrodes 40) of the electronic device.

Regarding the electronic device of the present disclosure, in a variant embodiment, the electronic component 11 does

not overlap the electronic component 12 when viewed from above. FIG. 41 shows the electronic devices in the variant embodiment. FIG. 41 is a top view of the electronic device and depicts the hermetic seal resin 20 with an imaginary line. Referring to FIG. 41, as mentioned above, when viewed from above, the electronic component 11 does not overlap the electronic component 12, but the electronic components 11, 12 are aligned in direction x. Referring to FIG. 41, when viewed from above, the electronic components 11, 12 either overlap or do not overlap.

Regarding the electronic device of the present disclosure, the present disclosure is not restrictive of the quantity of the electronic components 11 and the quantity of the electronic components 12. FIG. 42 shows the electronic devices in the variant embodiment. FIG. 42 is a top view of the electronic device and corresponds to FIG. 19 according to the second embodiment. Referring to FIG. 42, the electronic components 12 are in the number of two and are joined to the wiring layers 33, respectively. FIG. 42 shows two electronic components 12, but more than two electronic components 12 can be provided. The electronic components 11 can also be in the number of at least two.

Regarding the electronic device of the present disclosure, the first, second and third embodiments are not restrictive of the structure of the external electrode 40. FIG. 43 shows that the external electrode 40 comprises columnar conductor covering portions 41 and wiring layer covering portions 42. FIG. 44 shows that, in the electronic device A2 of the second embodiment, the external electrode 40 does not include the wiring layer covering portions 42 but comprises the columnar conductor covering portions 41. FIG. 43 and FIG. 44 are cross-sectional views of the electronic device according to this variant embodiment and correspond to the cross-sectional view of FIG. 20. Referring to FIG. 43, the columnar conductors 31 are not formed on the wiring layers 33, whereas the electronic component 11 and the electronic component 12 are not electrically connected inside the electronic device. The electronic component 11 is electrically connected to the columnar conductor covering portions 41 (external electrode 40) by the connecting portions 51, the wiring layers 32 and the columnar conductors 31. The electronic component 12 is electrically connected to the wiring layer covering portions 42 (external electrode 40) by the connecting portions 52 and the wiring layers 33. Therefore, in the electronic device shown in FIG. 43, the columnar conductor covering portions 41 are terminals electrically connected to the electronic component 11, and the wiring layer covering portions 42 are terminals electrically connected to the electronic component 12. Referring to FIG. 44, the first columnar conductor 31 penetrates in direction z from the first resin layer main surface 211 of the first resin layer 21 to the first resin layer inner surface 212, and the second columnar conductor 31 is formed on the wiring layers 33. The electronic component 11 is electrically connected to the columnar conductor covering portions 41 (external electrode 40) by the connecting portions 51, wiring layers 32 and first columnar conductor 31. The electronic component 12 is electrically connected to the columnar conductor covering portions 41 (external electrode 40) by the connecting portions 52, wiring layers 33, second columnar conductor 31, wiring layers 32 and first columnar conductor 31. Therefore, in the electronic device shown in FIG. 44, the wiring layer covering portions 42 (external electrode 40) are each a terminal electrically connected to the electronic component 11 and the electronic component 12.

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Regarding the electronic device of the present disclosure, the structure of the hermetic seal resin **20** is not restricted to the first, second and third embodiments, and thus the hermetic seal resin **20** can be formed by laminating the first resin layer **21**, the second resin layer **22**, and at least one resin layer to each other. Under the aforesaid condition, each resin layer has a conductor penetrating the resin layer, an electronic component covered by the resin layer, and wiring layers electrically connected to the electronic component, so as to achieve a mounting structure for mounting in more phases than the electronic devices **A2**, **A3**.

The aforesaid embodiment is not restrictive of the electronic device and the manufacturing method of the electronic device. The structural features of constituent elements of the electronic device of the present disclosure and specific process flows of the steps of the manufacturing method of the electronic device according to the present disclosure are subject to changes.

The electronic device of the present disclosure and the manufacturing method of the electronic device according to the present disclosure are illustrated by embodiments related to remarks as follows:

[Remark 1]

An electronic device, comprising:

- a first resin layer, having a first resin layer main surface and a first resin layer inner surface, the first resin layer main surface and the first resin layer inner surface face opposite sides in a first direction;
- a first conductor, having a first conductor main surface and a first conductor inner surface, the first conductor main surface and the first conductor inner surface face opposite sides in the first direction, and the first conductor penetrates the first resin layer in the first direction;
- a first wiring layer, straddling the first resin layer main surface and the first conductor main surface;
- a first electronic component, in the first direction having a first component main surface facing the same side as the first resin layer main surface and a first component inner surface facing the same side as the first resin layer inner surface, and electrically connected with the first wiring layer;
- a second resin layer, having a second resin layer main surface facing the same direction as the first resin layer main surface and a second resin layer inner surface being in contact with the first resin layer main surface, and covering the first wiring layer and the first electronic component; and
- an external electrode, disposed closer to the side where the first resin layer inner surface faces than the first resin layer and electrically connected to the first conductor.

[Remark 2]

Regarding the electronic device described in remark 1, a grinding mark is formed on the first resin layer main surface.

[Remark 3]

Regarding the electronic device described in remark 2, the first conductor main surface dents relative to the first resin layer main surface.

[Remark 4]

The electronic device described in any one of remark 1 through remark 3 further comprises a second wiring layer, having a second wiring layer main surface and a second wiring layer inner surface, the second wiring layer main surface and the second wiring layer inner surface face

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opposite sides in the first direction, and the second wiring layer inner surface being exposed from the first resin layer inner surface.

[Remark 5]

The electronic device described in remark 4 further comprises a second electronic component different from the first electronic component and electrically connected to the second wiring layers, wherein at least a part of the second electronic component is covered by the first resin layer.

[Remark 6]

Regarding the electronic device described in remark 5, the second electronic component has a second component main surface facing the same direction as the first component main surface, and the second component main surface and the first resin layer main surface are coplanar.

[Remark 7]

Regarding the electronic device described in any one of remark 1 through remark 6, the external electrode comprises a first conductor covering portion for covering the first conductor inner surface.

[Remark 8]

Regarding the electronic device described in any one of remark 4 through remark 6, the external electrode comprises a second wiring layer covering portion for covering a part of the second wiring layer inner surface.

[Remark 9]

The electronic device described in remark 8 further comprises a protective film for covering a portion of the second wiring layer inner surface, the portion being exposed from the external electrode.

[Remark 10]

Regarding the electronic device described in remark 8 or remark 9, the first conductor inner surface is in contact with the second wiring layer main surface.

[Remark 11]

The electronic device described in any one of remark 1 through remark 10 further comprises a conductive connecting layer for connecting the first electronic component with the first wiring layers;

wherein a part of the first wiring layer overlaps the first electronic component when viewed in the first direction, and

wherein the conductive connecting layer lies between the first component inner surface and the first wiring layer.

[Remark 12]

The electronic device described in any one of remark 1 through remark 11 further comprises a second conductor penetrating the second resin layer in the first direction and the second conductor is disposed on a periphery of the first electronic component when viewed in the first direction.

[Remark 13]

Regarding the electronic device described in remark 12, the second conductor and the first wiring layer are spaced apart from each other when viewed in the first direction.

[Remark 14]

Regarding the electronic device described in remark 13, the second conductor surrounds the first electronic component when viewed in the first direction.

[Remark 15]

Regarding the electronic device described in any one of remark 12 through remark 14, the second conductor has a second conductor main surface facing the same direction as the second resin layer main surface in the first direction, and the second conductor main surface is exposed from the second resin layer main surface.

[Remark 16]

The electronic device described in remark 15 further comprises a metal film overlapping the first electronic component and formed on the second resin layer main surface when viewed in the first direction.

[Remark 17]

Regarding the electronic device described in remark 16, the metal film is in contact with the second conductor main surface.

[Remark 18]

Regarding the electronic device described in any one of remark 15 through remark 17, the second conductor main surface dents relative to the second resin layer main surface.

[Remark 19]

Regarding the electronic device described in any one of remark 1 through remark 18, the first electronic component is a semiconductor component which comprises a semiconductor.

[Remark 20]

A manufacturing method of an electronic device, comprising:

- a supporting substrate preparing step, preparing a supporting substrate having a substrate main surface and a substrate inner surface, the substrate main surface and the substrate inner surface face opposite sides in a first direction;
- a first conductor forming step, forming a first conductor on the substrate main surface;
- a first resin layer forming step, forming a first resin layer for covering the first conductor;
- a first resin layer grinding step, grinding the first resin layer in the first direction from a side which the substrate main surface faces to a side which the substrate inner surface faces such that a portion of the first conductor is exposed from the first resin layer, so as to respectively form a first conductor main surface and a first resin layer main surface, the first conductor main surface and the first resin layer main surface face the same side as the substrate main surface in the first direction;
- a first wiring layer forming step, forming a first wiring layer straddling the first resin layer main surface and the first conductor main surface;
- a first electronic component mounting step, electrically connecting a first electronic component on the first wiring layer;
- a second resin layer forming step, forming a second resin layer for covering the first wiring layers and the first electronic component;
- a supporting substrate removing step, removing the supporting substrate to expose a first resin layer inner surface facing opposite side with the first resin layer main surface in the first direction; and
- an external electrode forming step, forming an external electrode, the external electrode is disposed closer to the side where the first resin layer inner surface faces than the first resin layer, and the external electrode is electrically connected to the first conductor.

[Remark 21]

Regarding the method described in remark 20, after the supporting substrate preparing step and before the first conductor forming step, further comprises: a second wiring layer forming step, forming a second wiring layer for covering a part of the substrate main surface, and in the first conductor forming step, forming the first conductor on the second wiring layer.

[Remark 22]

The method described in remark 21 further comprises a second electronic component mounting step for electrically connecting a second electronic component on the second wiring layer.

[Remark 23]

The method described in any one of remark 20 through remark 22 after the first resin layer grinding step and before the second resin layer forming step, further comprises a second conductor forming step, forming a second conductor on a part of the first resin layer.

What is claimed is:

1. An electronic device, comprising:

- a first resin layer having a first resin layer main surface and a first resin layer inner surface, the first resin layer main surface and the first resin layer inner surface face opposite sides in a first direction;
- a first conductor having a first conductor main surface and a first conductor inner surface, the first conductor main surface and the first conductor inner surface face opposite sides in the first direction, and the first conductor penetrates the first resin layer in the first direction;
- a first wiring layer formed adjacent to the first resin layer main surface and connected to the first conductor main surface;
- a first electronic component having a first component main surface facing same side as the first resin layer main surface in the first direction and a first component inner surface facing same side as the first resin layer inner surface, and electrically connected with the first wiring layer;
- a second resin layer having a second resin layer main surface facing same direction as the first resin layer main surface and a second resin layer inner surface being in contact with the first resin layer main surface, and covering the first wiring layer and the first electronic component;
- an external electrode disposed closer to a side where the first resin layer inner surface faces than the first resin layer, and electrically connected to the first conductor; and
- a second conductor penetrating the second resin layer in the first direction, wherein the second conductor is disposed on a periphery of the first electronic component when viewed in the first direction.

2. The electronic device of claim 1, wherein a grinding mark is formed on the first resin layer main surface.

3. The electronic device of claim 2, wherein the first conductor main surface dents relative to the first resin layer main surface.

4. The electronic device of claim 1, further comprising: a second wiring layer having a second wiring layer main surface and a second wiring layer inner surface, the second wiring layer main surface and the second wiring layer inner surface facing opposite sides in the first direction, and the second wiring layer inner surface being exposed from the first resin layer inner surface.

5. The electronic device of claim 4, further comprising: a second electronic component is different from the first electronic component, wherein the second electronic component is electrically connected to the second wiring layer, and at least a part of the second electronic component is covered by the first resin layer.

6. The electronic device of claim 5, wherein the second electronic component has a second component main surface

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facing same direction as the first component main surface, and the second component main surface and the first resin layer main surface are coplanar.

7. The electronic device of claim 1, wherein the external electrode comprises a first conductor covering portion for covering the first conductor inner surface. 5

8. The electronic device of claim 4, wherein the external electrode comprises a second wiring layer covering portion for covering a part of the second wiring layer inner surface.

9. The electronic device of claim 8, further comprising: 10
a protective film covering a portion of the second wiring layer inner surface, the portion being exposed from the external electrode.

10. The electronic device of claim 8, wherein the first conductor inner surface is in contact with the second wiring layer main surface. 15

11. The electronic device of claim 1, further comprising: a conductive connecting layer connecting the first electronic component with the first wiring layer, wherein

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a part of the first wiring layer overlaps the first electronic component when viewed in the first direction, and wherein the conductive connecting layer lies between the first component inner surface and the first wiring layer.

12. The electronic device of claim 1, wherein the second conductor is spaced apart from the first wiring layer when viewed in the first direction.

13. The electronic device of claim 12, wherein the second conductor surrounds the first electronic component when viewed in the first direction.

14. The electronic device of claim 1, wherein the second conductor has a second conductor main surface facing a same direction as the second resin layer main surface in the first direction, and the second conductor main surface is exposed from the second resin layer main surface.

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